

## CH 系列 IC 常用封装尺寸

版本: 4

<https://wch.cn>

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| · LQFP144 (LQFP144-20*20) .....               | 5 0 |
| QFN 类 .....                                   | 5 1 |
| · QFN/DFN 命名规则参考 .....                        | 5 1 |
| · QFN10 (QFN10-1.4*1.8*0.55-0.4) .....        | 5 2 |
| · QFN10F15X2 (QFN10-1.5*2*0.55-0.4) .....     | 5 3 |
| · QFN12 (QFN12-2*2*0.5-0.4) .....             | 5 4 |
| · QFN14 (QFN14-2*2*0.55-0.4) .....            | 5 5 |
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| · QFN16 (QFN16-3*3-0.5) .....                 | 5 8 |
| · QFN16L (QFN16L-3*3-0.5) .....               | 5 9 |
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| · QFN16X4 (QFN16-4*4-0.65) .....              | 6 1 |
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| · QFN20X4 (QFN20-4*4-0.5) .....               | 6 6 |
| · QFN20X25X45 (QFN20-2.5*4.5*0.75-0.5) .....  | 6 7 |
| · QFN20C2 (QFN20C-2*2*0.55-0.32) .....        | 6 8 |
| · QFN24 (QFN24-4*4-0.5) .....                 | 6 9 |
| · QFN24X3 (QFN24-3*3*0.55-0.4) .....          | 7 0 |
| · QFN24C2X3 (QFN24C-2*3-0.35) .....           | 7 1 |
| · QFN26C3 (QFN26C-3*3*0.75-0.4) .....         | 7 2 |
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| · QFN28X5 (QFN28-5*5-0.5) .....               | 7 4 |
| · QFN30C3 (QFN30C-3*3-0.35) .....             | 7 5 |
| · QFN32 (QFN32-4*4-0.4) .....                 | 7 6 |
| · QFN32X5 (QFN32-5*5-0.5) .....               | 7 7 |
| · QFN32C28X45 (QFN32C-2.8*4.5*0.75-0.4) ..... | 7 8 |
| · QFN36C4 (QFN36C-4*4*0.75-0.4) .....         | 7 9 |
| · QFN36X4 (QFN36-4*4-0.35) .....              | 8 0 |
| · QFN36X6 (QFN36-6*6-0.5) .....               | 8 1 |
| · QFN40 (QFN40-5*5-0.4) .....                 | 8 2 |
| · QFN40C4 (QFN40C-4*4-0.35) .....             | 8 3 |
| · QFN40X6 (QFN40-6*6-0.5) .....               | 8 4 |
| · QFN42X35X9 (QFN42-3.5*9*0.75-0.5) .....     | 8 5 |
| · QFN42C35X9 (QFN42C-3.5*9*0.75-0.5) .....    | 8 6 |
| · QFN46C5 (QFN46C-5*5*0.75-0.4) .....         | 8 7 |
| · QFN48 (QFN48-5*5*0.75-0.35) .....           | 8 8 |

|                                           |       |
|-------------------------------------------|-------|
| . QFN48-5*5*0.55-0.35 .....               | 8 9   |
| . QFN48X6 (QFN48-6*6-0.4) .....           | 9 0   |
| . QFN48X7 (QFN48-7*7-0.5) .....           | 9 1   |
| . QFN52X35X9 (QFN52-3.5*9*0.75-0.4) ..... | 9 2   |
| . QFN56 (QFN56-7*7-0.4) .....             | 9 3   |
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| . QFN60X6 (QFN60-6*6-0.35) .....          | 9 6   |
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| . QFN68 (QFN68-8*8-0.4) .....             | 9 9   |
| . QFN68X7 (QFN68-7*7-0.35) .....          | 1 0 0 |
| . QFN76 (QFN76-9*9-0.4) .....             | 1 0 1 |
| . QFN80 (QFN80-9*9-0.4) .....             | 1 0 2 |
| . QFN88 (QFN88-10*10-0.4) .....           | 1 0 3 |
| . QFN88X9 (QFN88-9*9-0.35) .....          | 1 0 4 |
| . QFN96 (QFN96-10*10-0.35) .....          | 1 0 5 |
| . QFN128 (QFN128-12.3*12.3-0.35) .....    | 1 0 6 |
| DFN 类.....                                | 1 0 7 |
| . DFN6X1.2 (DFN6-1.2*1.2*0.45-0.4) .....  | 1 0 7 |
| . DFN6 (DFN6-2*2-0.65) .....              | 1 0 8 |
| . DFN8 (DFN8-2*2-0.5) .....               | 1 0 9 |
| . DFN10 (DFN10-2*2-0.4) .....             | 1 1 0 |
| . DFN10X3 (DFN10-3*3-0.5) .....           | 1 1 1 |
| SOT 类.....                                | 1 1 2 |
| . SOT23-3 .....                           | 1 1 2 |
| . SOT23-5、TSOT23-5L、SOT25 .....           | 1 1 3 |
| . SOT23-6、TSOT23-6L、SOT26 .....           | 1 1 4 |
| . SOT23-8、TSOT23-8L、SOT28 .....           | 1 1 5 |
| . SOT363 (SC70-6L) .....                  | 1 1 6 |
| . SOT353 (SC70-5L) .....                  | 1 1 7 |

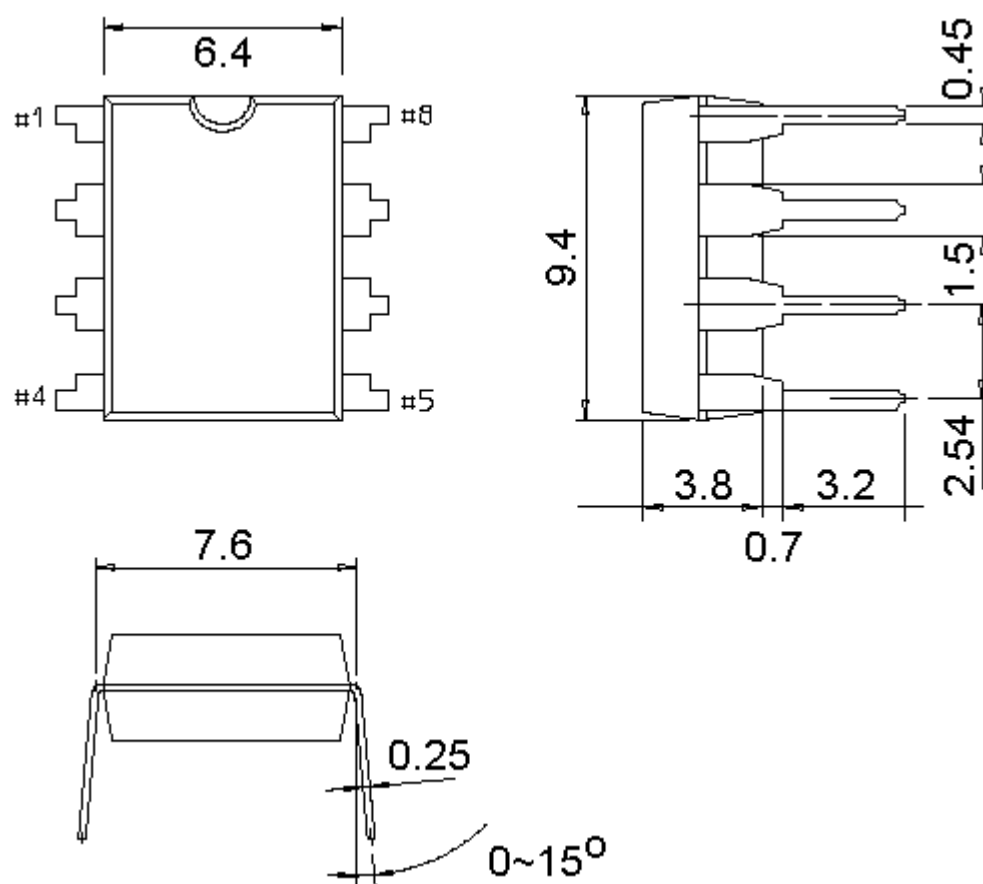
说明：所有尺寸标注的单位都是 mm（毫米），

引脚节距（中心间距）总是标称值，没有误差值；

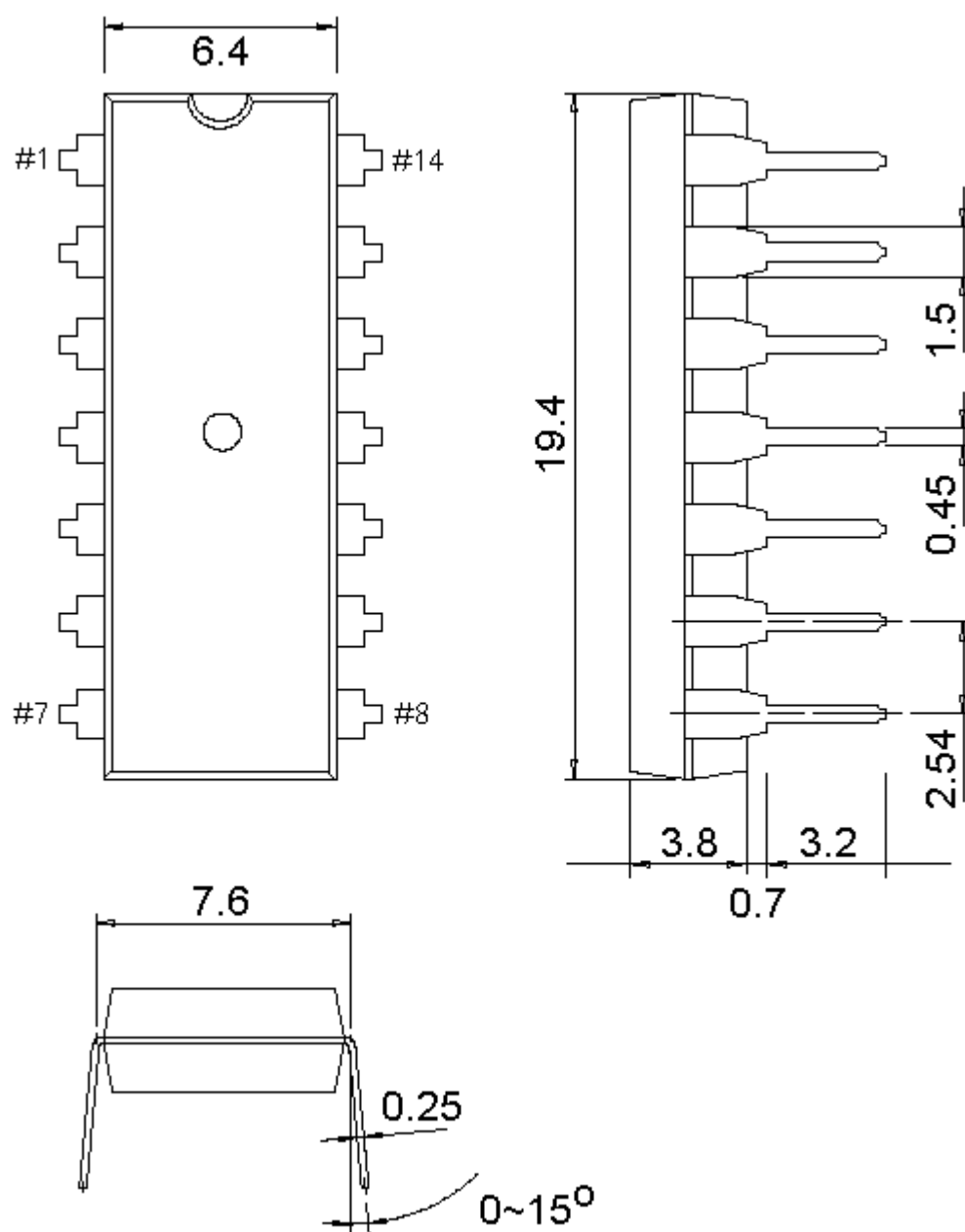
除此之外的尺寸误差不大于  $\pm 0.2\text{mm}$  或者  $\pm 10\%$  两者中的较大值。

## DIP 类 (PDIP)

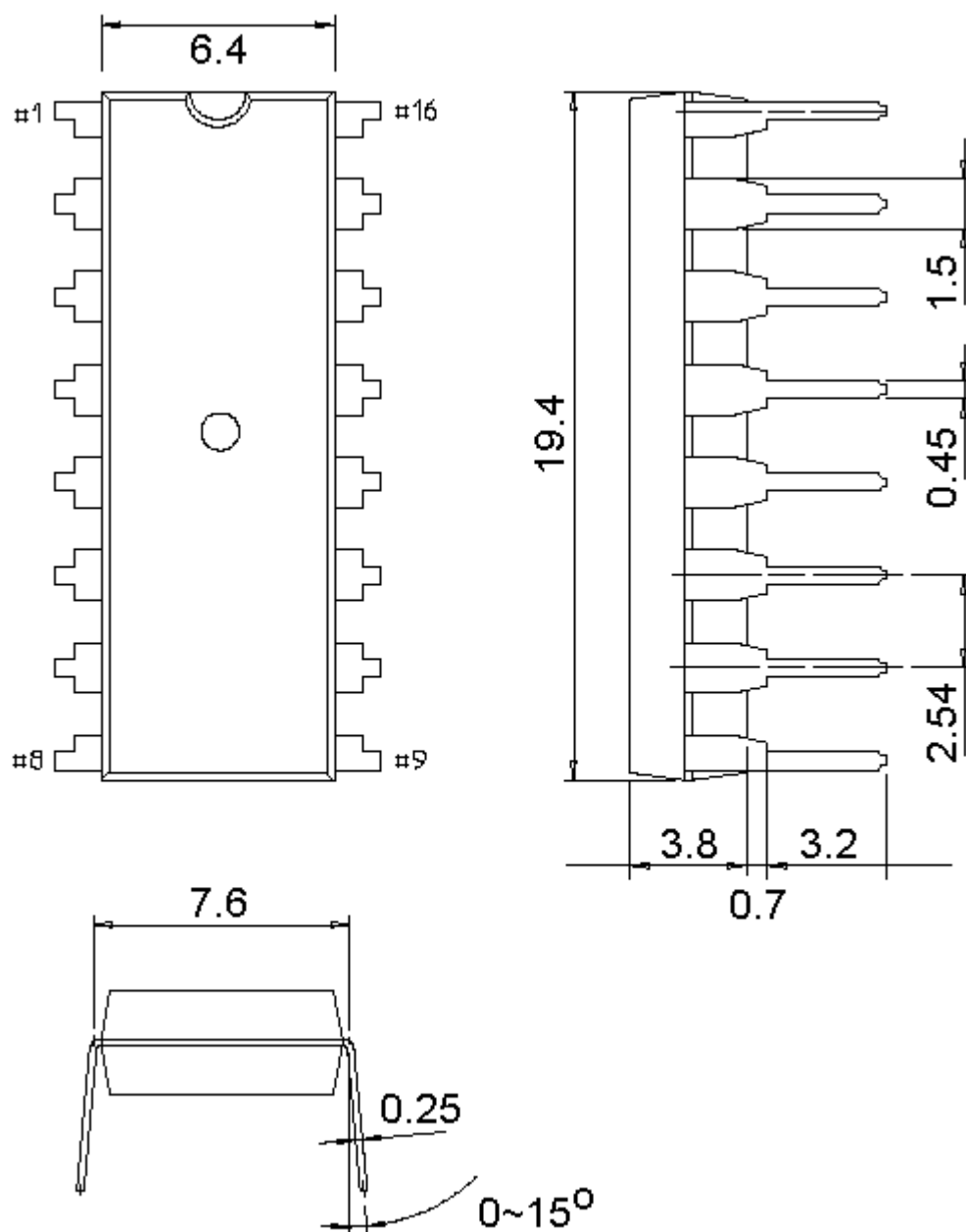
## . DIP8



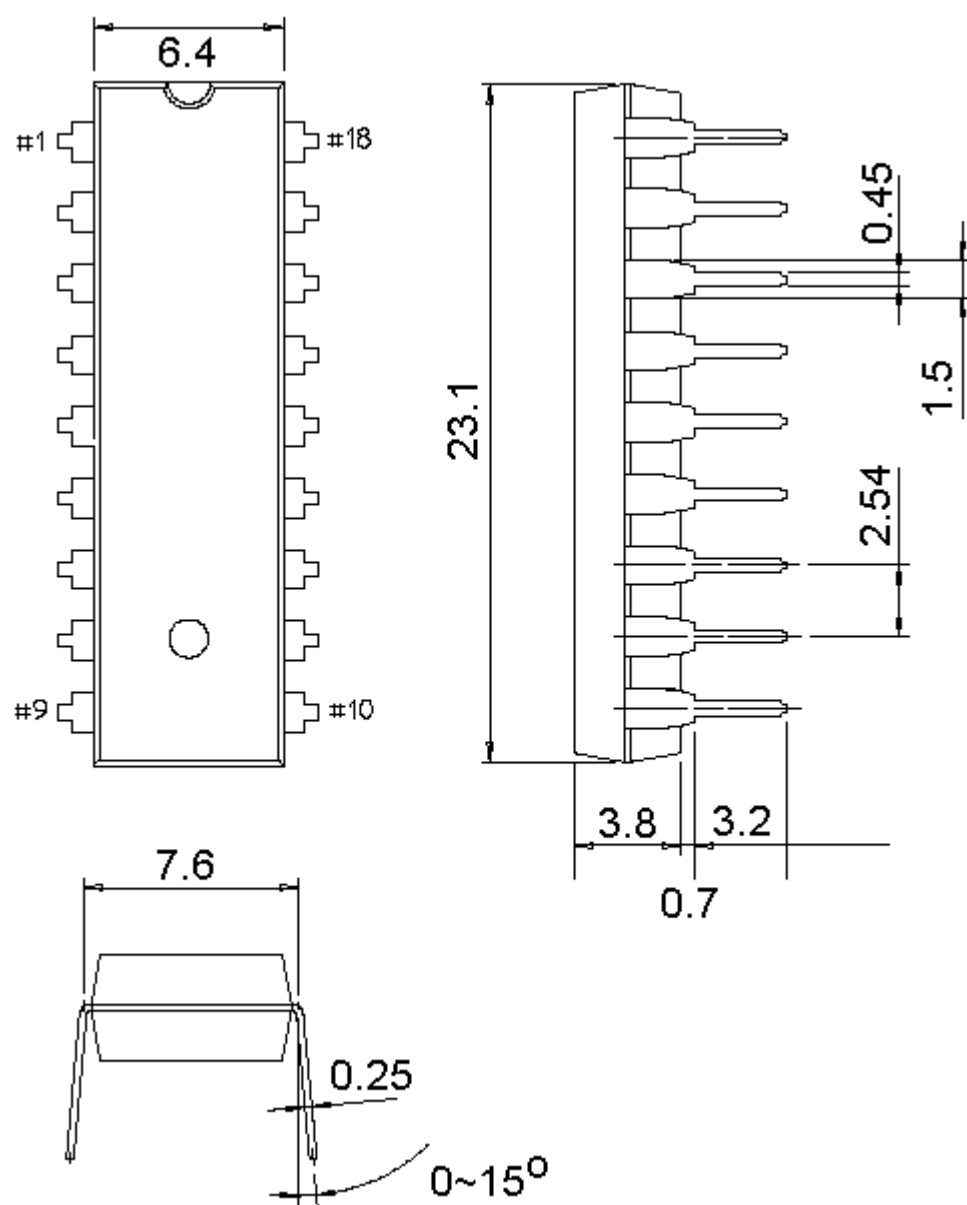
## . DIP14



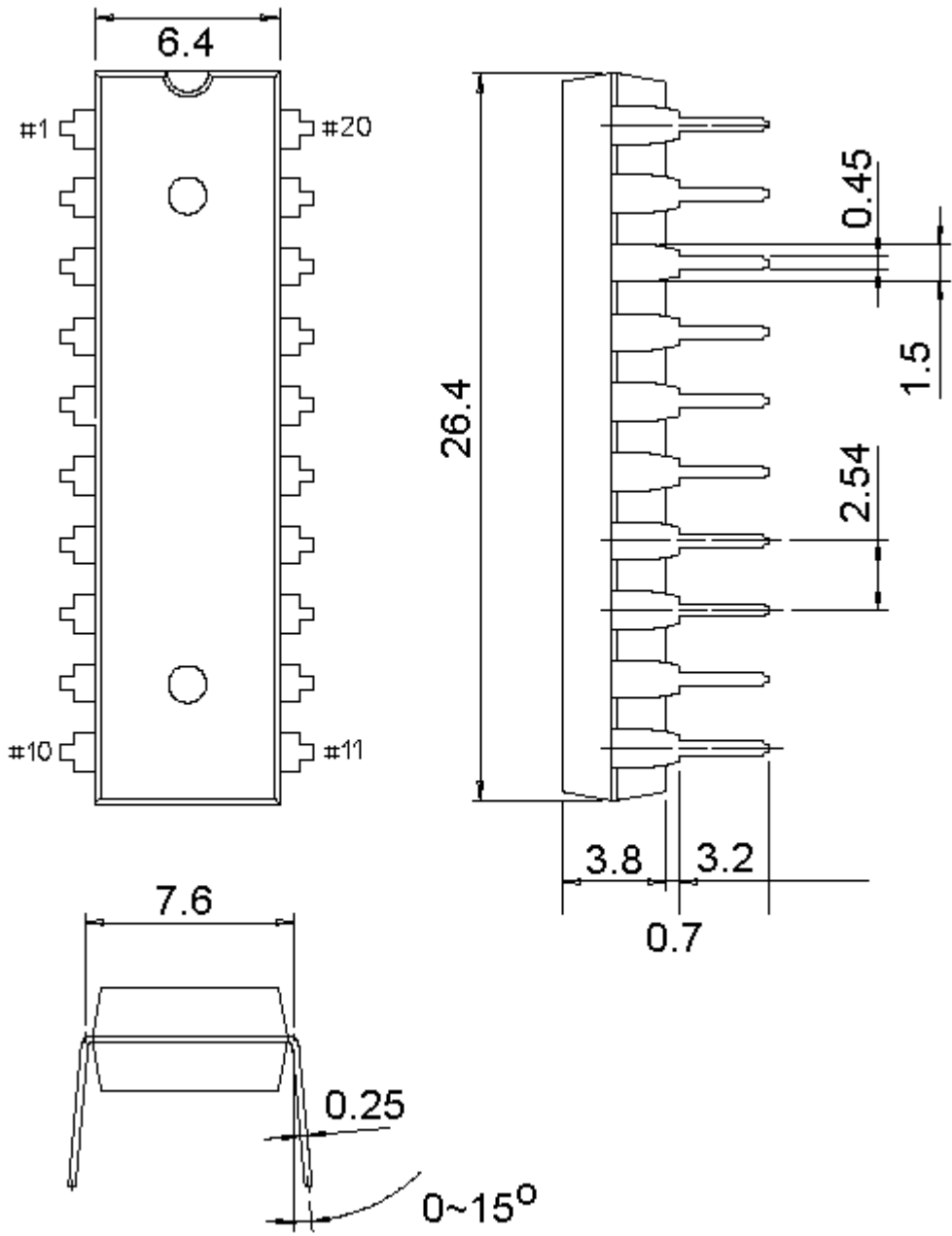
## . DIP16



## . DIP18

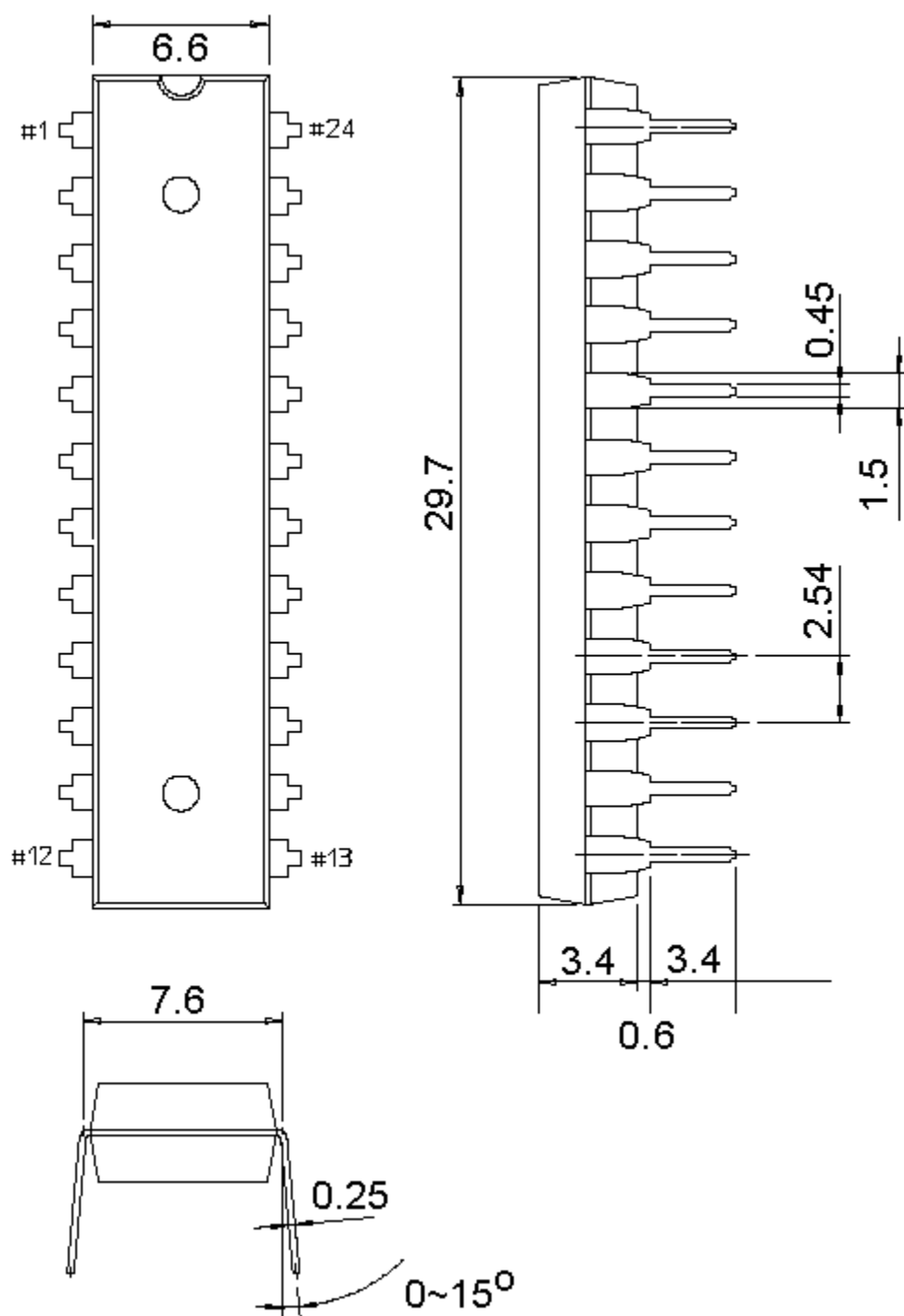


. DIP20

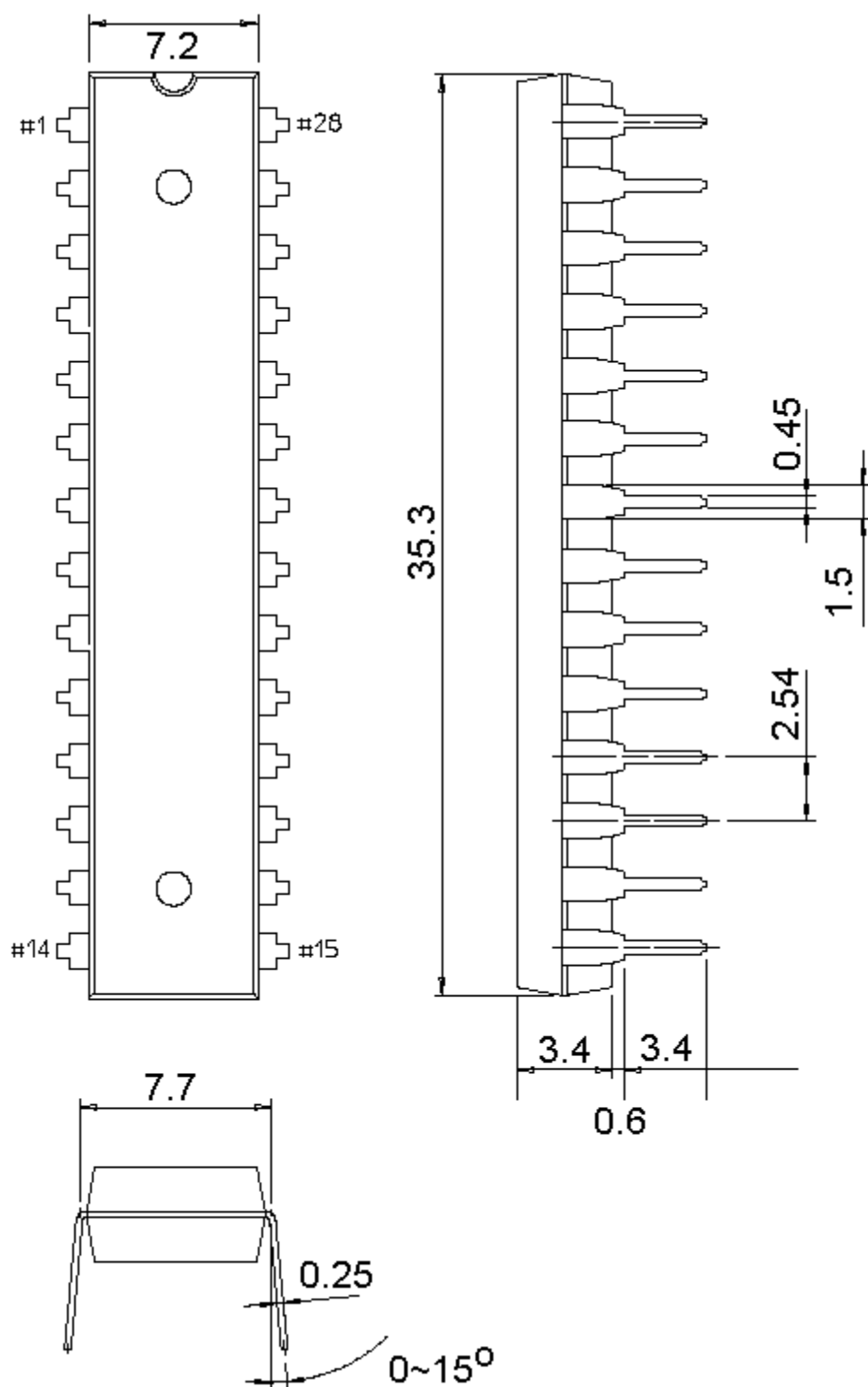




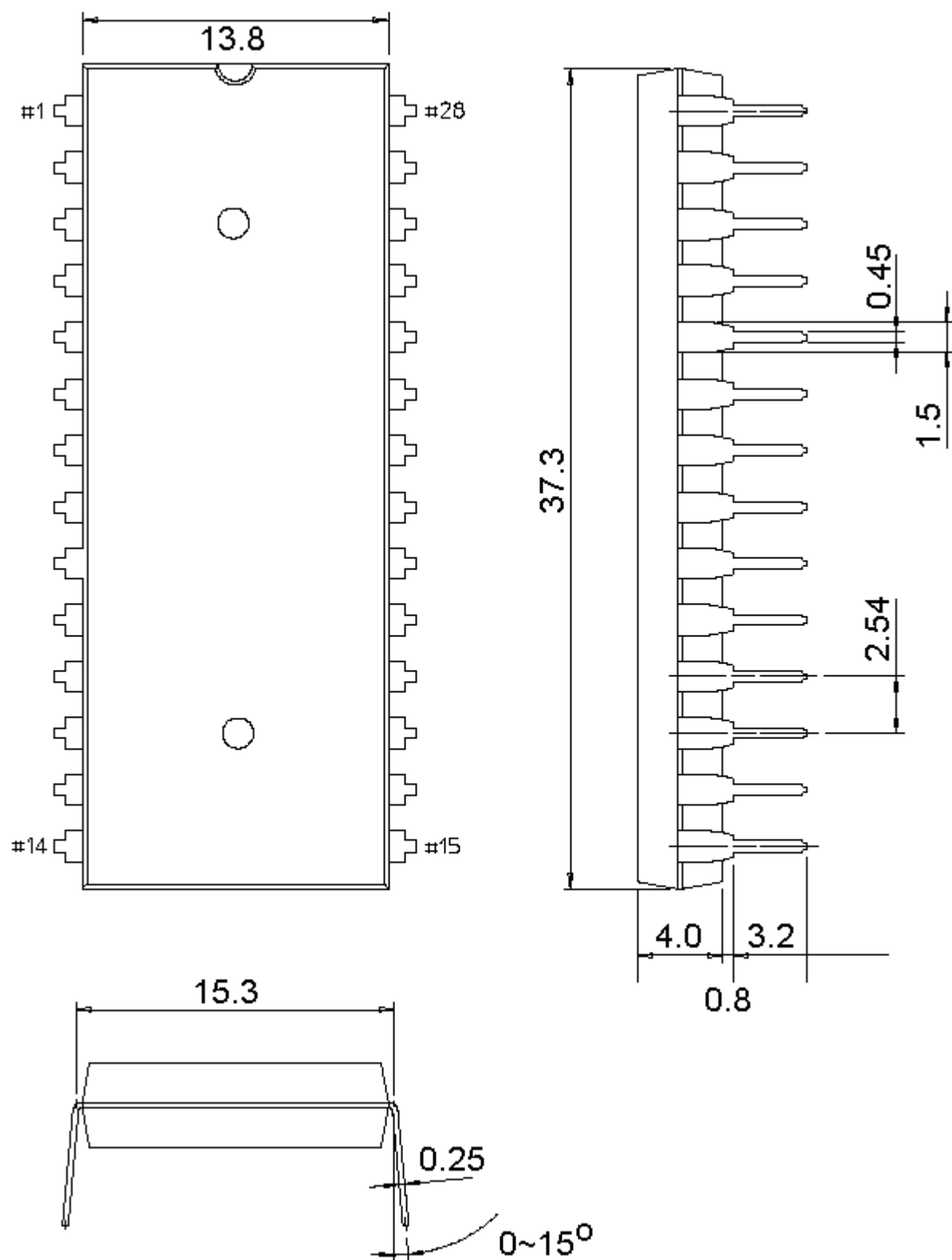
## . DIP24S (SKDIP24、DIP24-300mil)



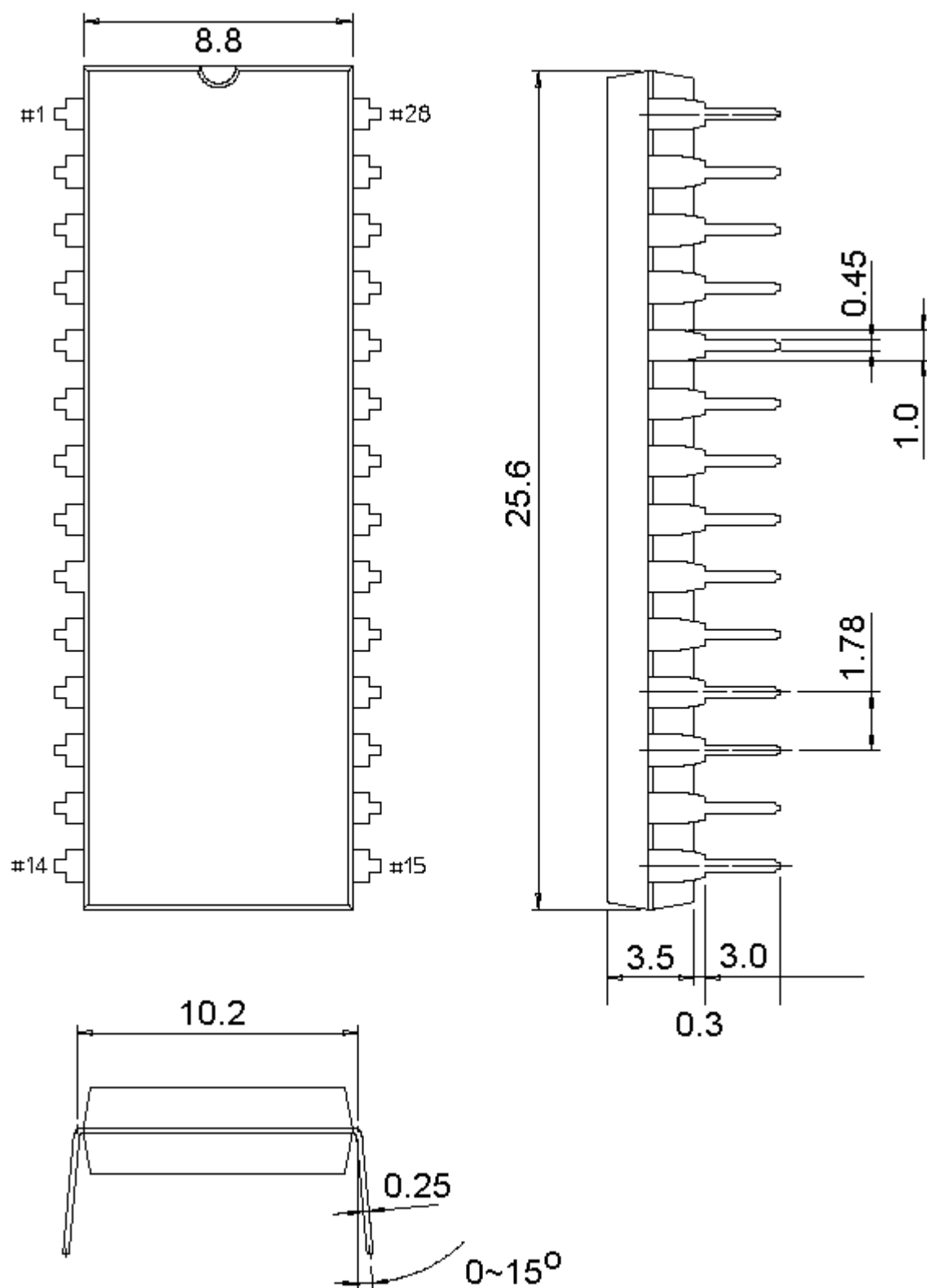
## . DIP28S (SKDIP28、DIP28-300mil)



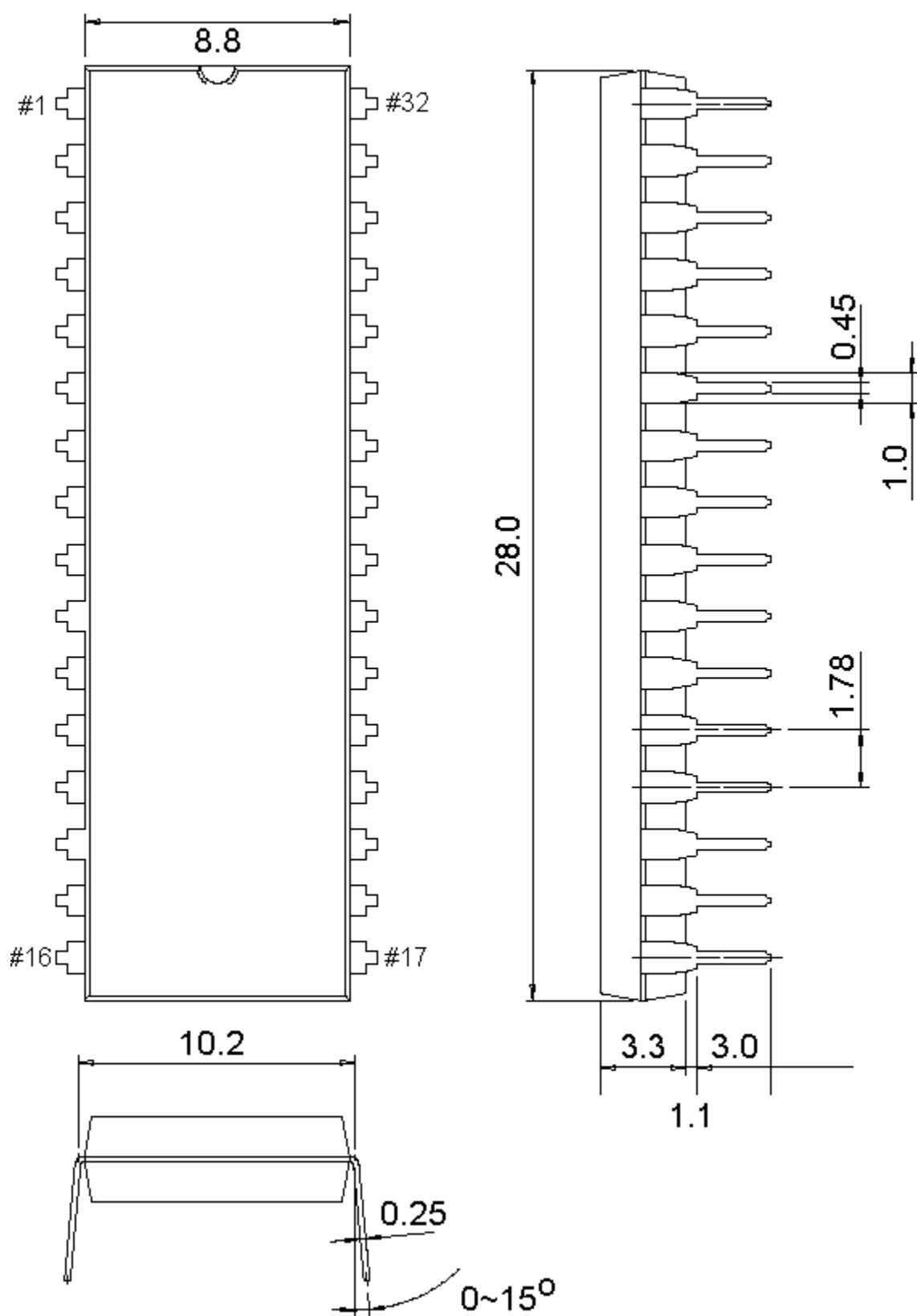
## . DIP28 (DIP28W、DIP28-600mil)



## . SDIP28 (SDIP28-400mil)



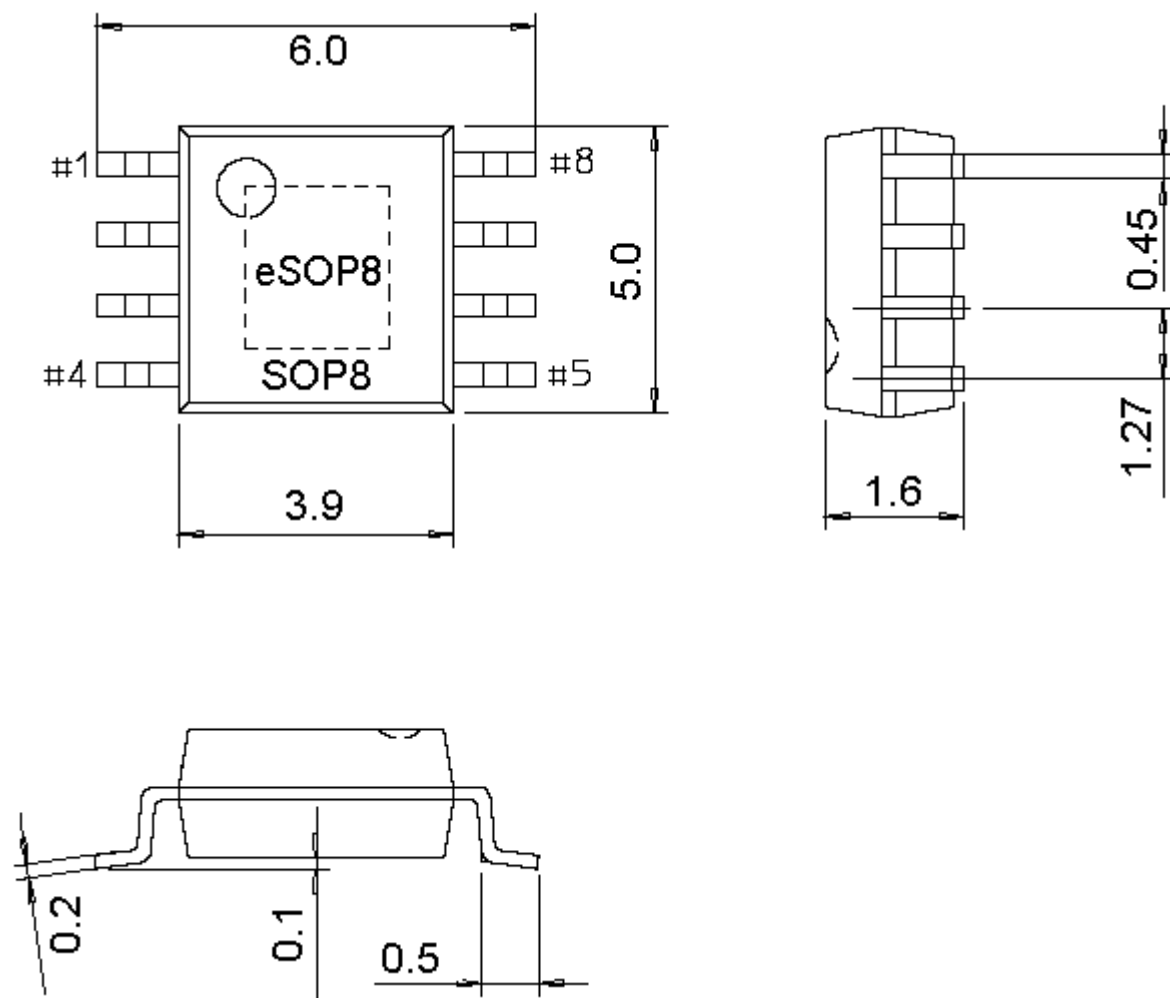
## . SDIP32 (SDIP32-400mil)



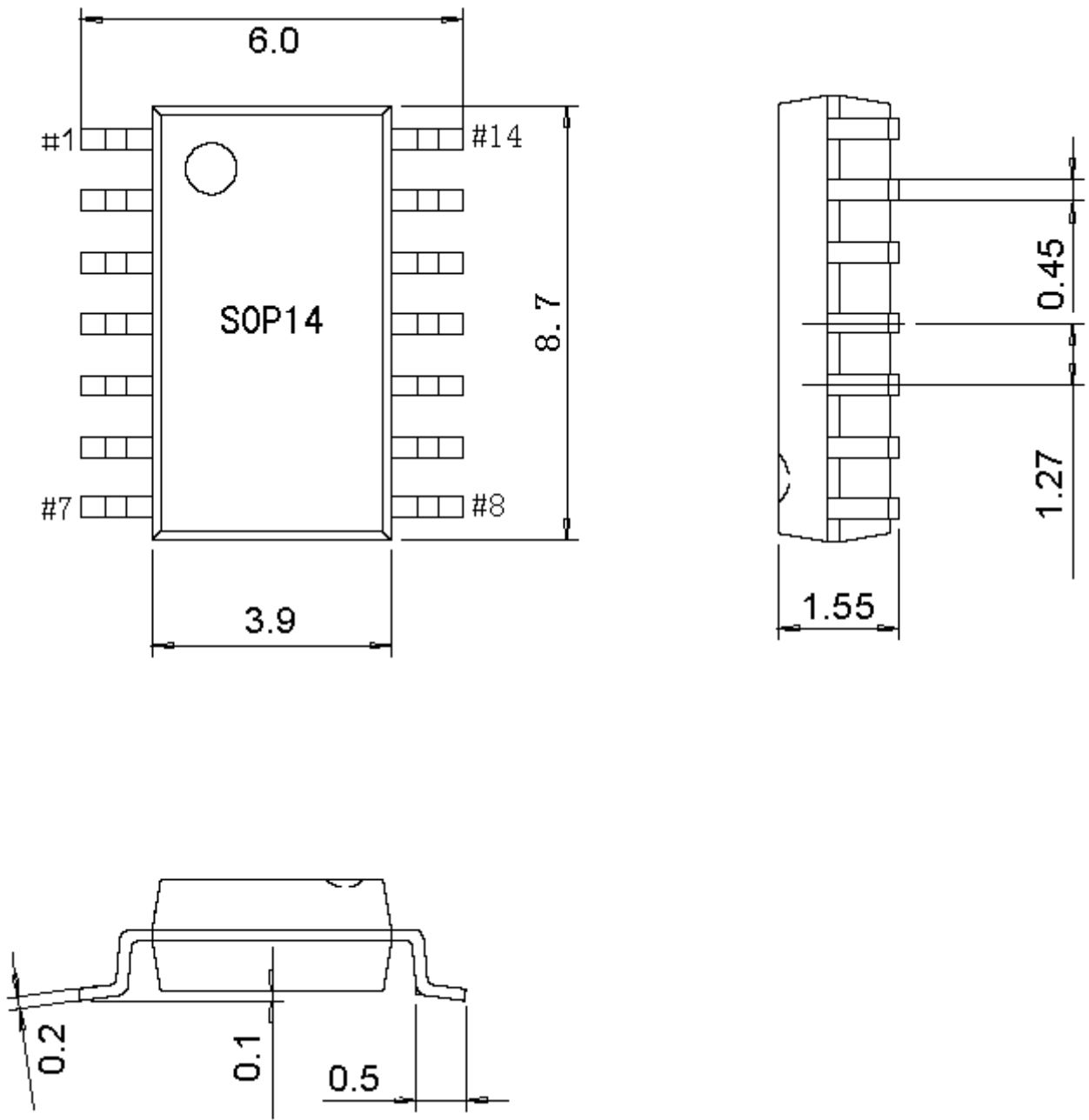
## SOP 类 (S01C)

## . SOP8、eSOP8

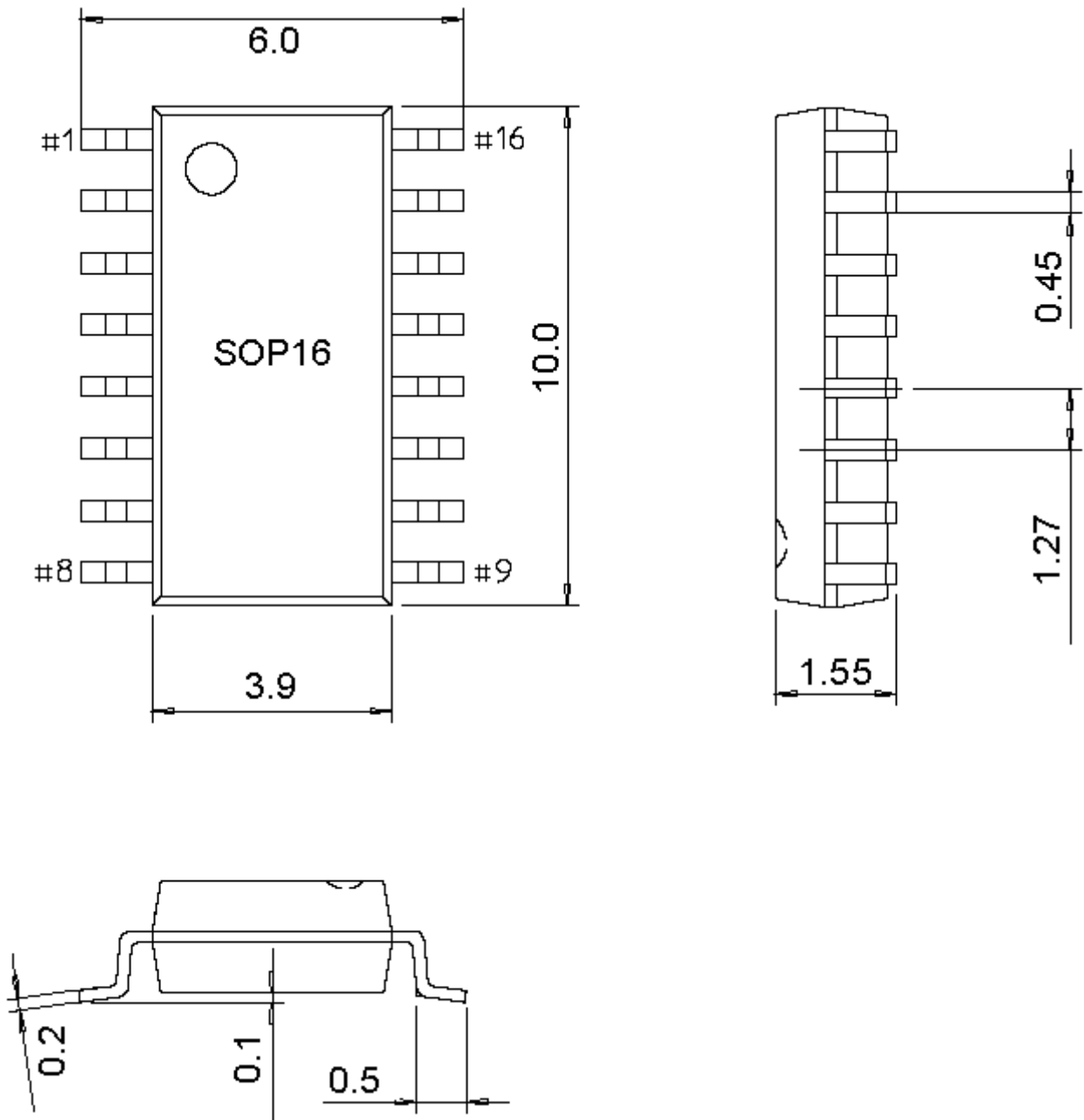
eSOP8 Exposed Pad: 2.1\*3.3、2.1\*2.1



SOP14

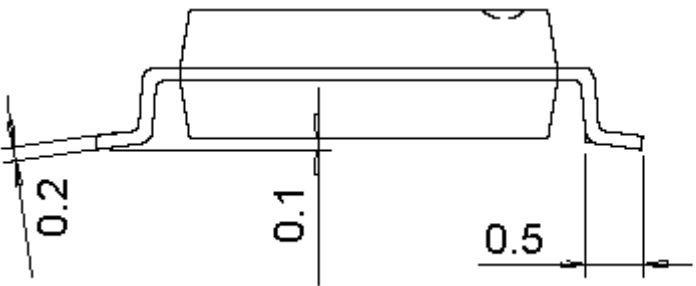
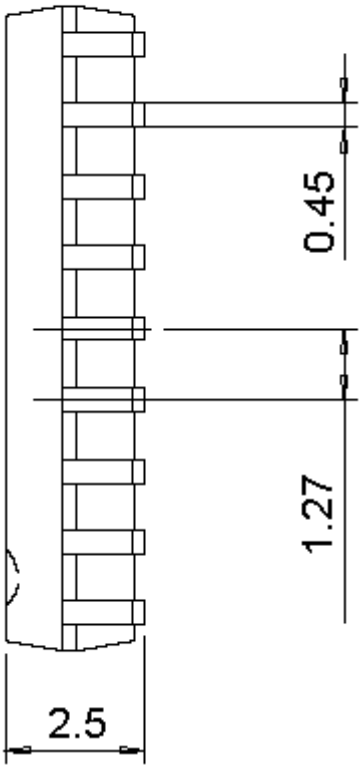
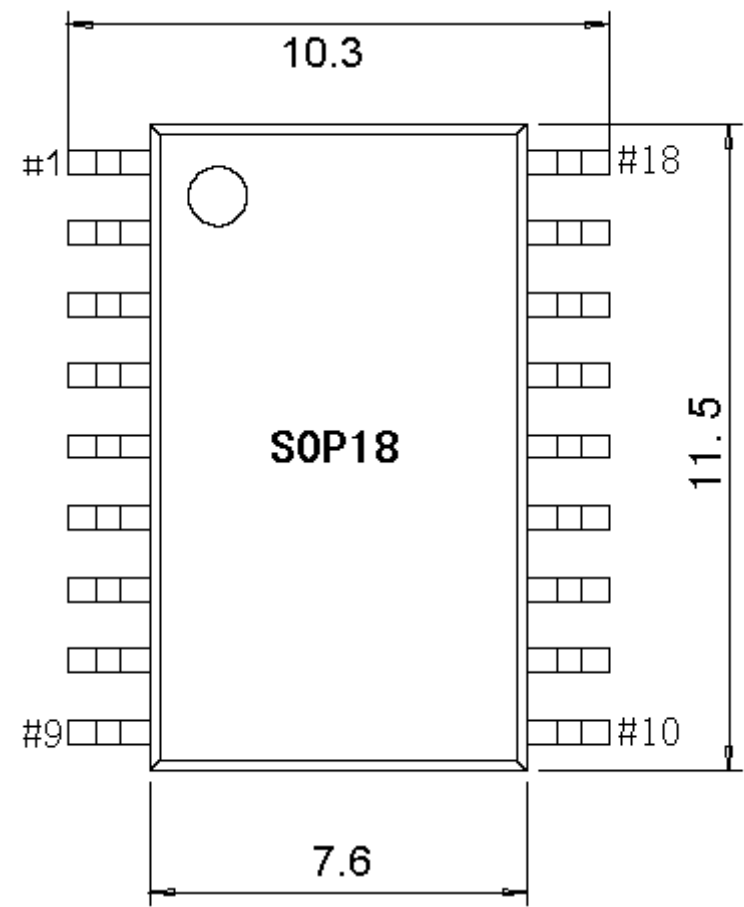


· SOP16 (SOP16-150mil)

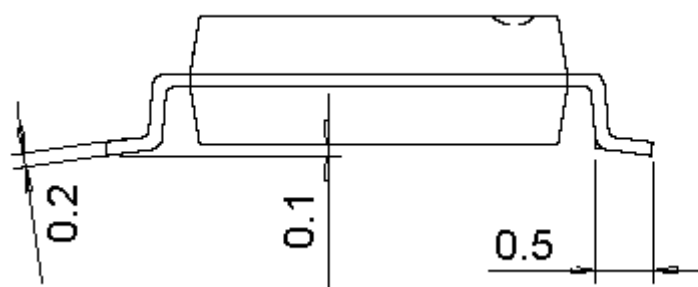
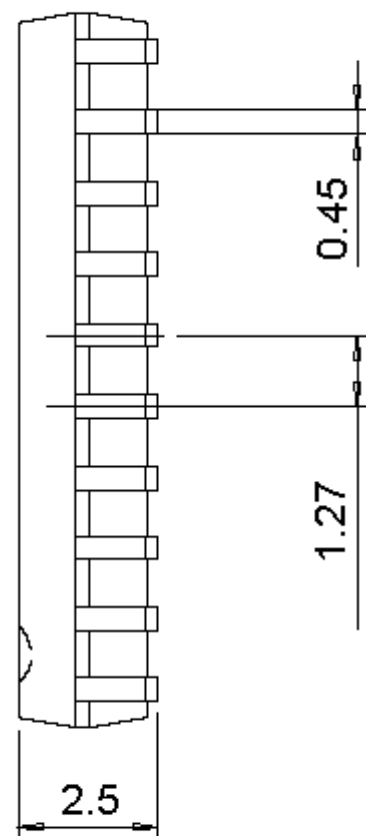
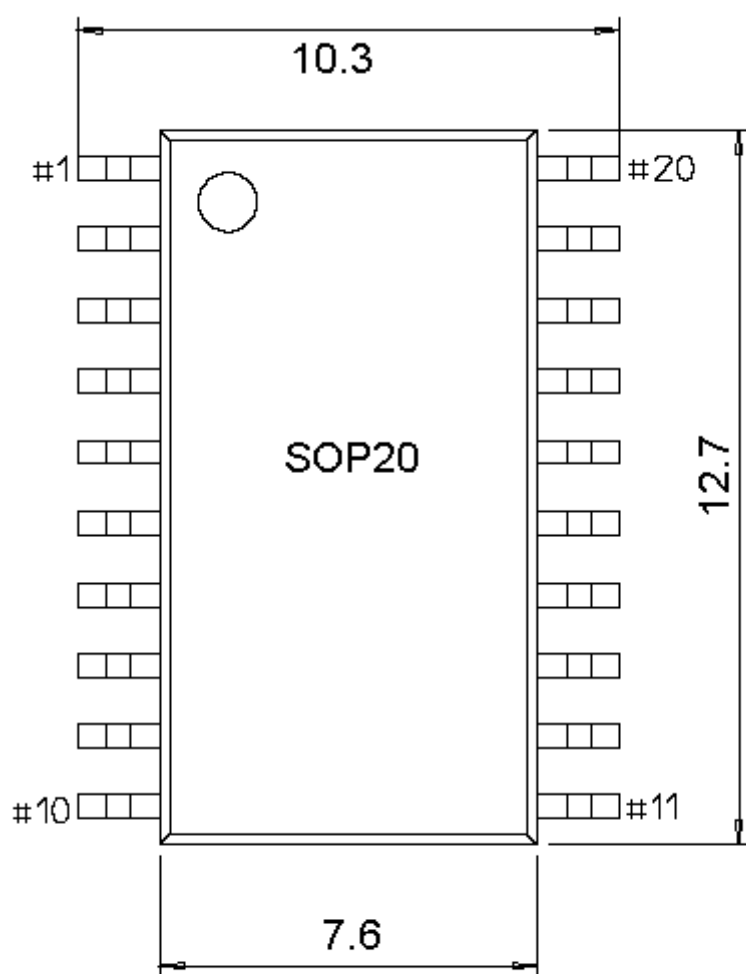




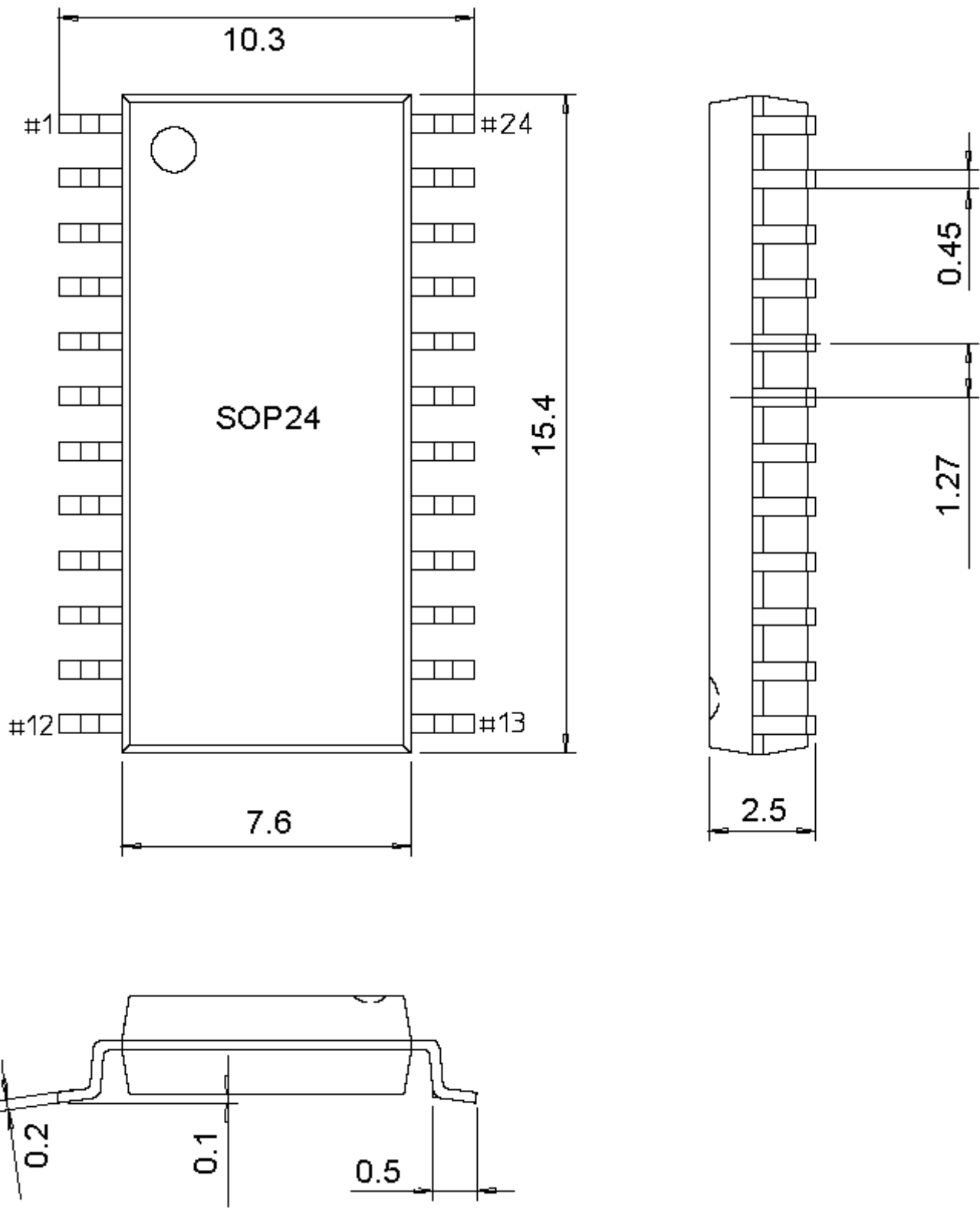
SOP18



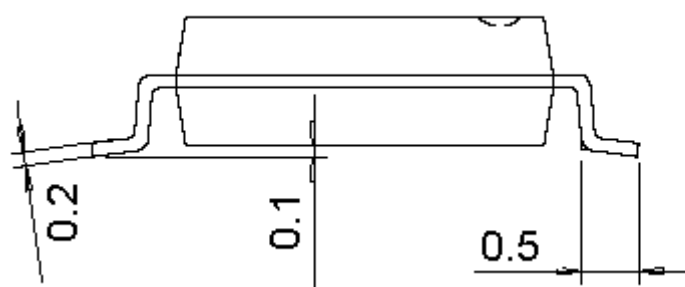
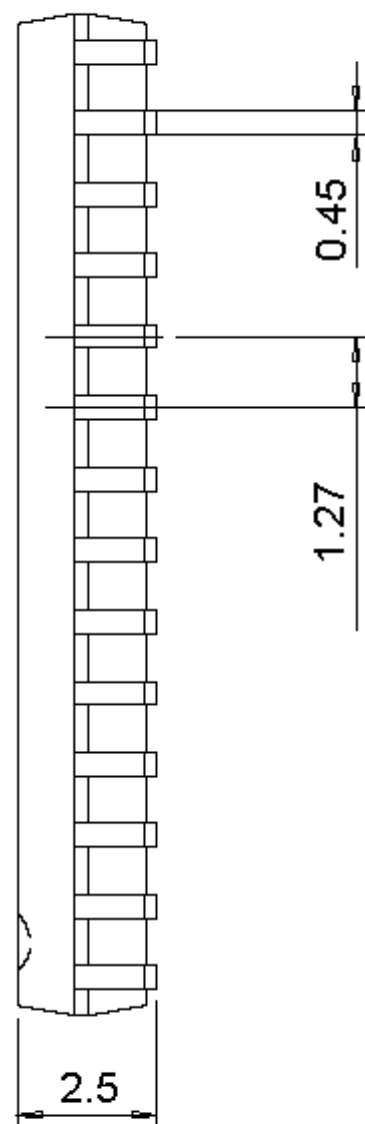
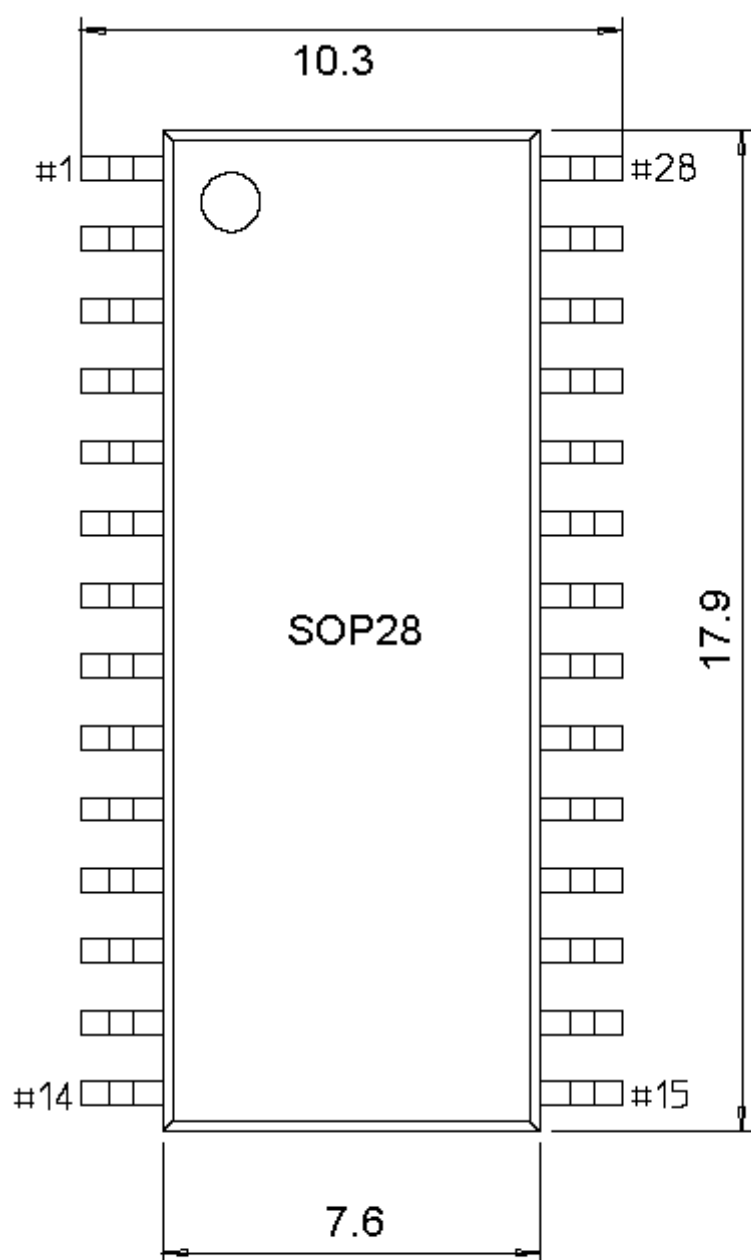
## SOP20



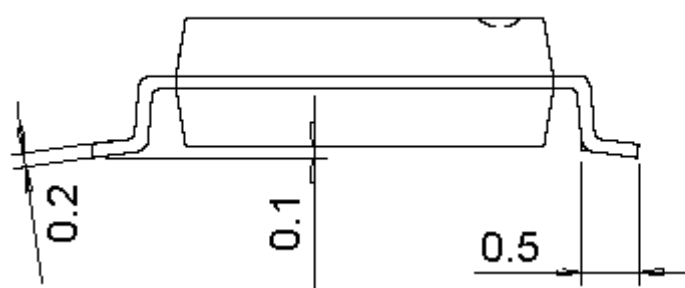
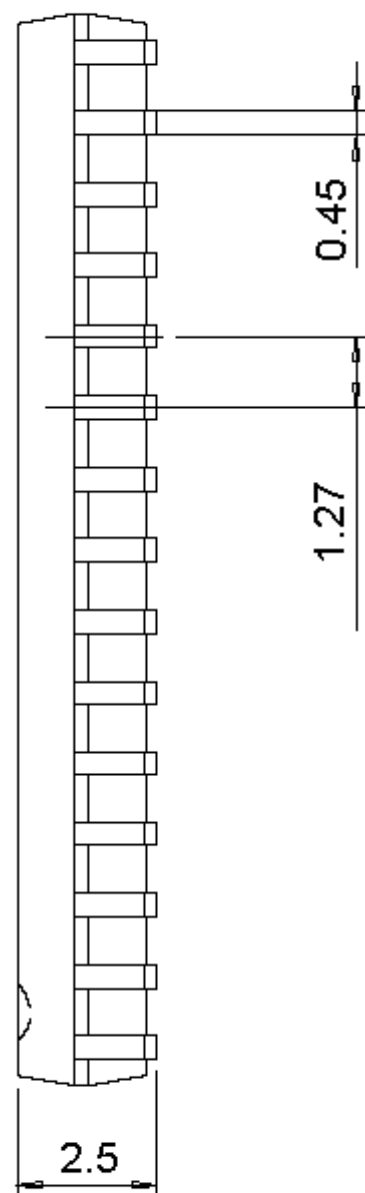
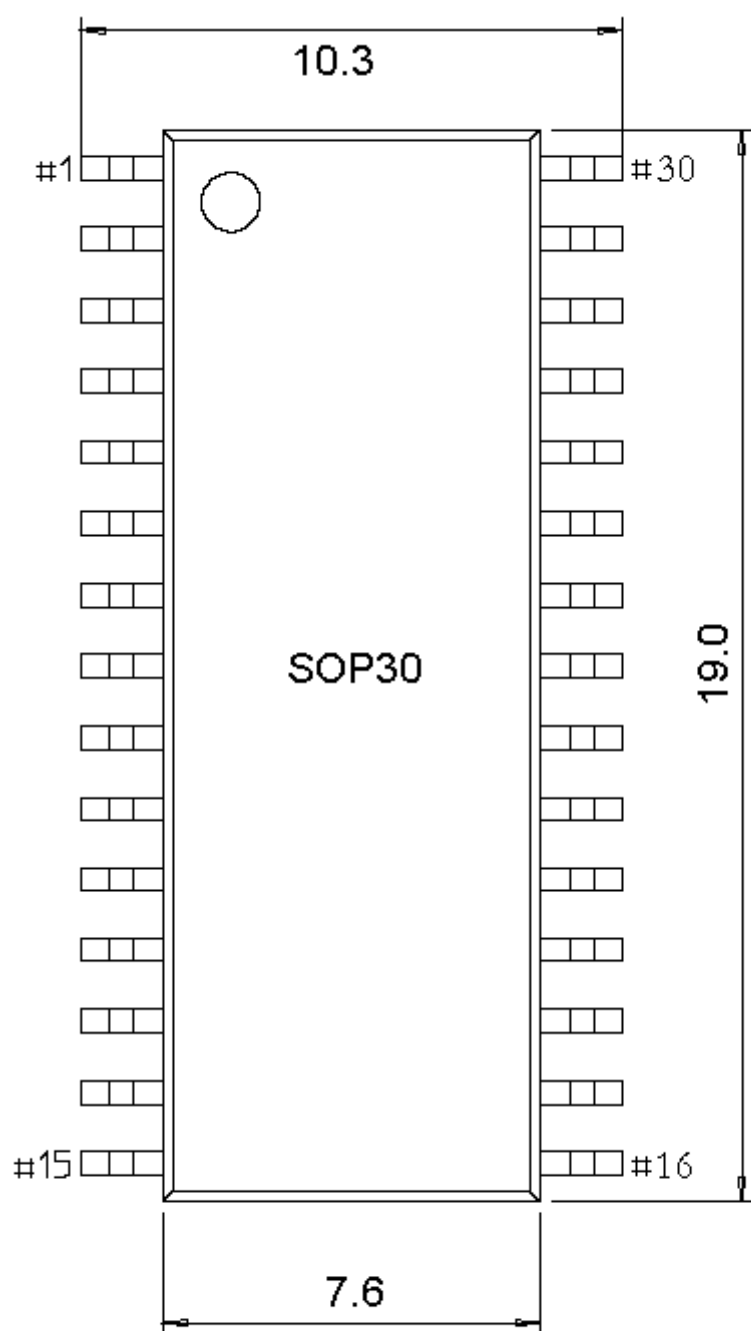
SOP24



## SOP28



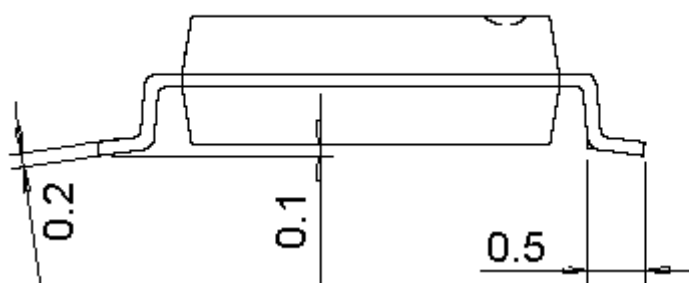
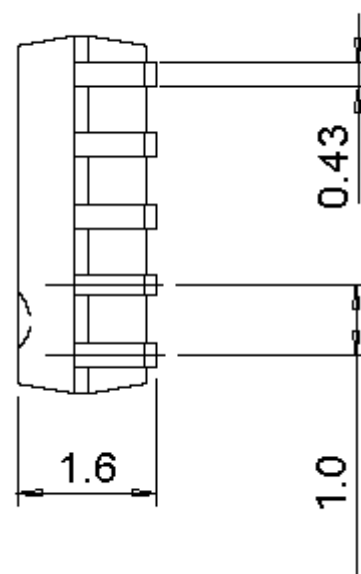
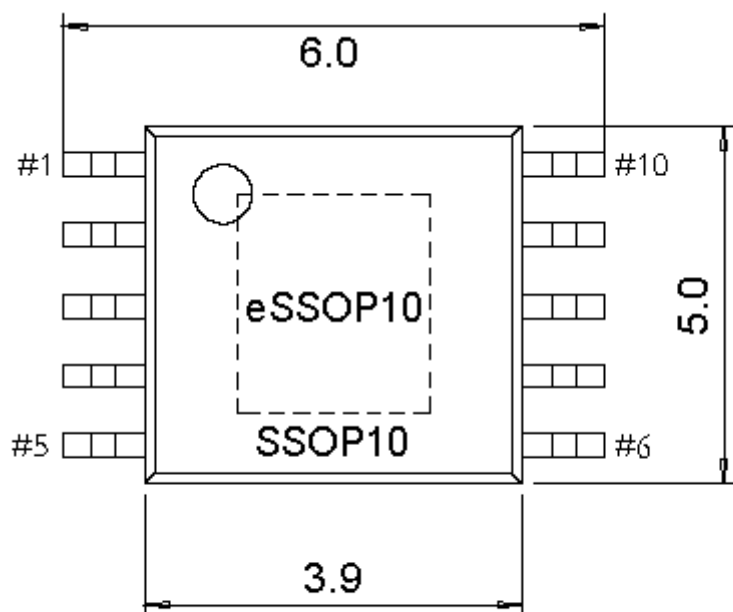
SOP30



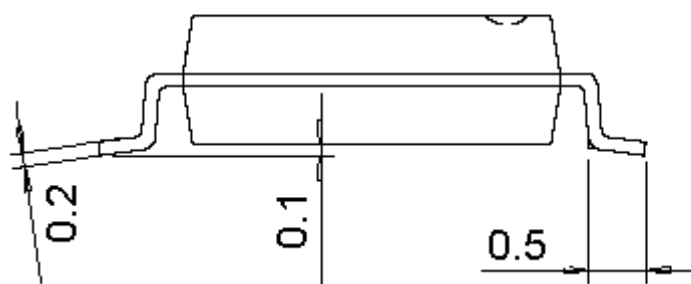
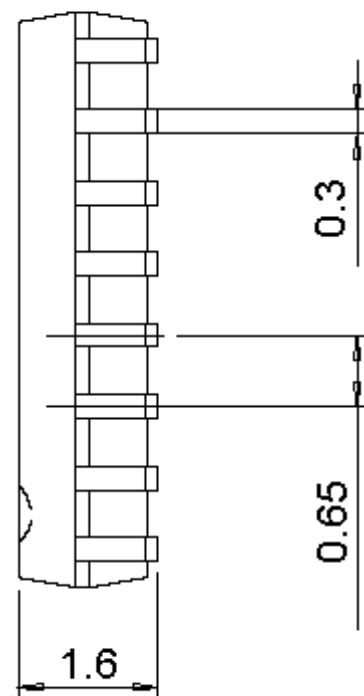
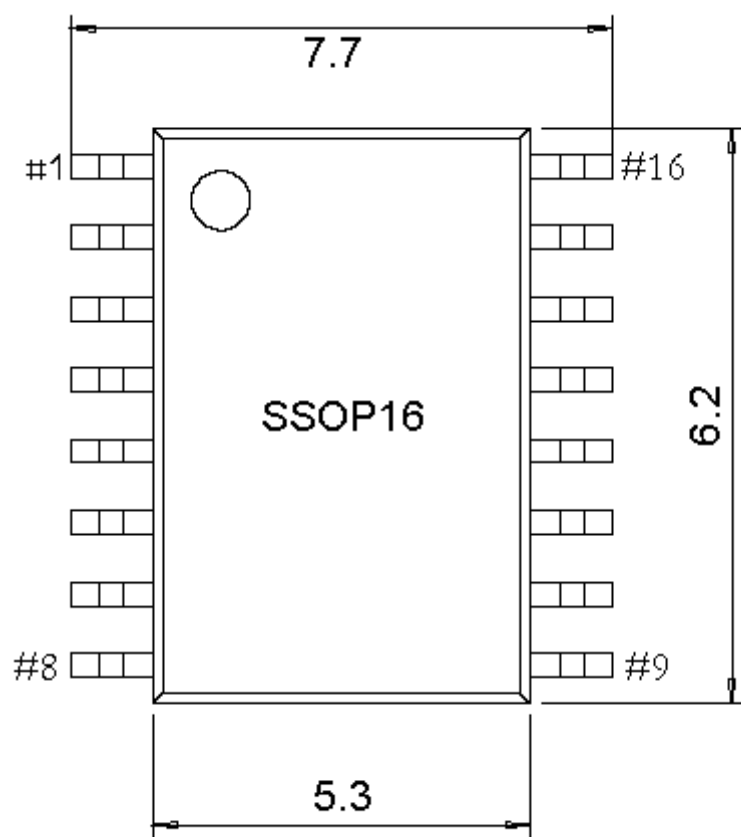
## SSOP 类

. SSOP10 (SSOP10-150mil-1.00)、eSSOP10

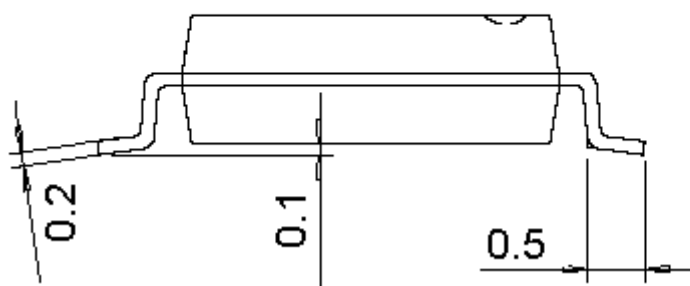
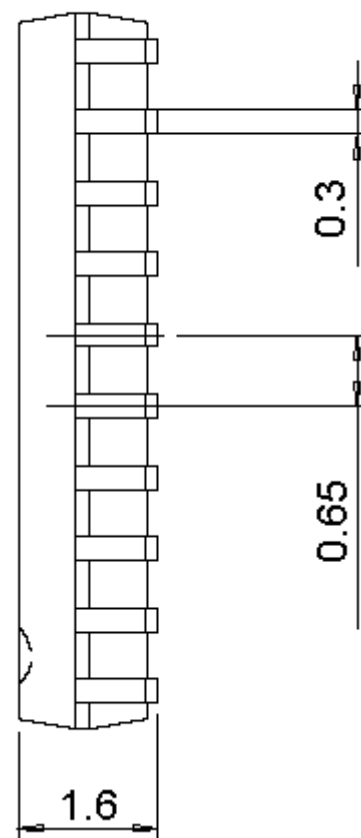
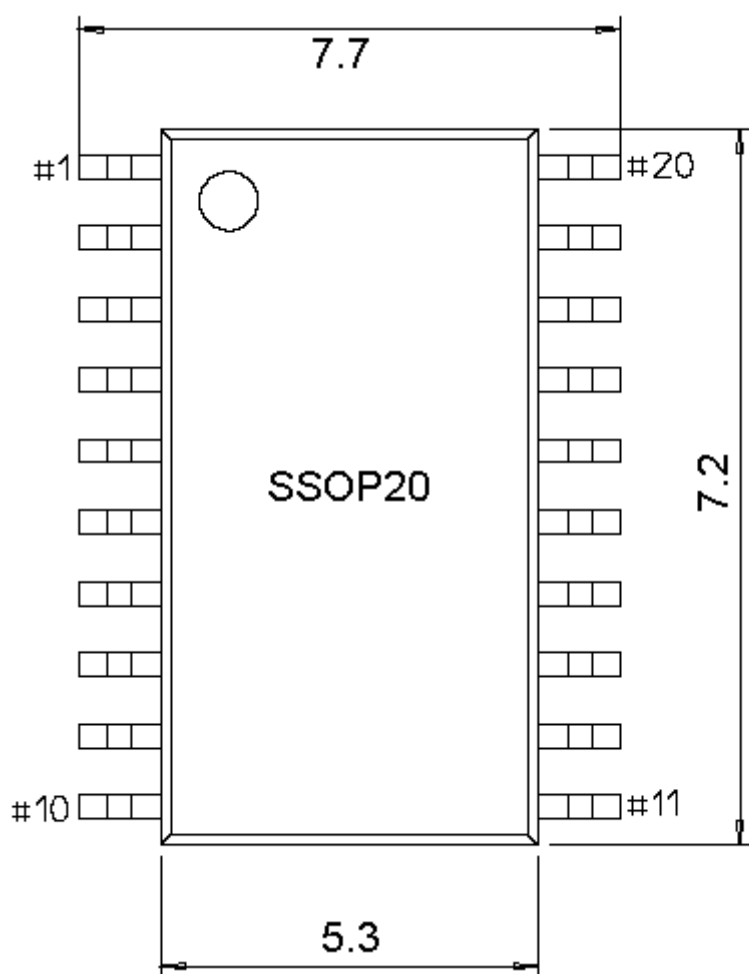
eSSOP10 Exposed Pad: 2.1\*3.3



## . SSOP16 (SSOP16-209mil-0.65)

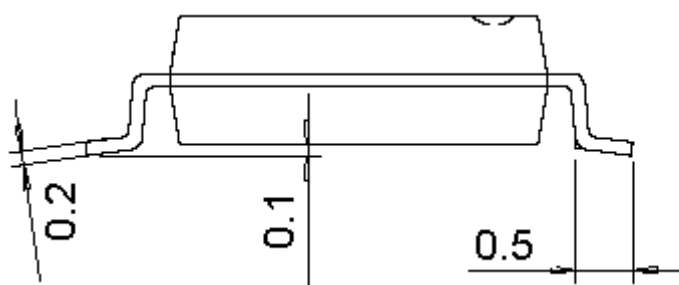
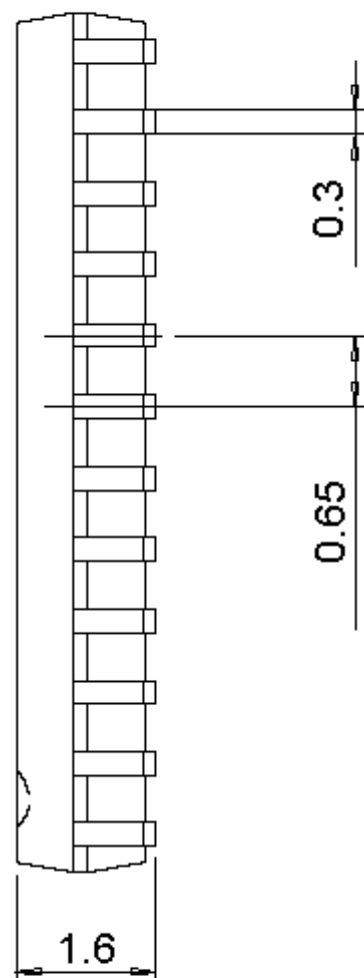
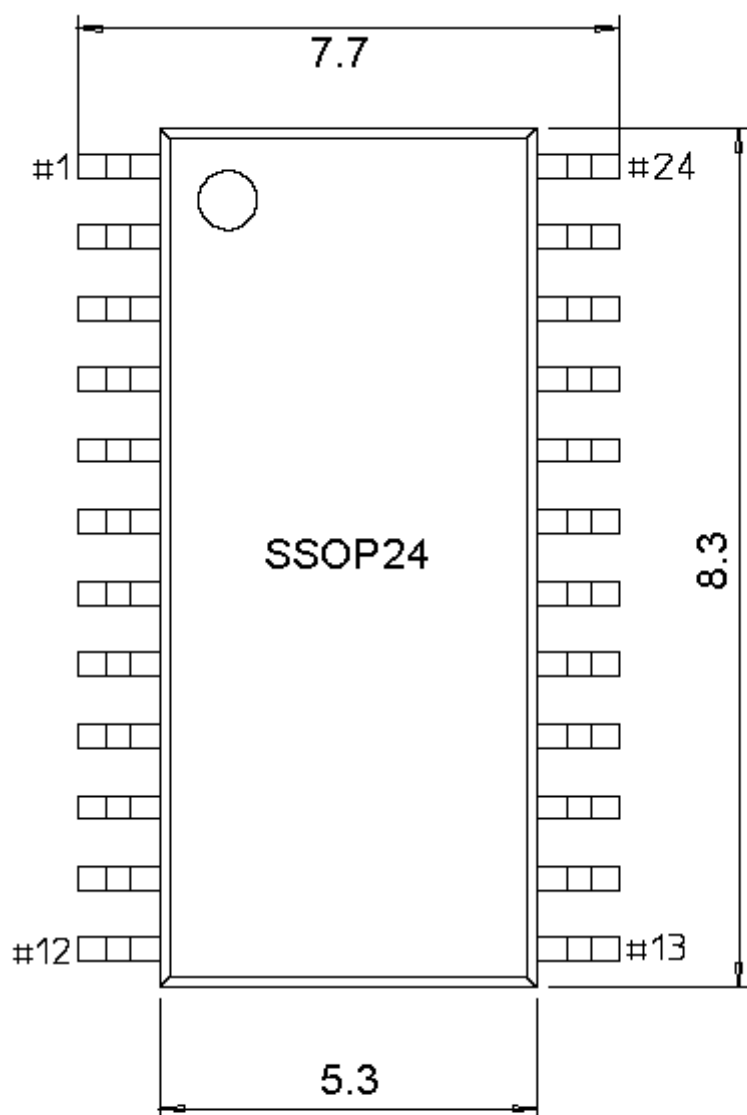


## . SSOP20 (SSOP20-209mil-0.65)

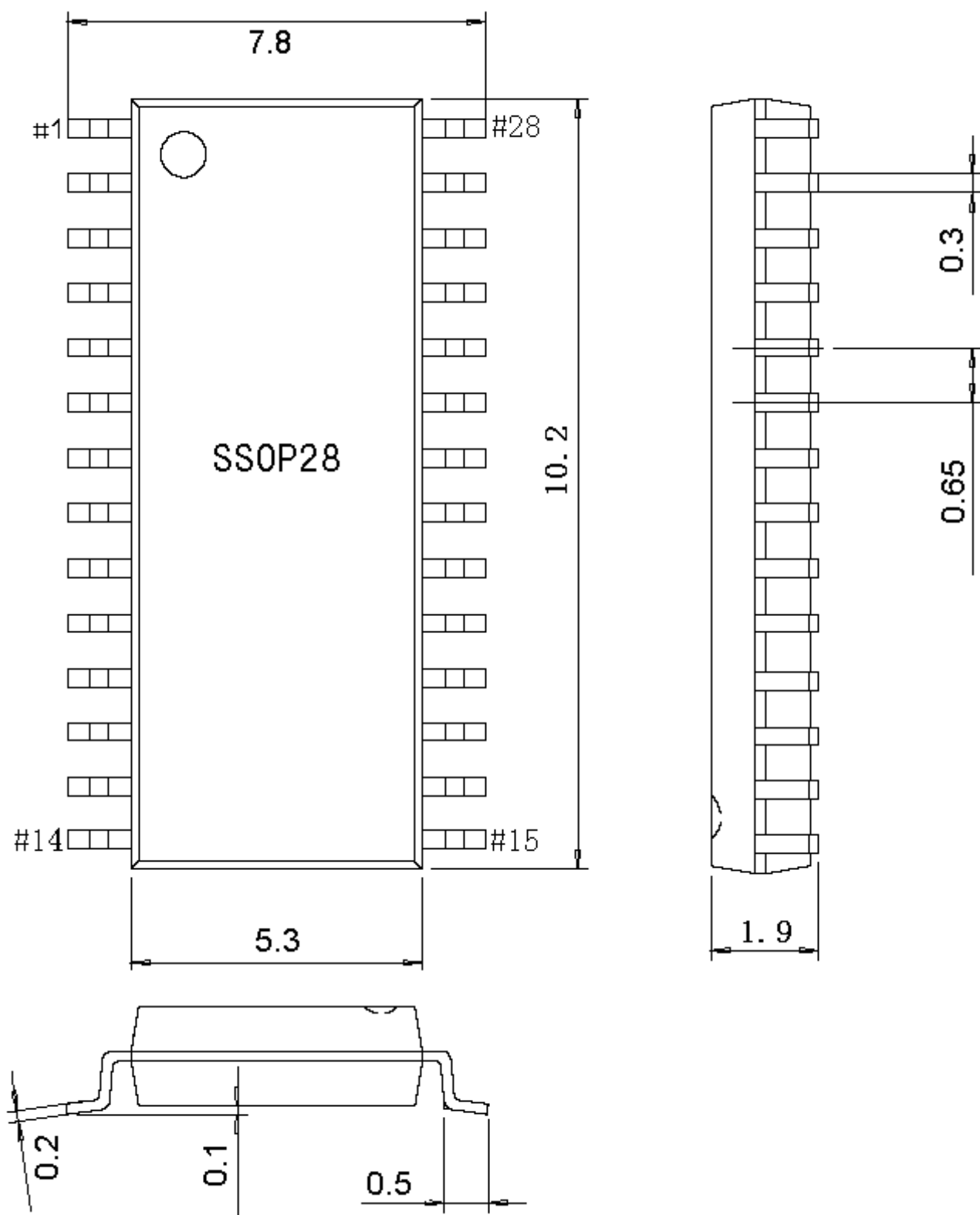




## . SSOP24 (SSOP24-209mil-0.65)

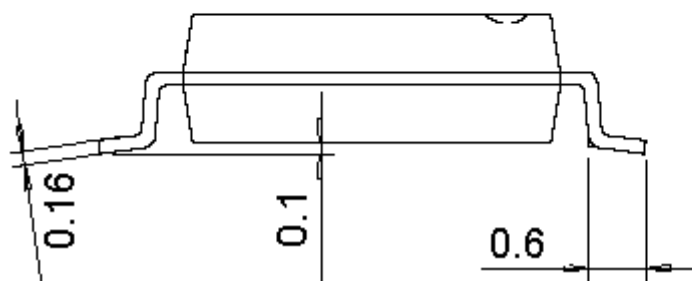
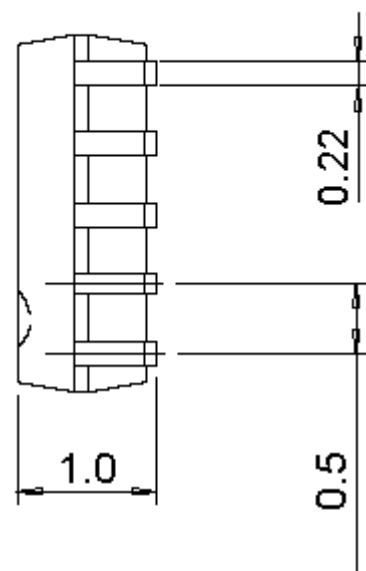
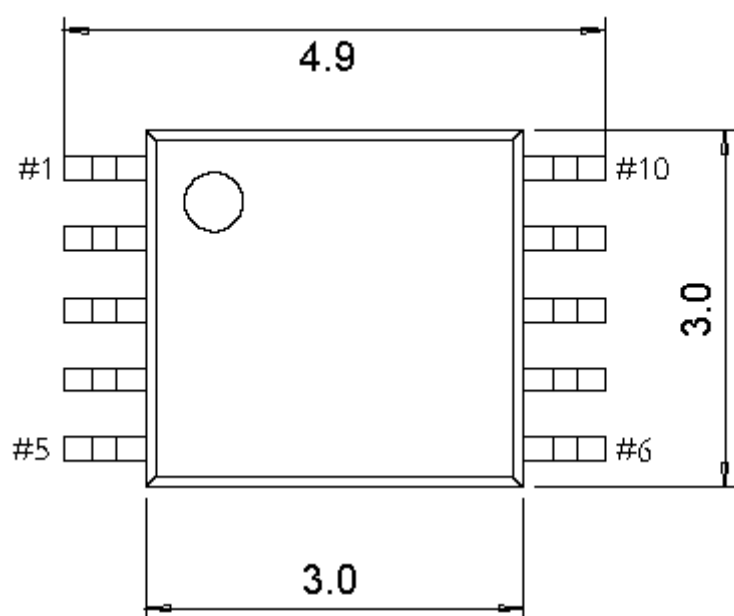


## . SS0P28 (SS0P28-209mil-0.65)



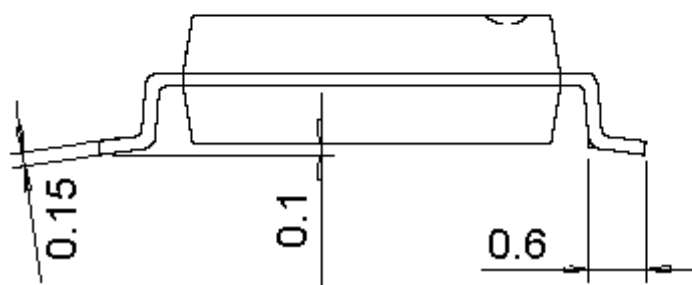
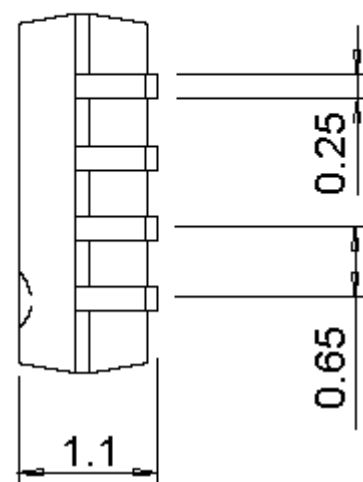
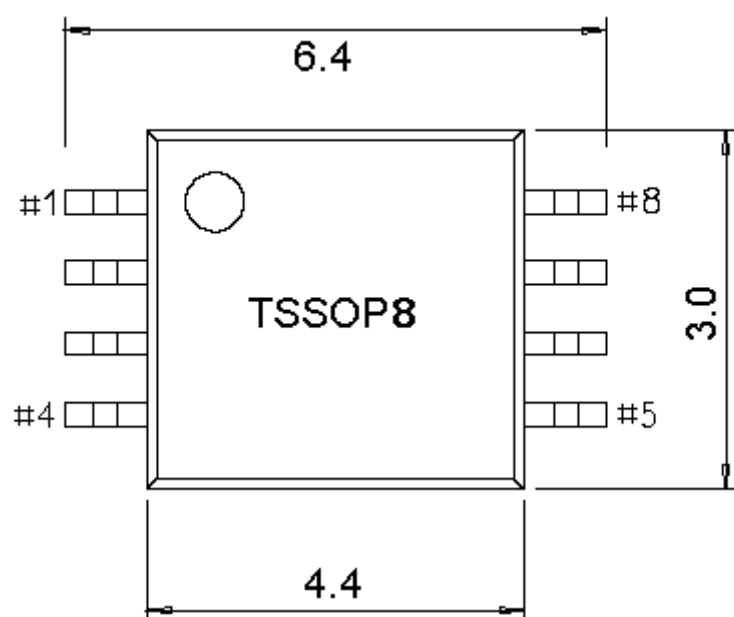
## MSOP 类

## . MSOP10



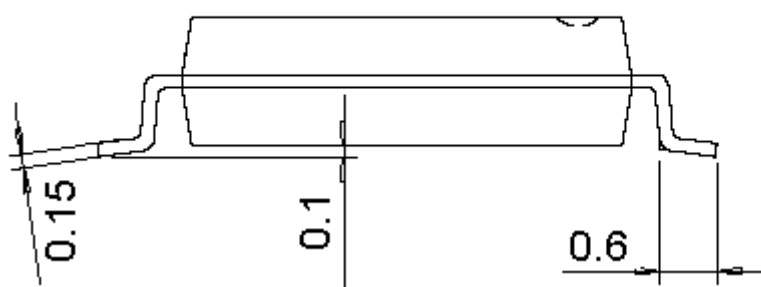
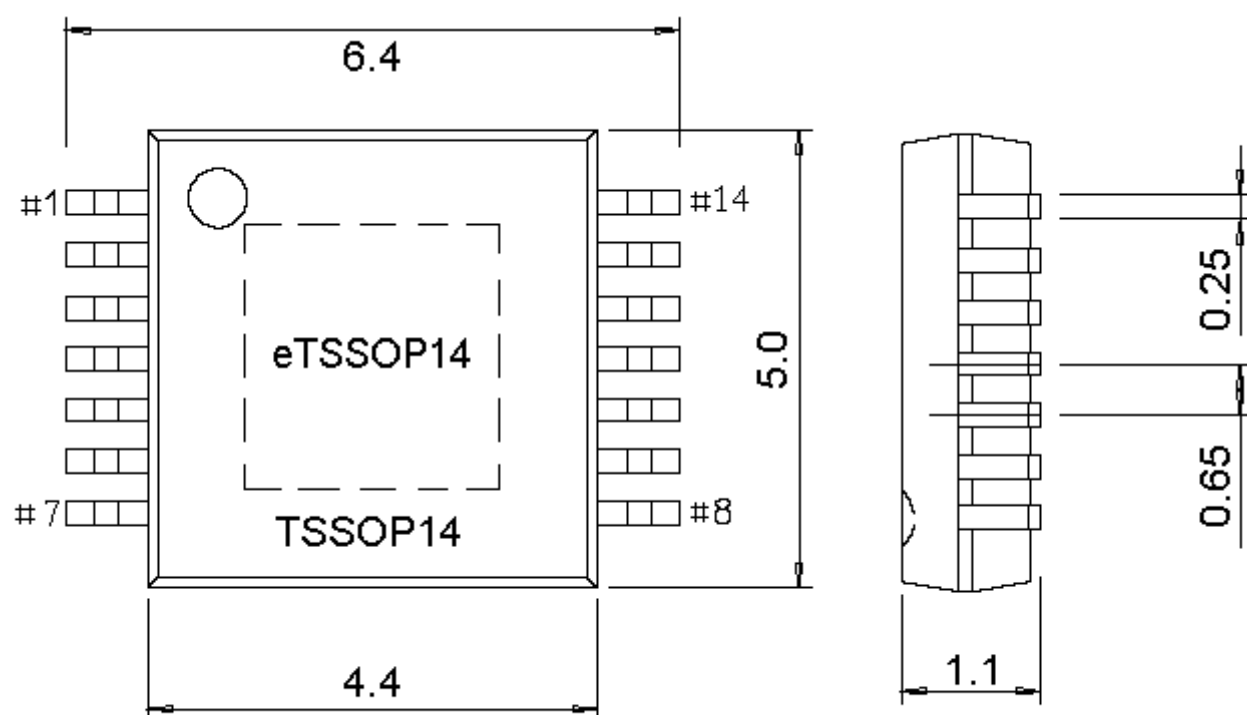
## TSSOP 类 (4.4mm, 173mil)

## . TSSOP8



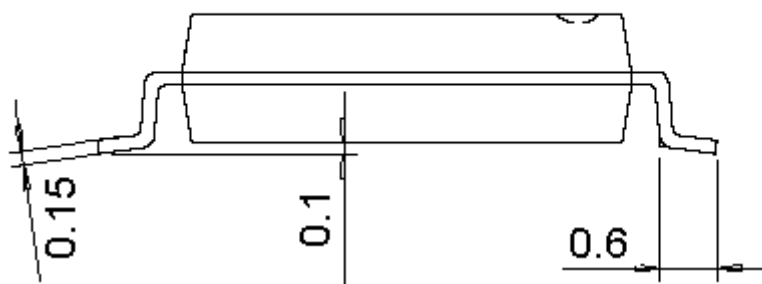
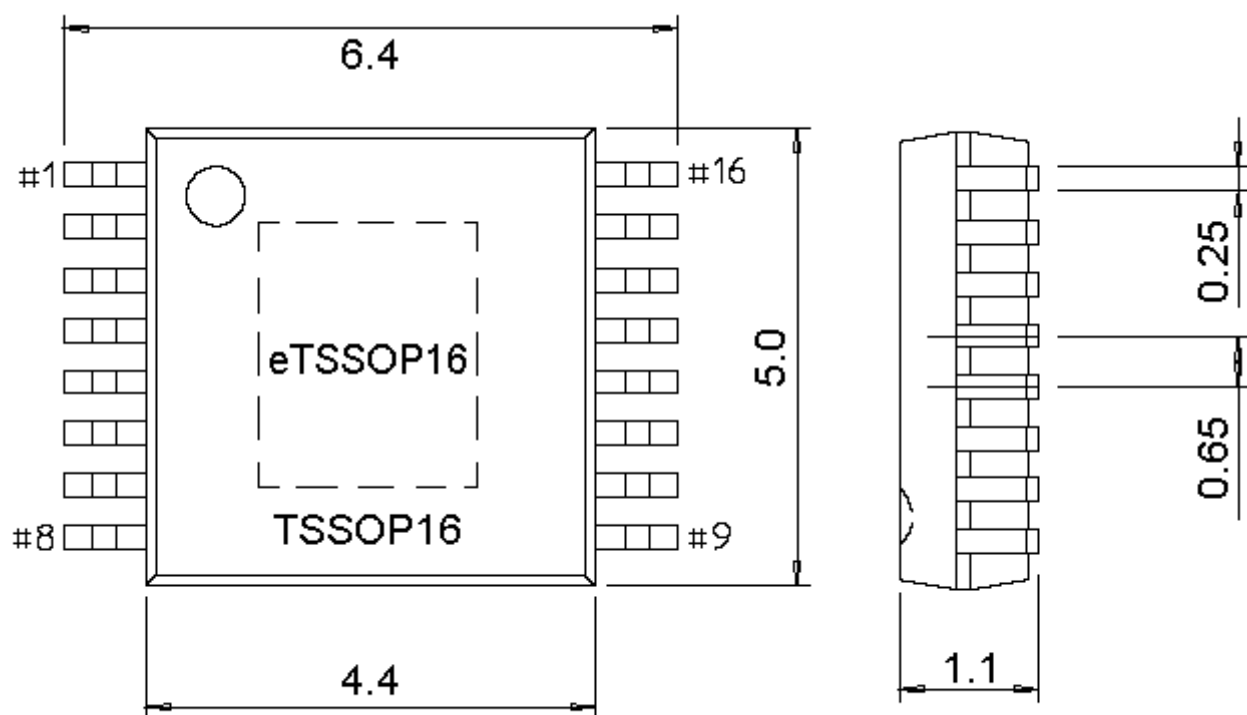
## . TSSOP14、eTSSOP14

eTSSOP14 Exposed Pad: 3.0\*3.0



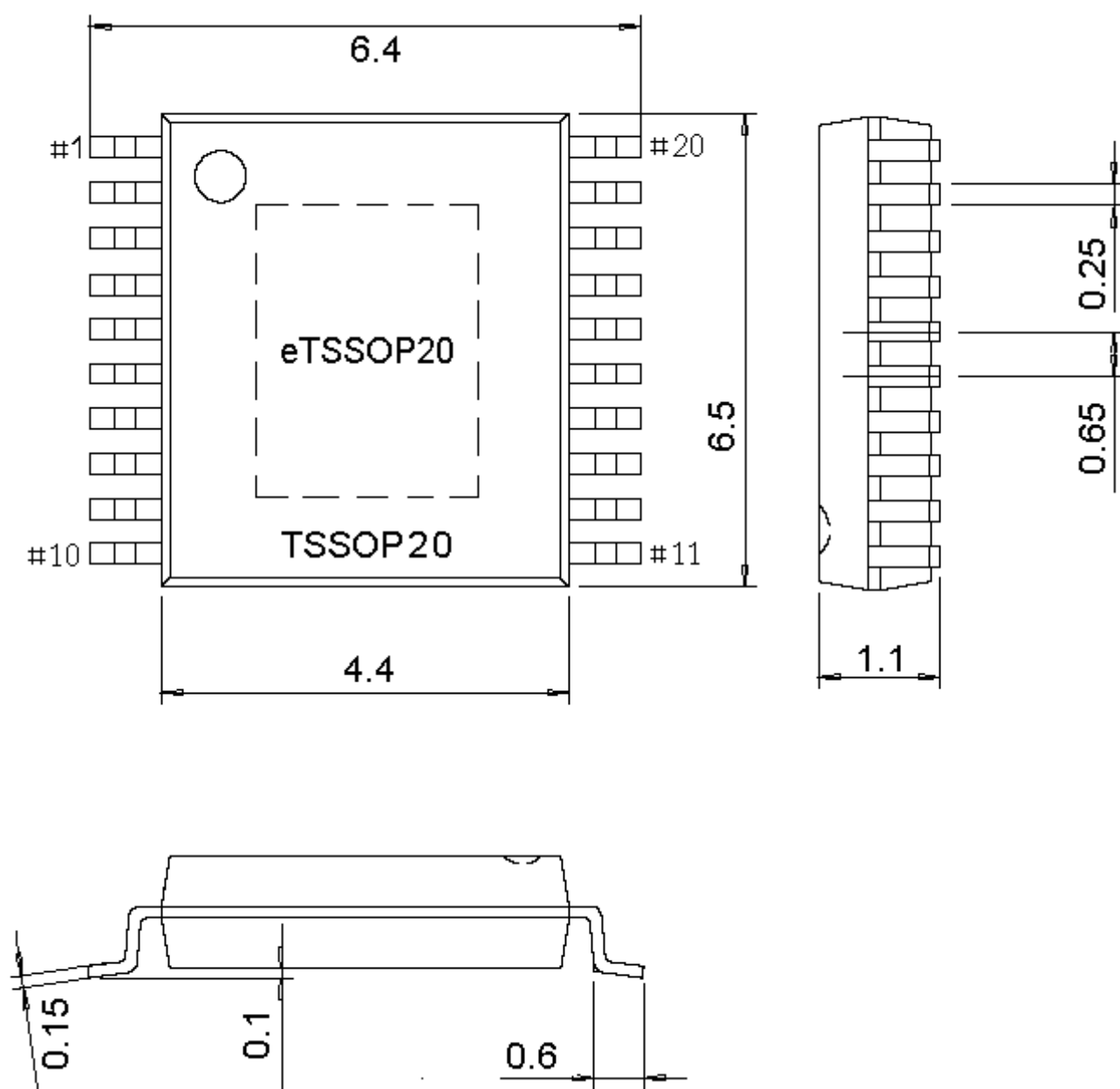
## . TSSOP16、eTSSOP16

eTSSOP16 Exposed Pad: 2.8\*3.0

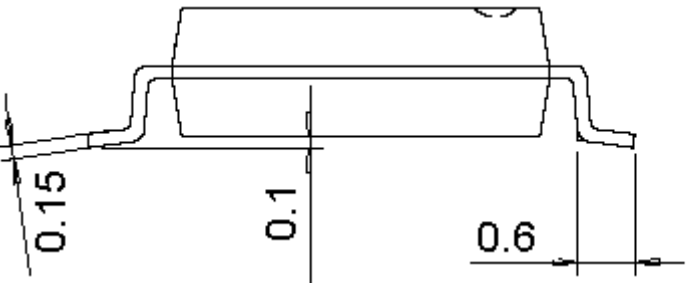
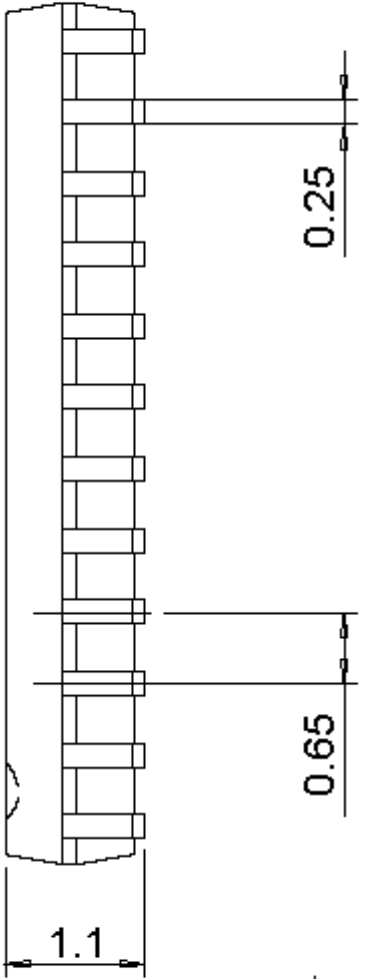
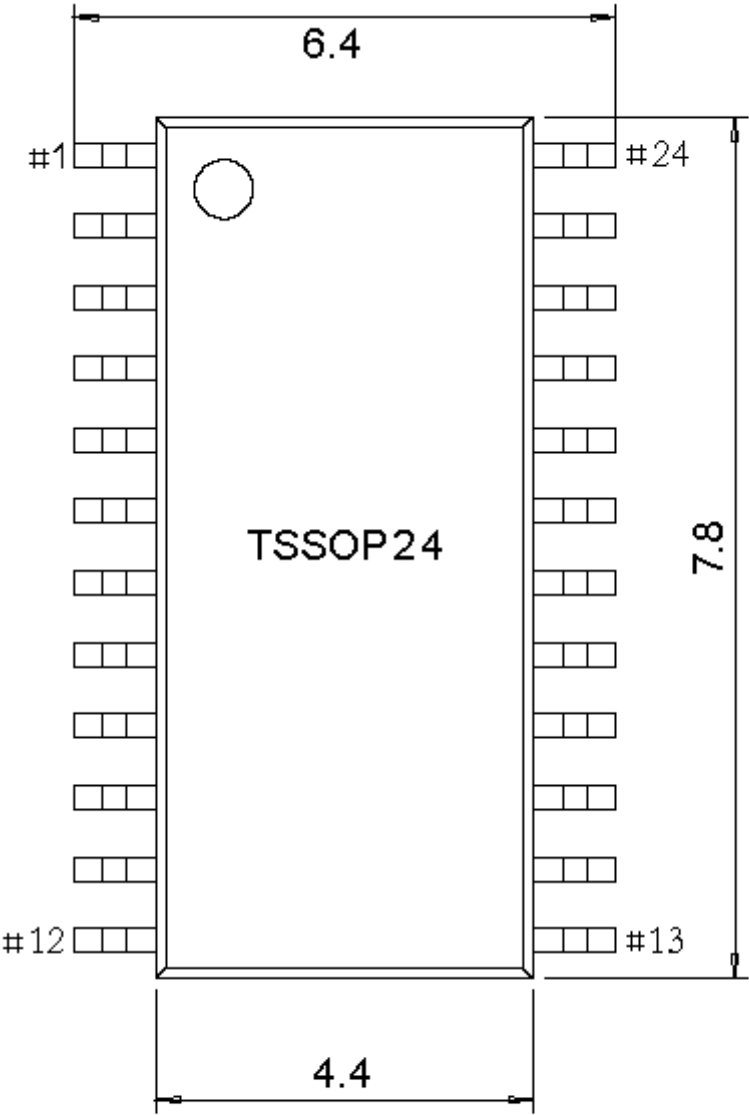


## . TSSOP20、eTSSOP20

eTSSOP20 Exposed Pad: 3.0\*4.2

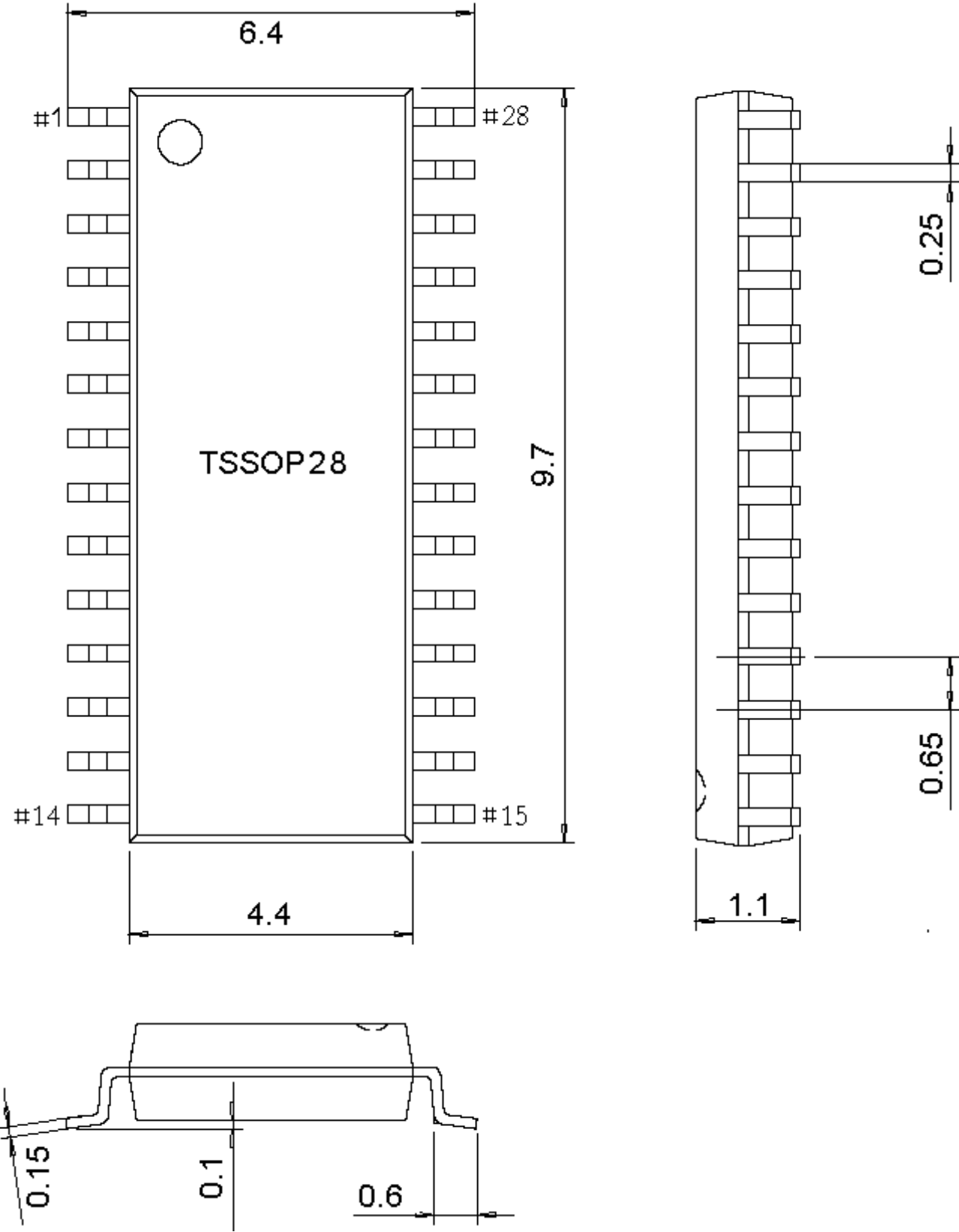


TSSOP24



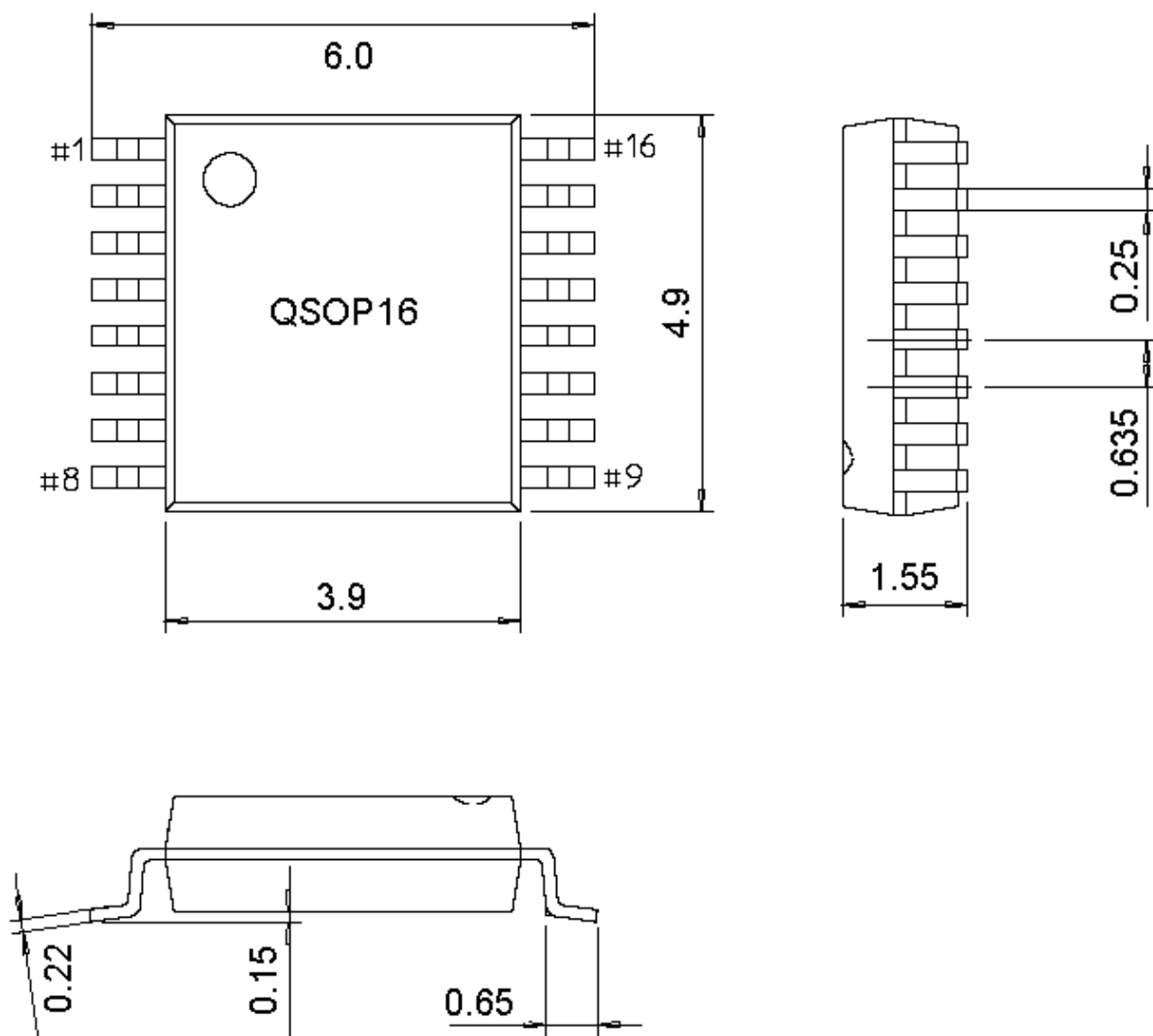


TSSOP28

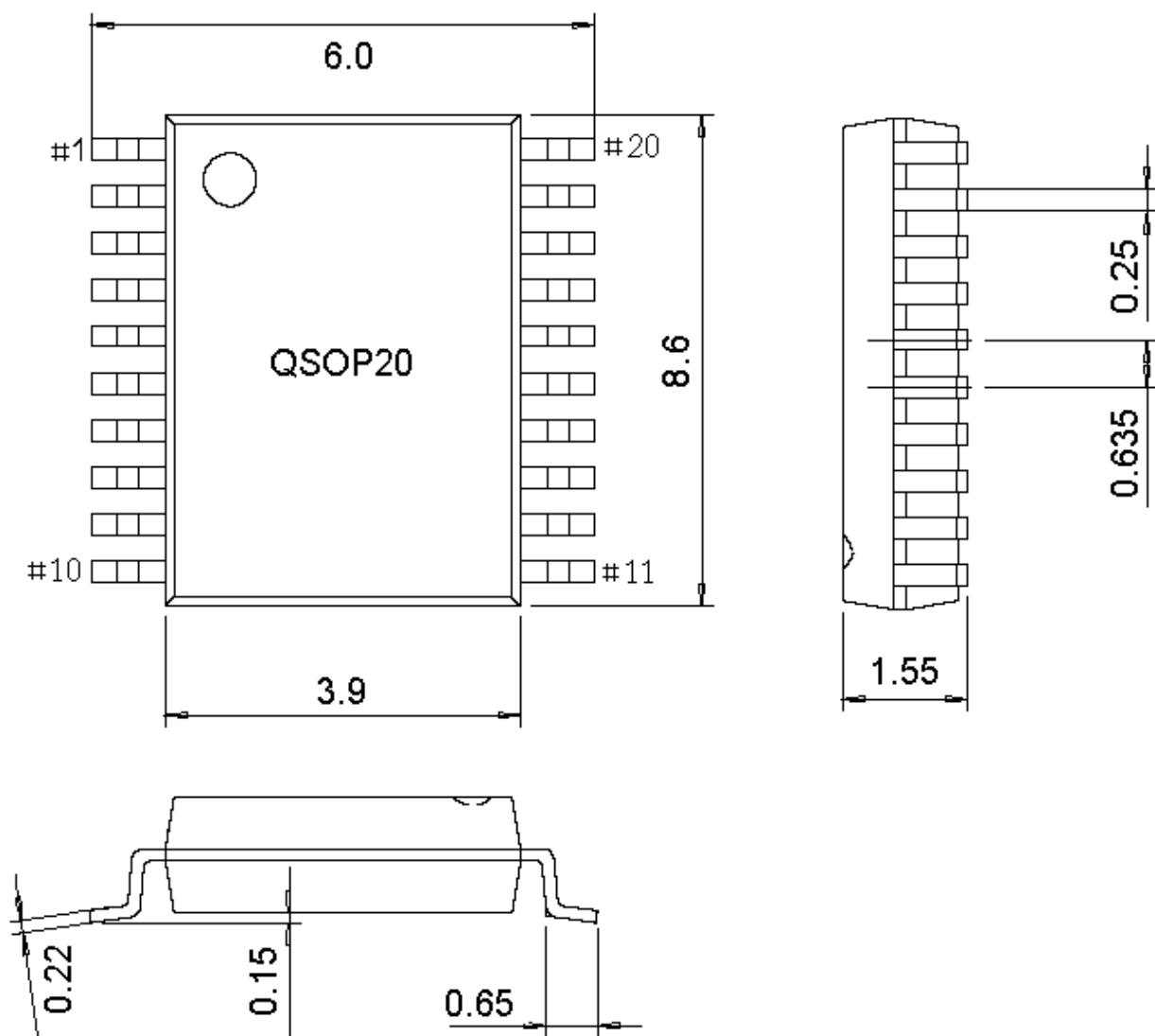


## QSOP 类 (3.9mm, 150mil) (SSOP-150)

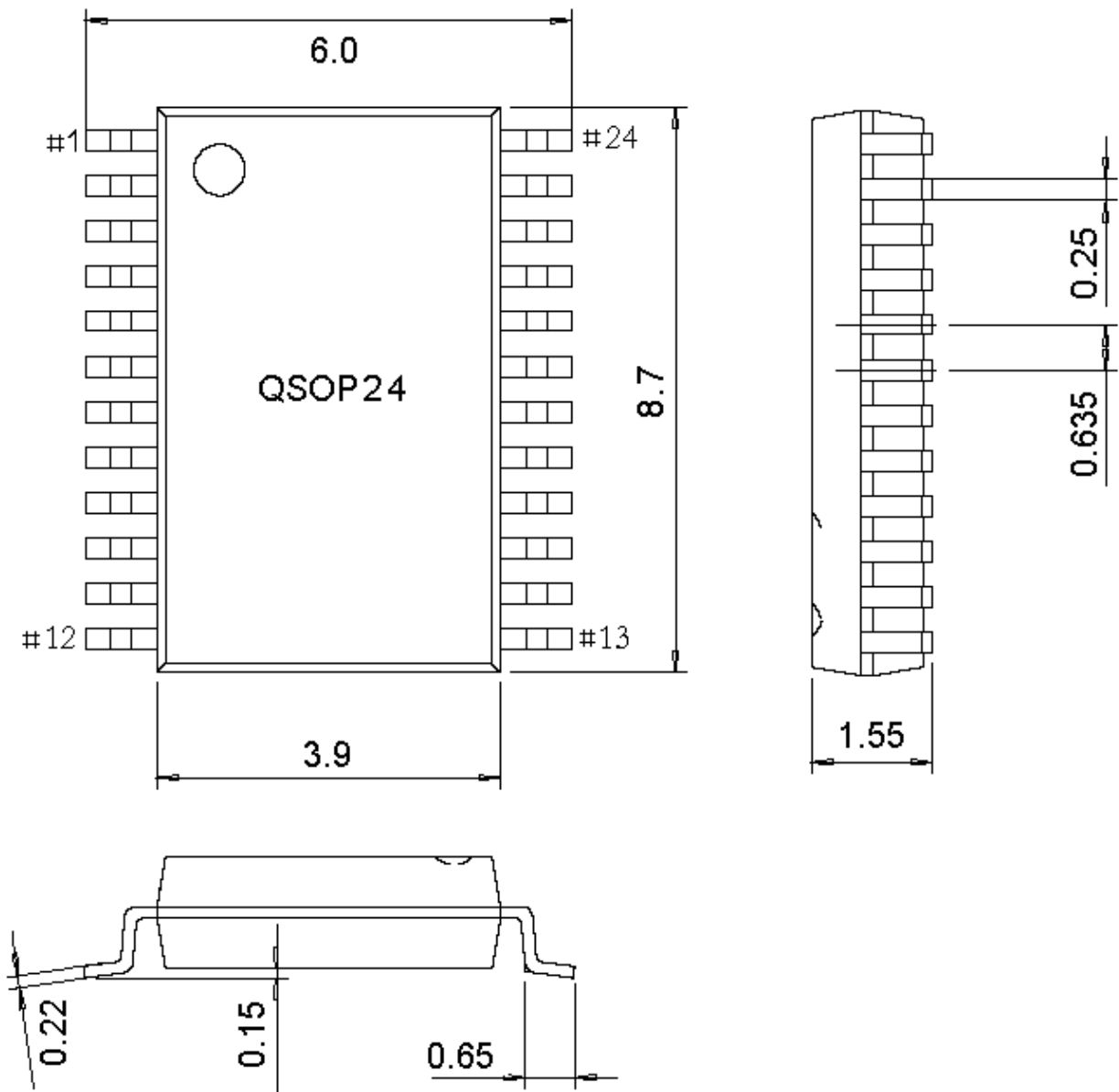
. QSOP16 (SSOP16-150mil-0.635)



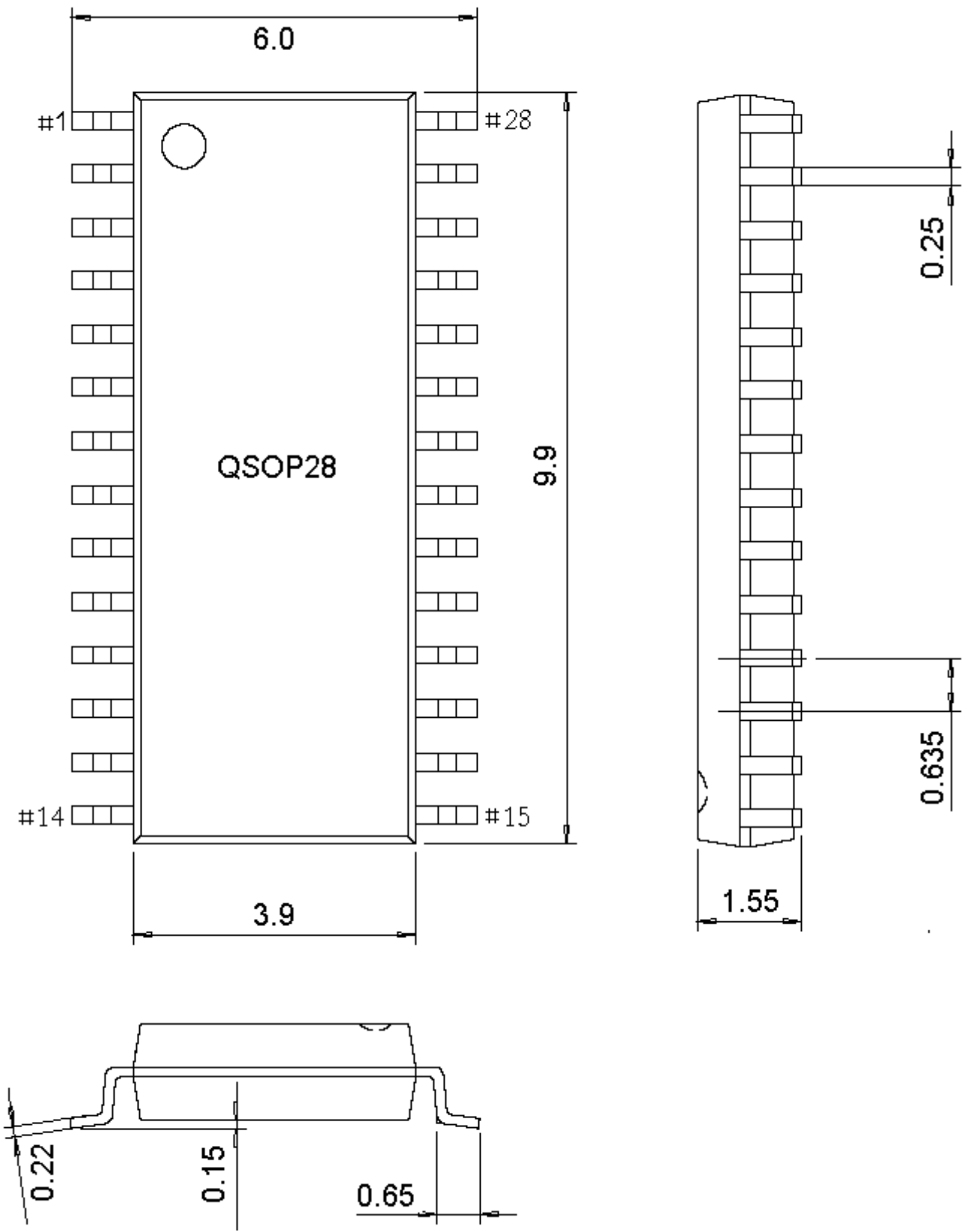
## . QSOP20 (SSOP20-150mil-0.635)



· QSOP24 (SSOP24-150mil-0.635)

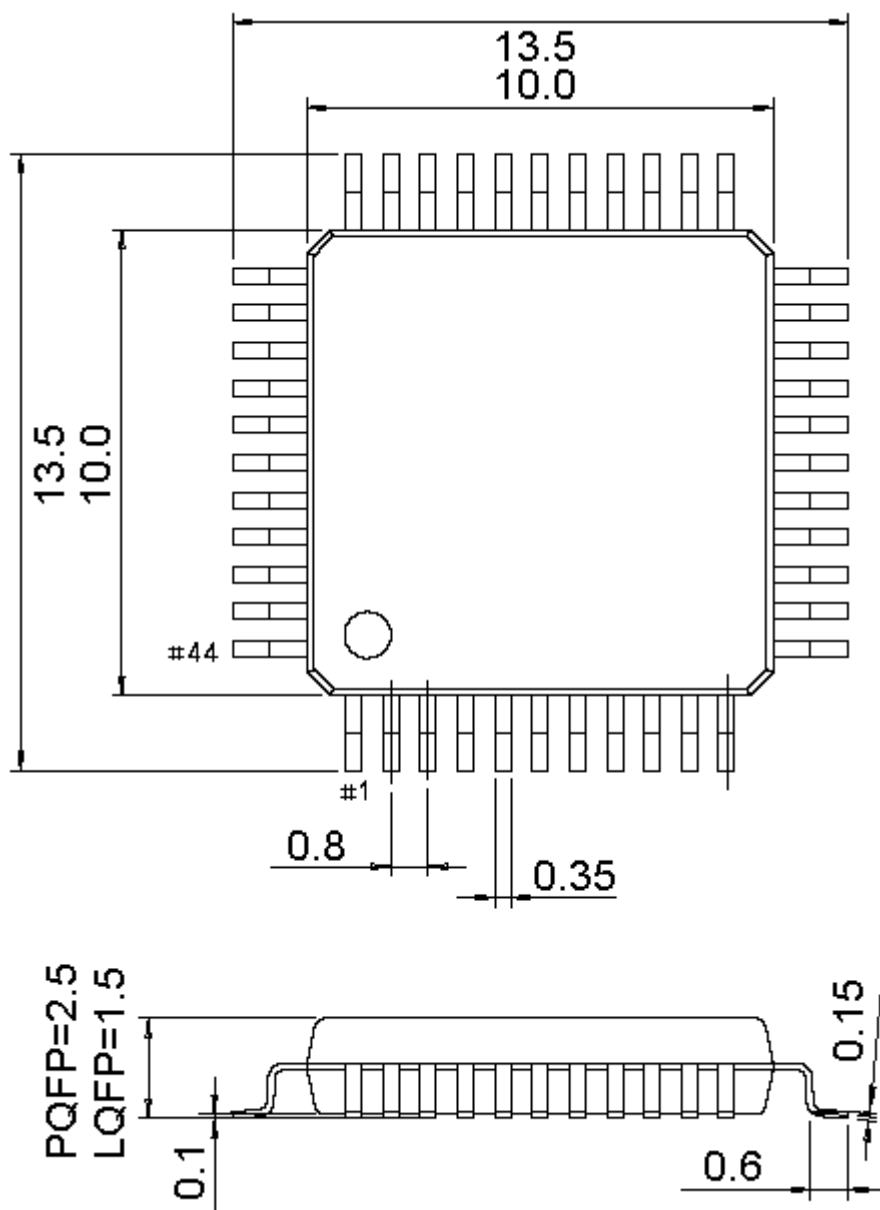


QSOP28 (SSOP28-150mil-0.635)

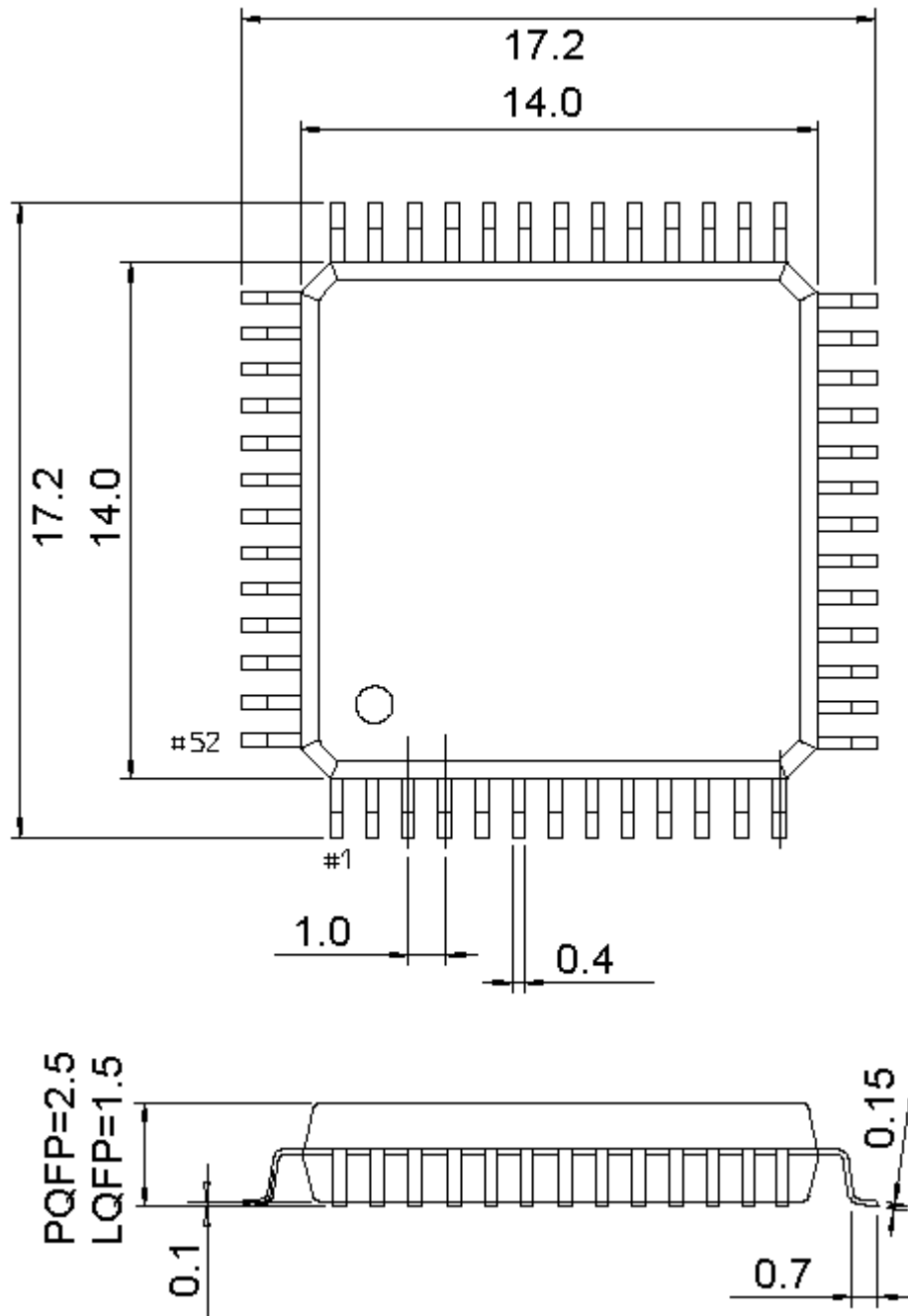


### QFP 类 (PQFP、LQFP、TQFP)

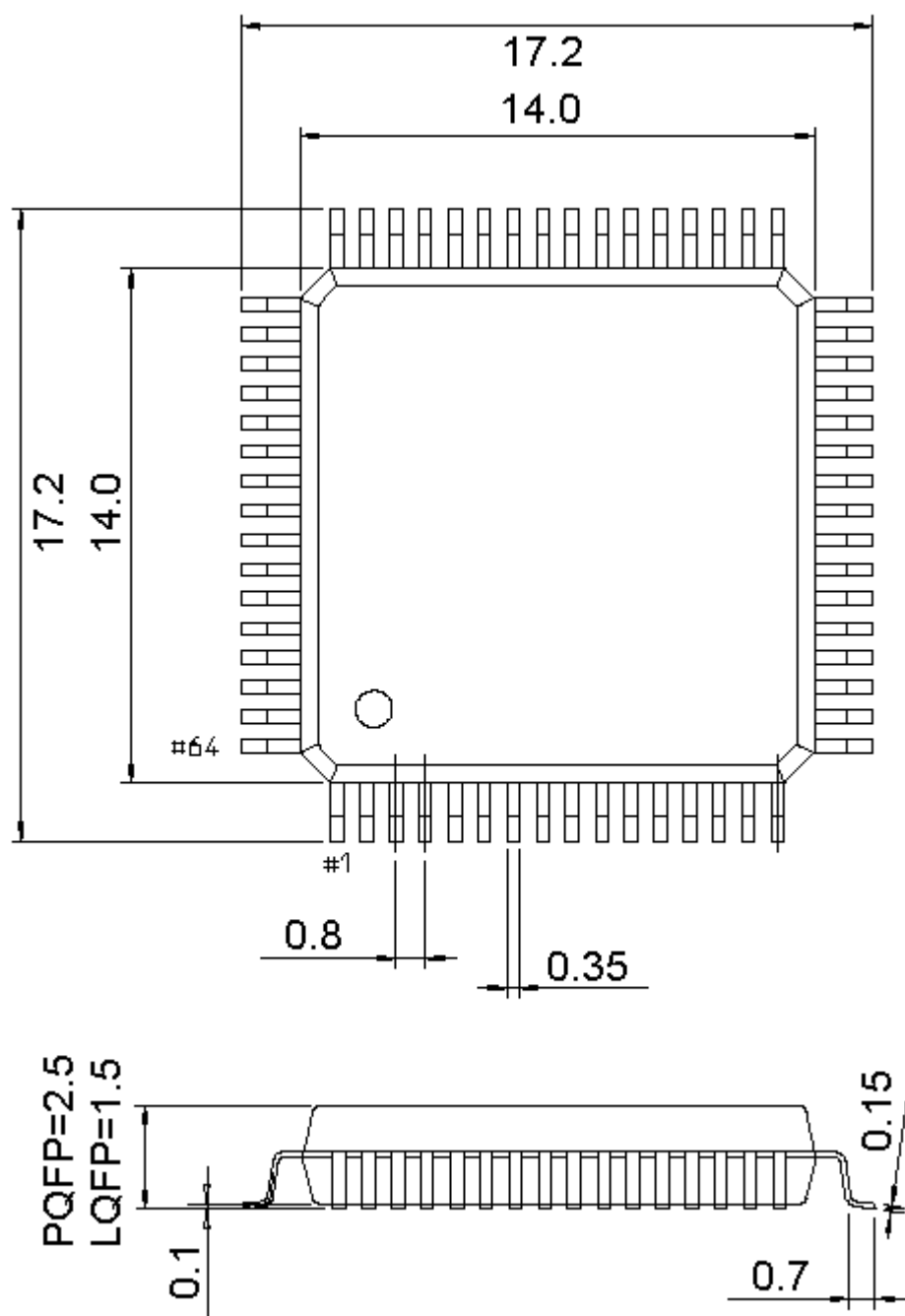
. PQFP44 (QFP44)



· PQFP52 (QFP52)

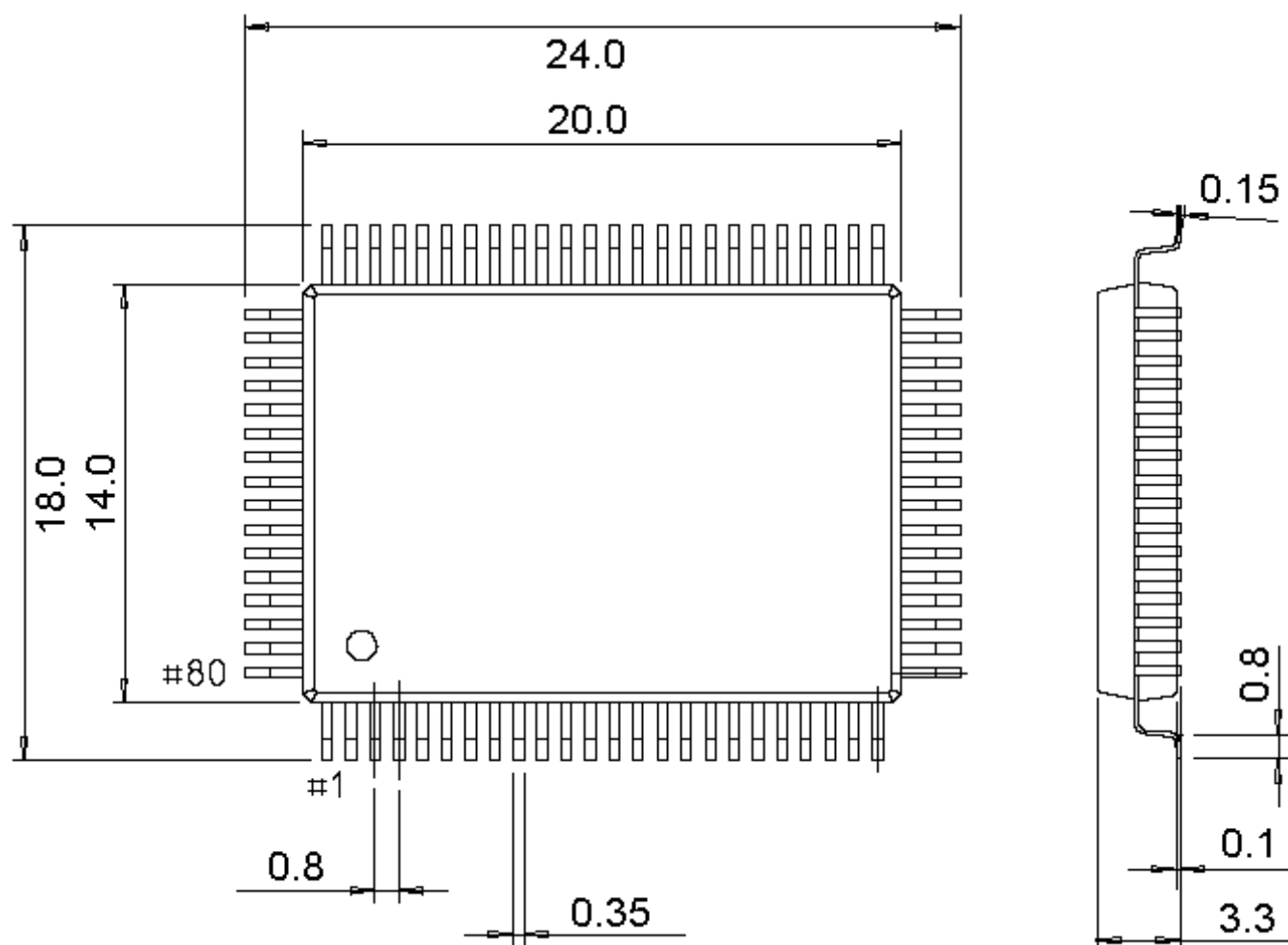


## . PQFP64 (QFP64)、LQFP64-14\*14

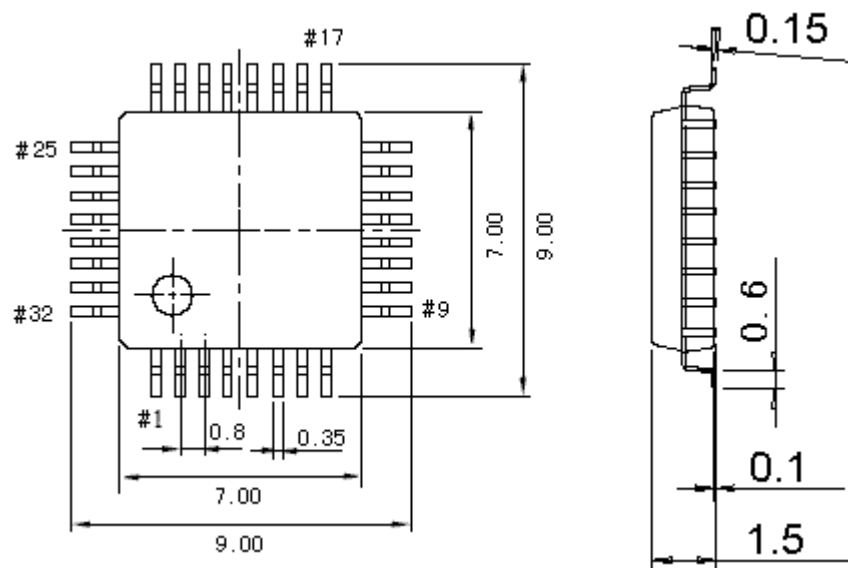




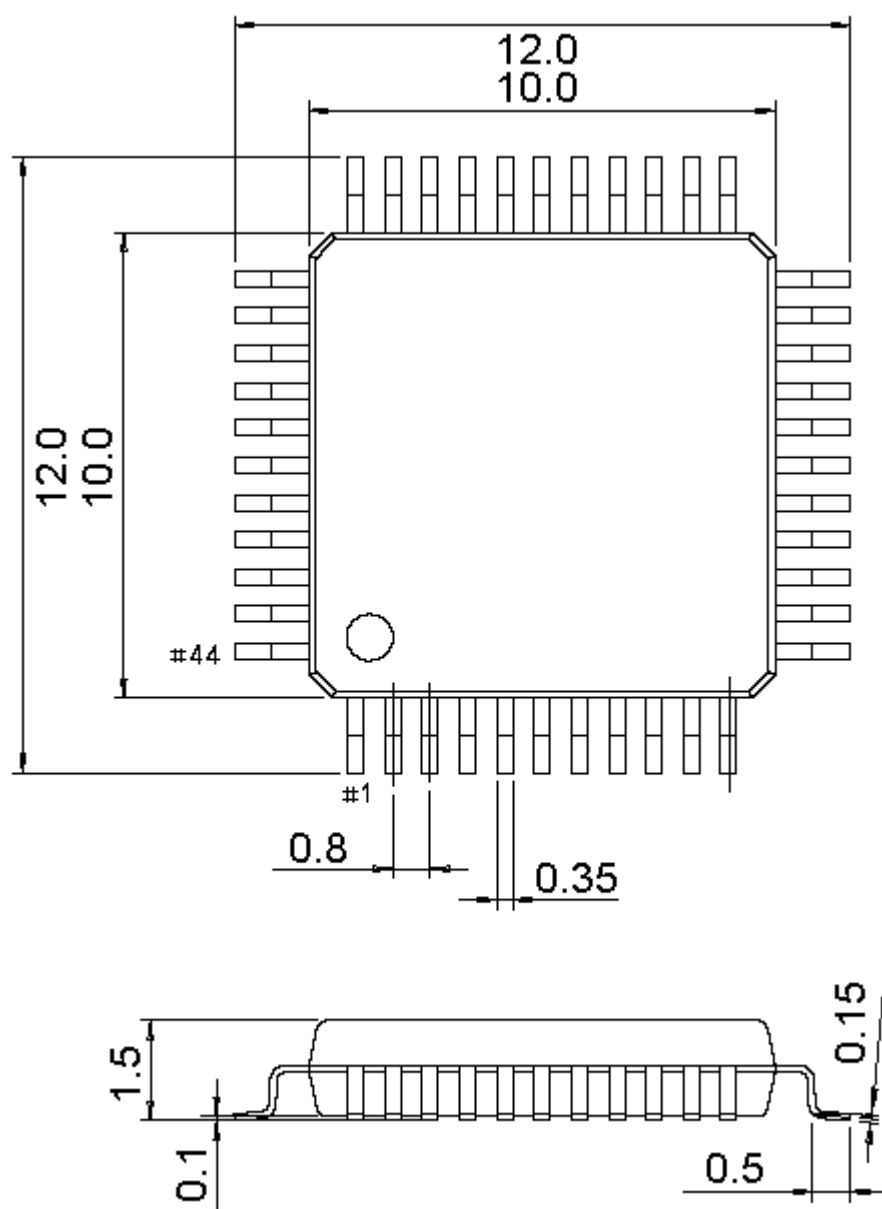
## . PQFP80 (QFP80)



## . LQFP32 (LQFP32-7\*7)

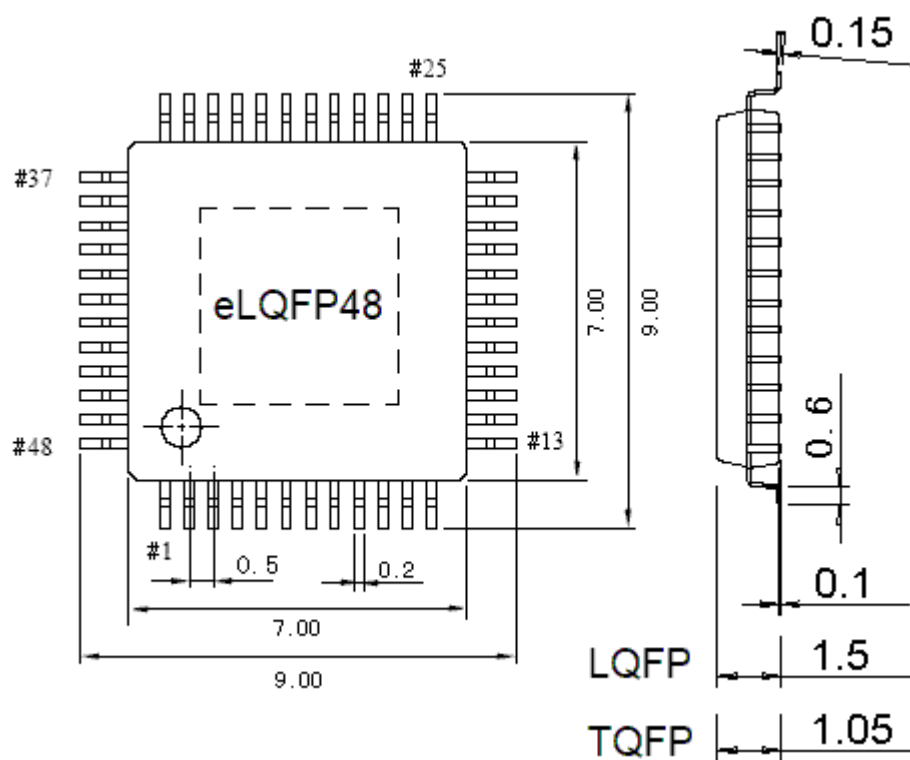


. LQFP44 (LQFP44-10\*10)



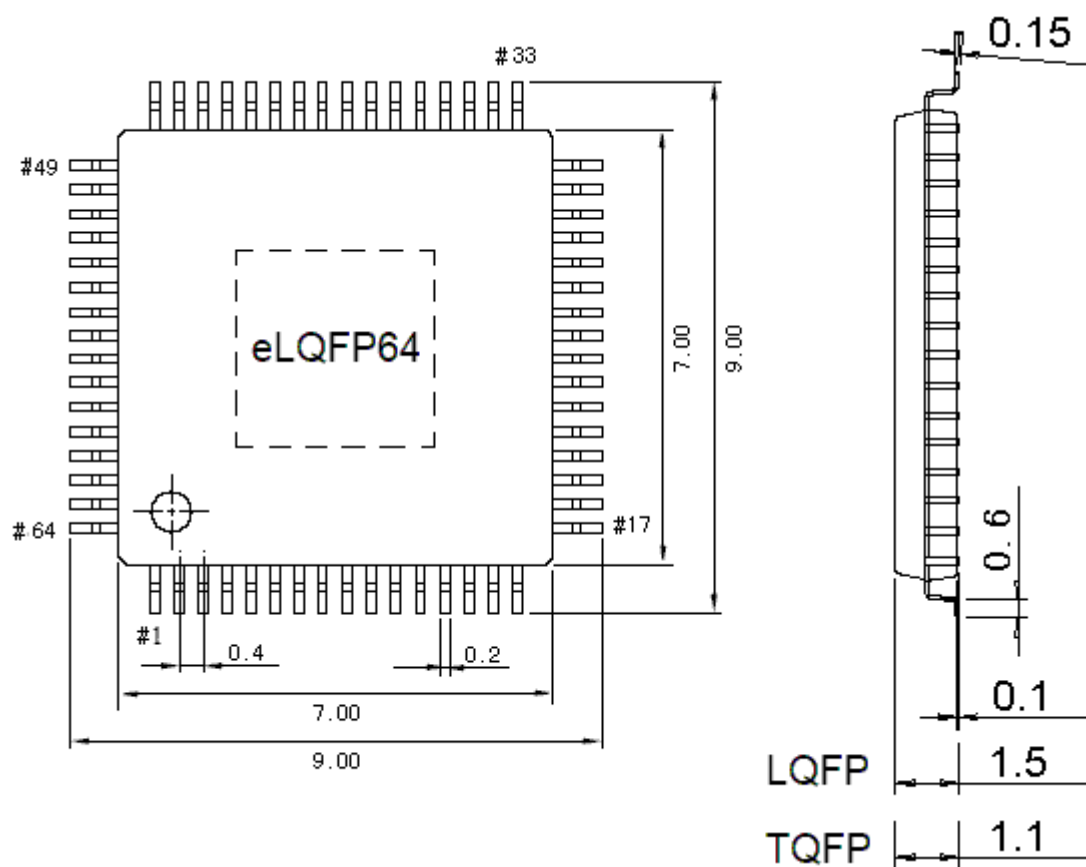
. LQFP48 (LQFP48-7\*7)、TQFP48、eLQFP48

eLQFP48 Exposed Pad: 3.2\*3.2

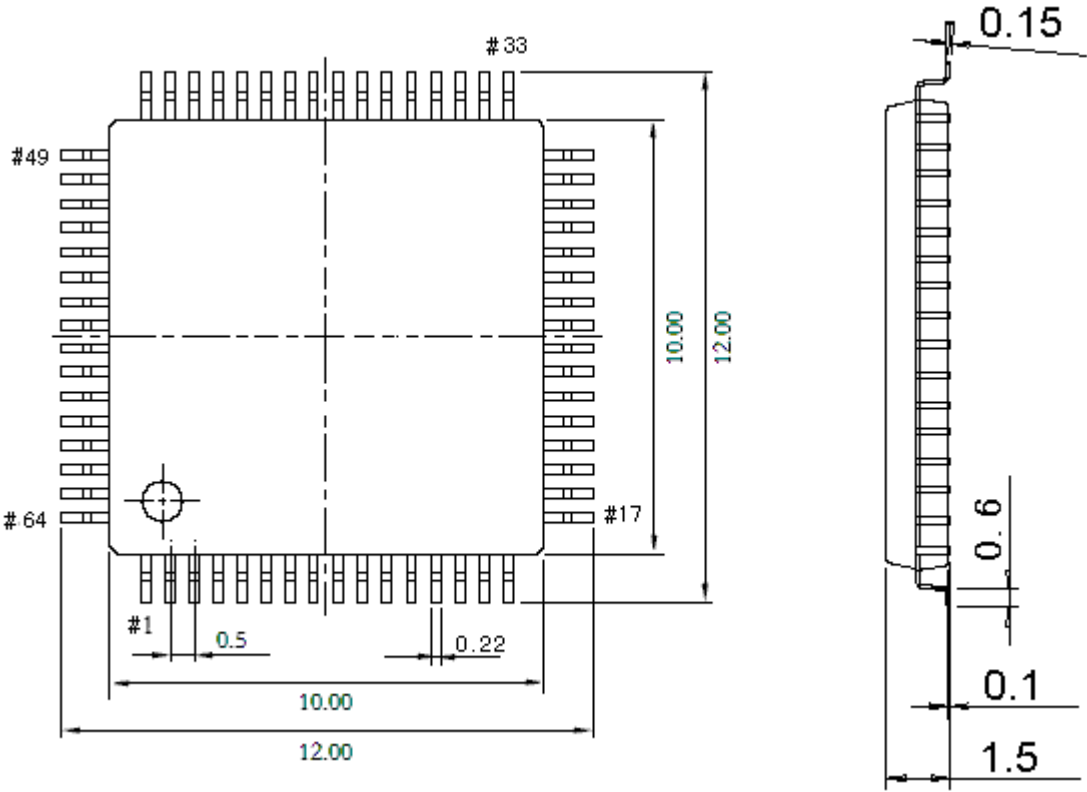


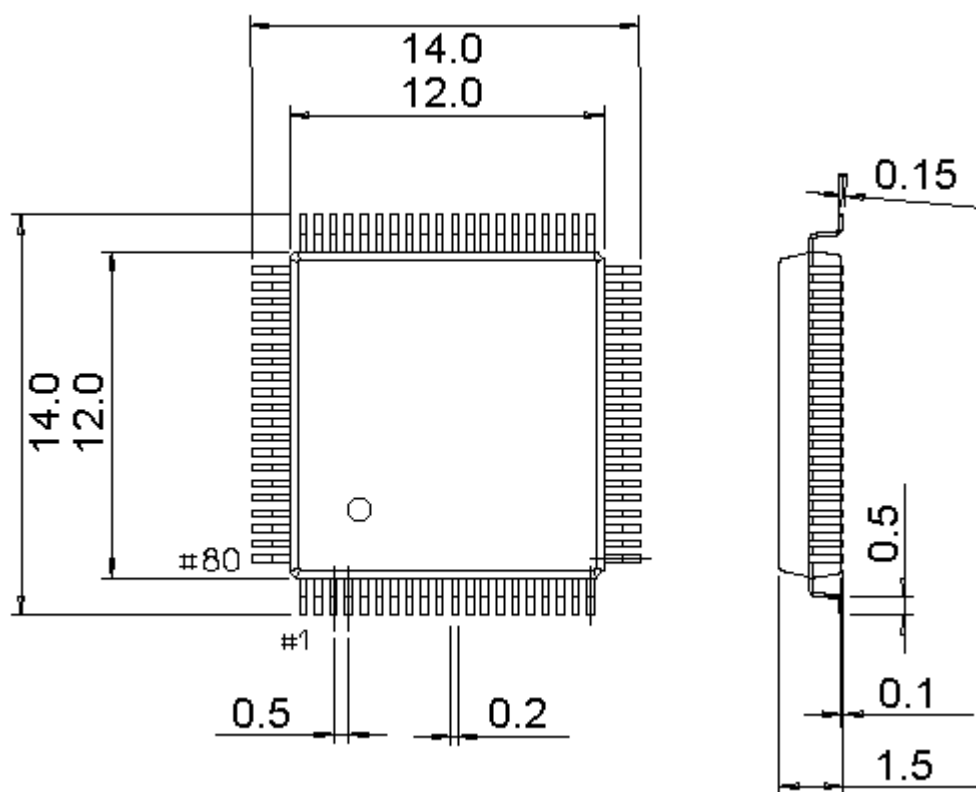
. LQFP64 (LQFP64-7\*7)、TQFP64、eLQFP64

eLQFP64 Exposed Pad: 3.2\*3.2

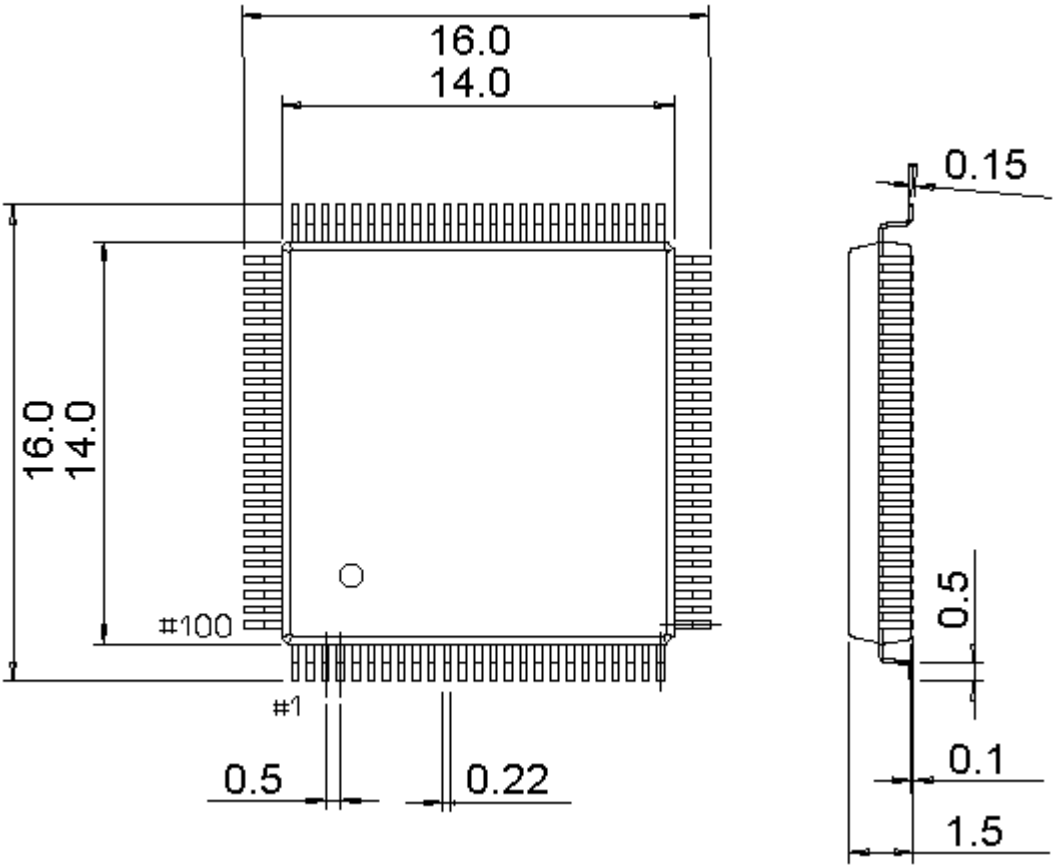


· LQFP64M (LQFP64-10\*10)



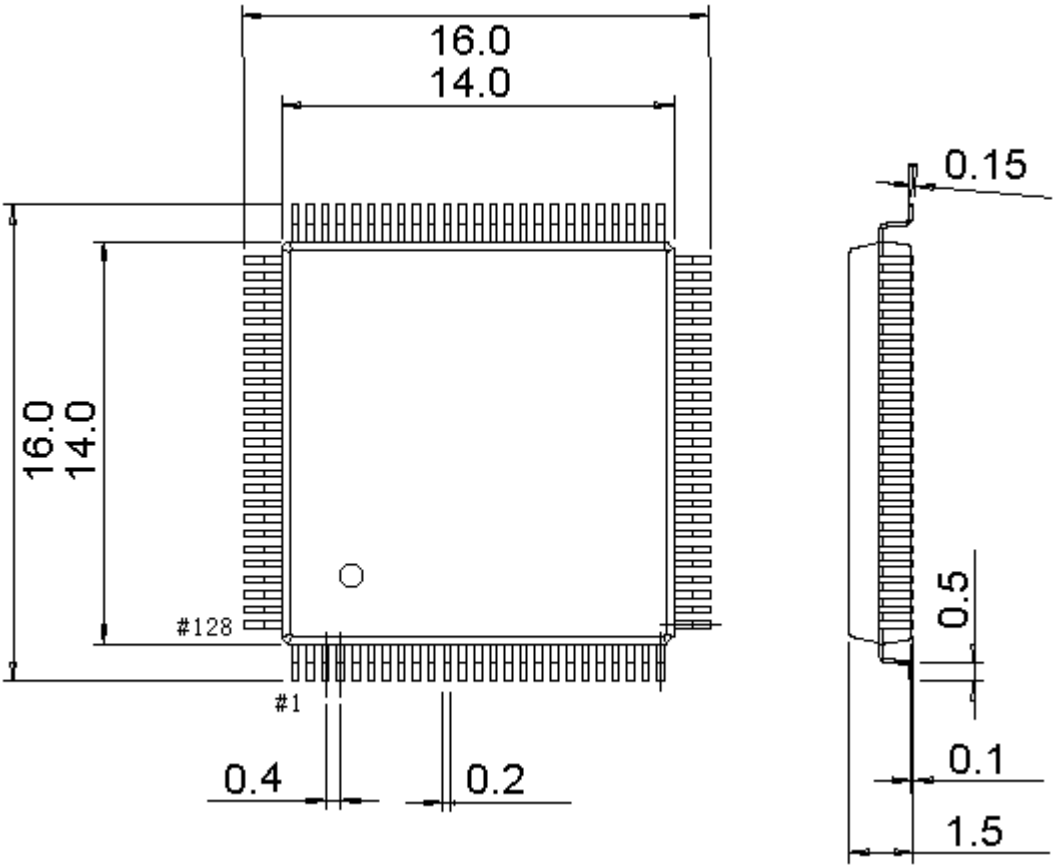


· LQFP100 (LQFP100-14\*14)

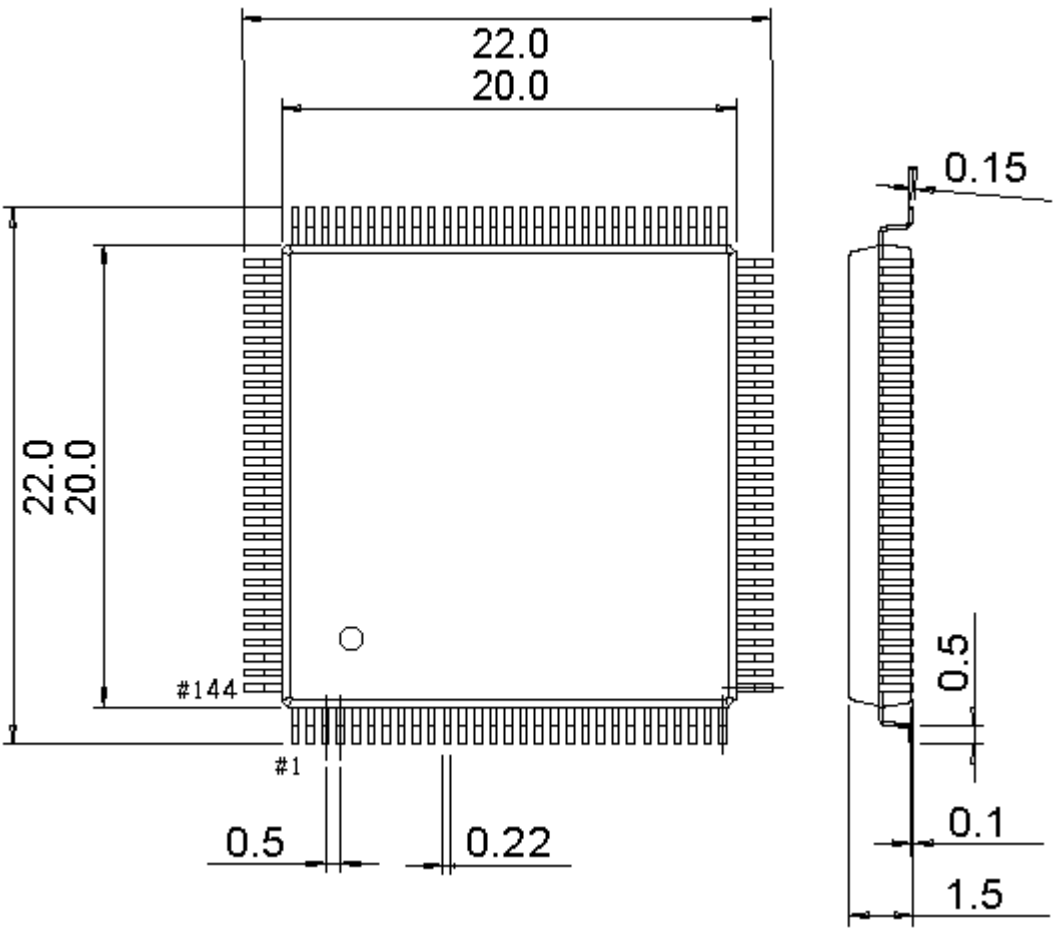




· LQFP128 (LQFP128-14\*14)



· LQFP144 (LQFP144-20\*20)



## QFN 类

### QFN/DFN 命名规则参考

#### 标准命名

QFN ab - d \* e \* t - p

数字 ab: QFN 封装 4 边累计引脚数, 目前最多为 128。

数字 d 和 e: d 和 e 代表 QFN 外形的长和宽 (1#引脚所在边)。

数字 t: t 代表 QFN 外形的厚度, 0.75 和 0.55 较为常用, 另有 0.85、0.5、0.4 等。

数字 p: p 代表引脚节距 pitch, 即相邻引脚的中心的间距, 如有两种则取其小, 0.4 和 0.5 较为常用, 另有 0.35、0.65、0.45、0.3 等。

#### 参考简化命名

QFN ab c d X e

数字 ab: QFN 封装 4 边累计引脚数, 目前最多为 128。

字母 c: 无 c 代表通用 QFN, 有 Exposed Pad, 默认尺寸, 无数字 d 和 e;

字母 X 代表通用 QFN, 有 Exposed Pad, 非默认尺寸, 至少有数字 d;

字母 L 代表通用 QFN, 有 Exposed Pad, 引脚长度 L (或 EPAD 大小) 不同;

字母 C 代表 WCH 设计的通用 QFN, 有 Exposed Pad, 至少有数字 d;

字母 F 代表 WCH 定制的专用 QFN 且至少有数字 d, 或通用 QFN, 无 Exposed Pad;

字母 E 代表 WCH 定制的专用 QFN, 有 Exposed Pad, 至少有数字 d。

数字 d 和 e: d 和 e 代表 QFN 外形的长和宽 (1#引脚所在边);

无 d 无 e 代表默认尺寸, 也是常用尺寸;

有 d 无 e 代表尺寸为 d\*d 正方形;

如果数字超过 13 则需除以 10, 例如 45 表示 4.5。

默认尺寸通常是指 pitch=0.4 时的最小正方形尺寸, 也是常用尺寸:

QFN12、QFN14 默认为 2\*2 正方形;

QFN16、QFN20 默认为 3\*3 正方形;

QFN24、QFN28、QFN32 默认为 4\*4 正方形;

QFN36、QFN40、QFN44 默认为 5\*5 正方形。

#### 以 QFN20 为例:

QFN20: 通用 QFN, 有 Exposed Pad, 默认尺寸 (QFN20 默认为 3\*3);

QFN20X4: 通用 QFN, 有 Exposed Pad, 尺寸为 4\*4;

QFN20X25X45: 通用 QFN, 有 Exposed Pad, 尺寸为 2.5\*4.5;

QFN20L: 通用 QFN, 有 Exposed Pad, 默认尺寸, 引脚长度 L 不同;

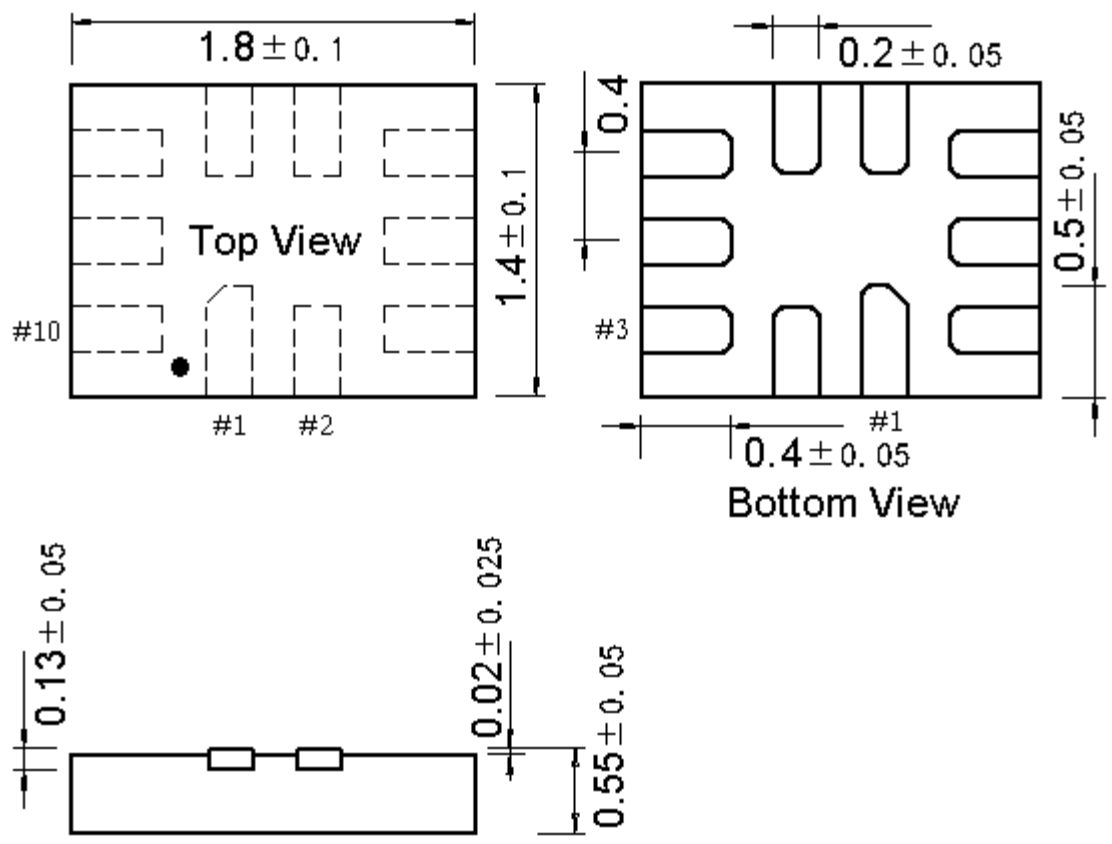
QFN20F: 通用 QFN, 无 Exposed Pad, 默认尺寸;

QFN20C2: WCH 设计的通用 QFN, 有 Exposed Pad, 尺寸为 2\*2;

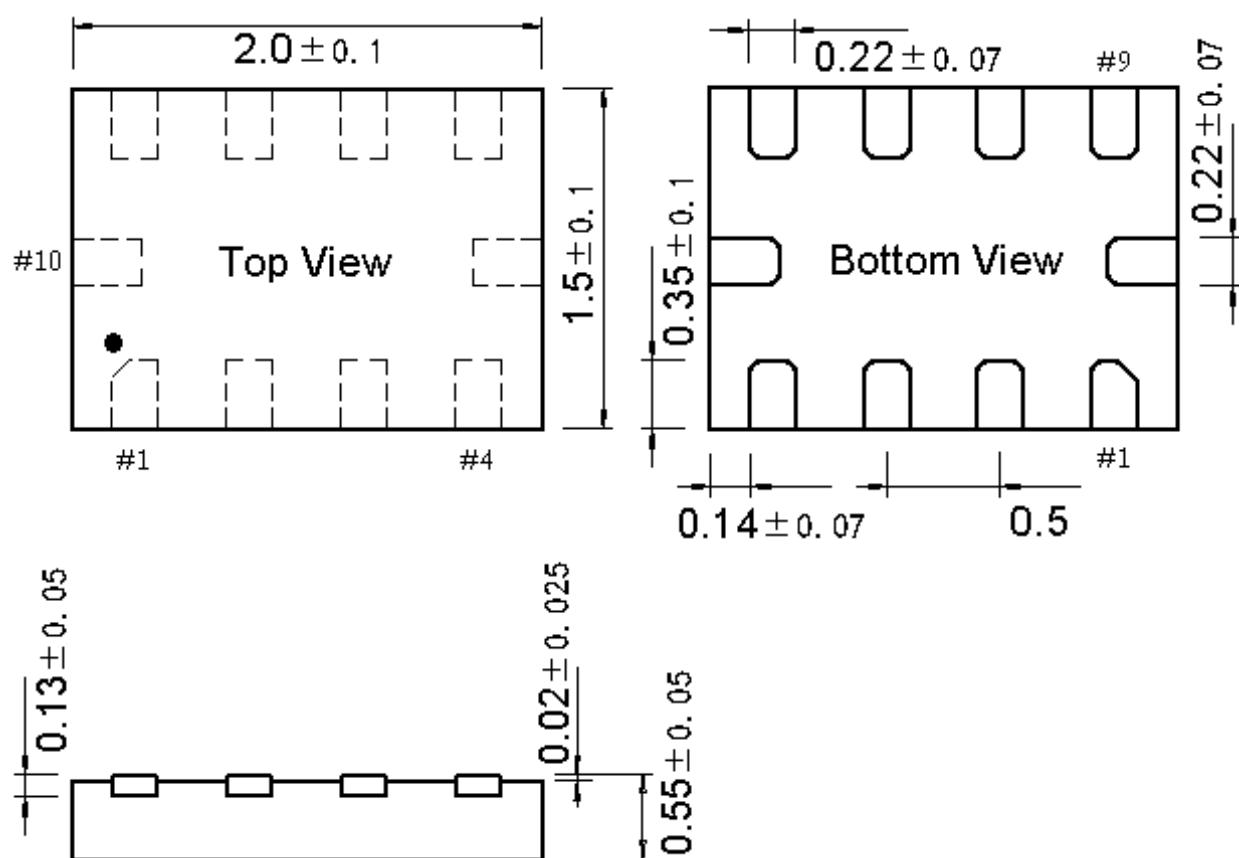
QFN20F3: WCH 定制的专用 QFN, 无 Exposed Pad, 尺寸为 3\*3;

QFN20E3X4: WCH 定制的专用 QFN, 有 Exposed Pad, 尺寸为 3\*4。

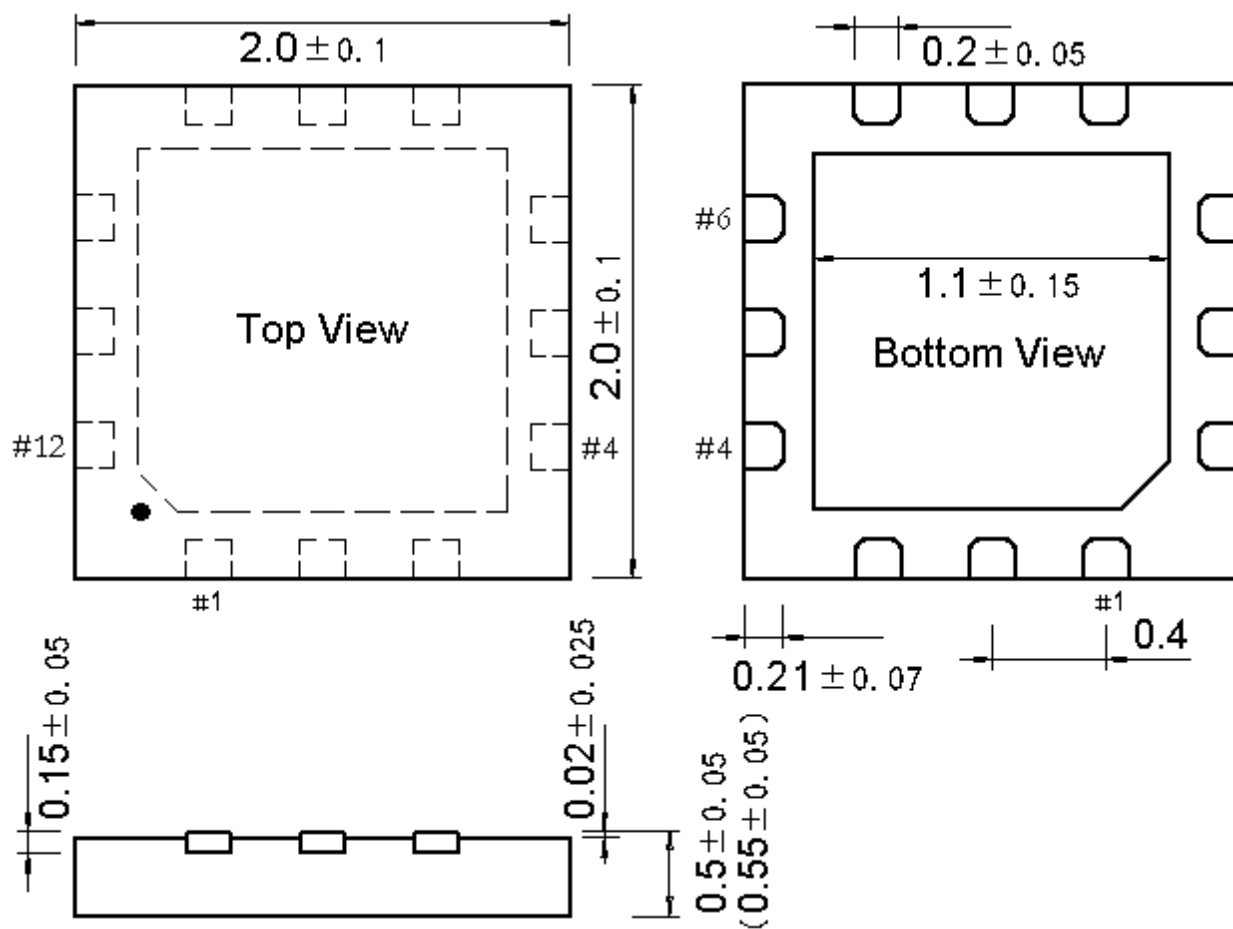
· QFN10 (QFN10-1.4\*1.8\*0.55-0.4)



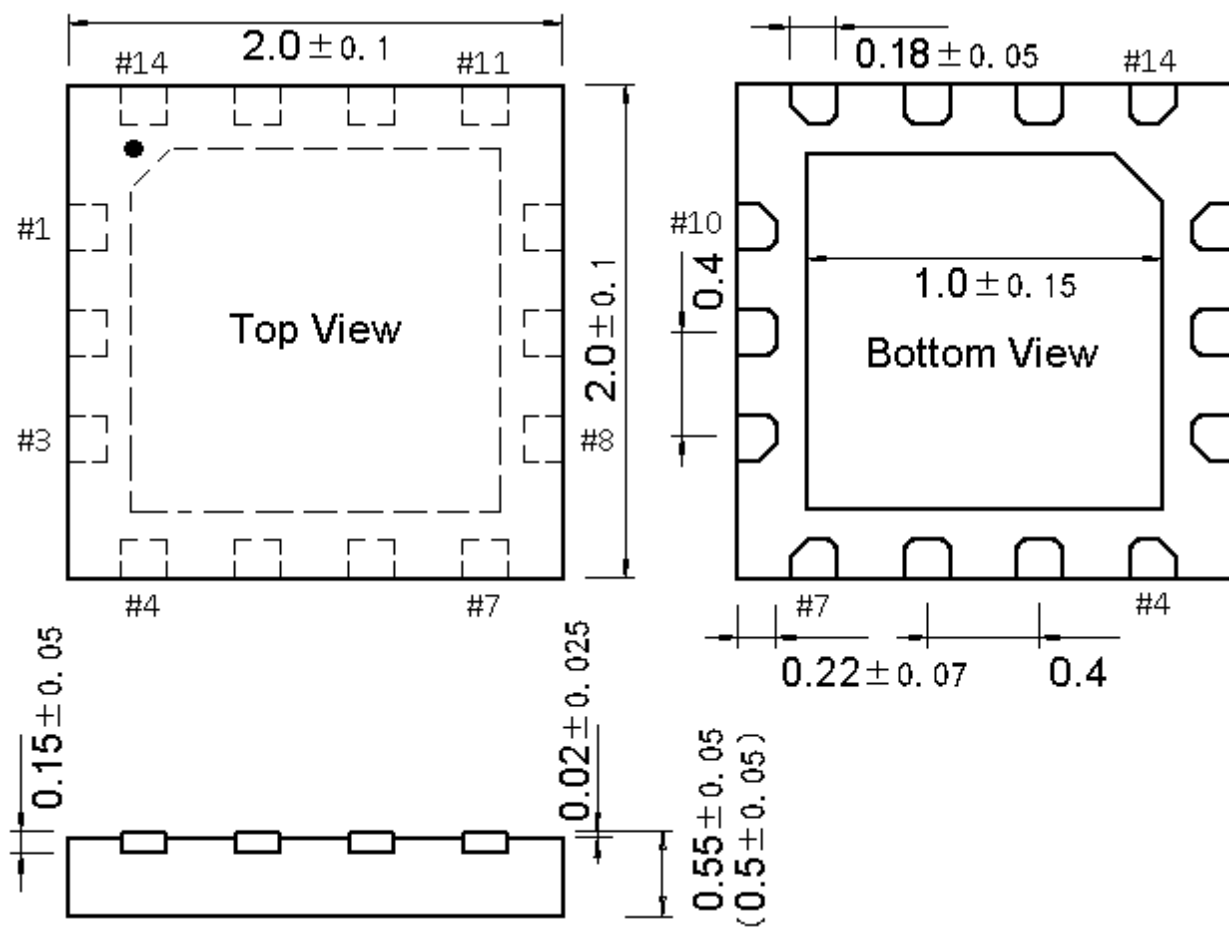
· QFN10F15X2 (QFN10-1.5\*2\*0.55-0.4)



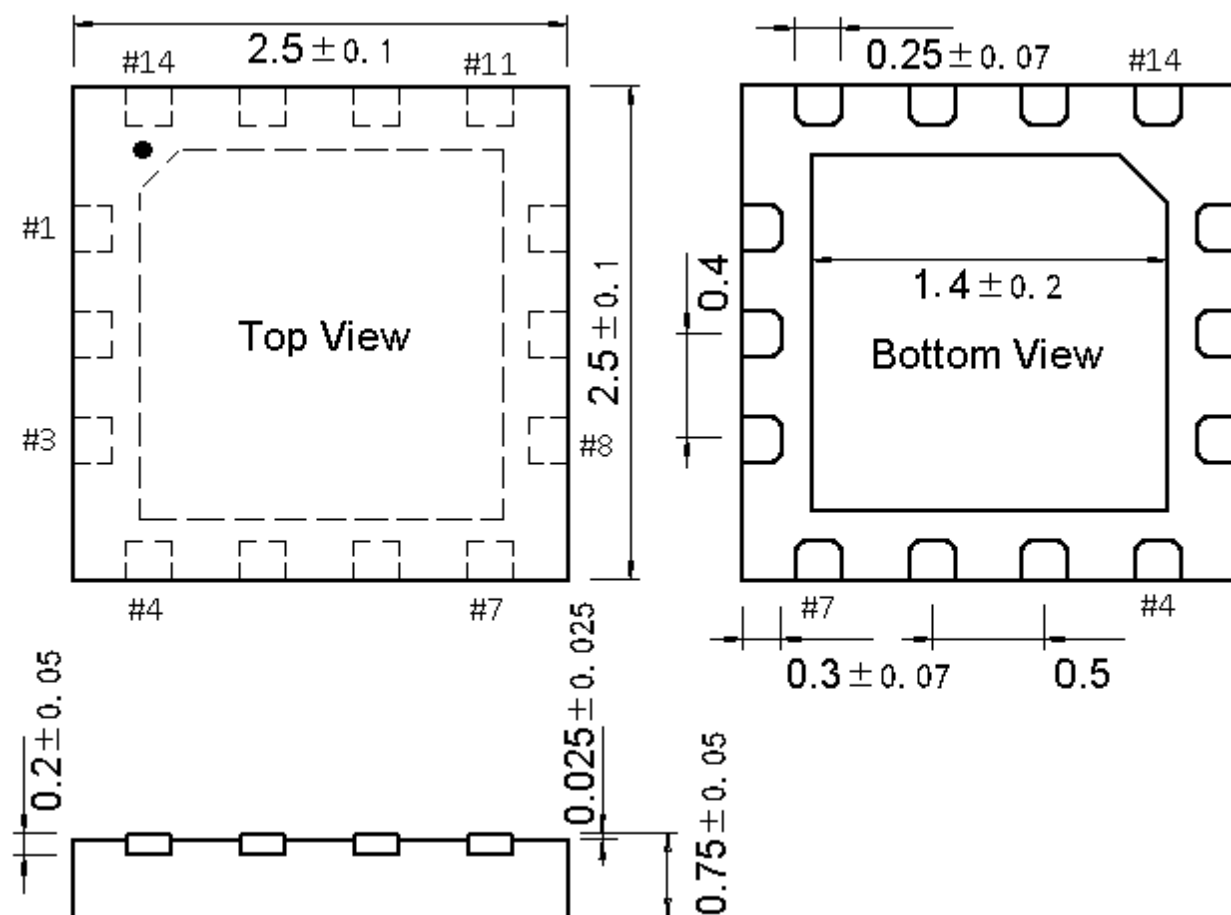
## . QFN12 (QFN12-2\*2\*0.5-0.4)



· QFN14 (QFN14-2\*2\*0.55-0.4)

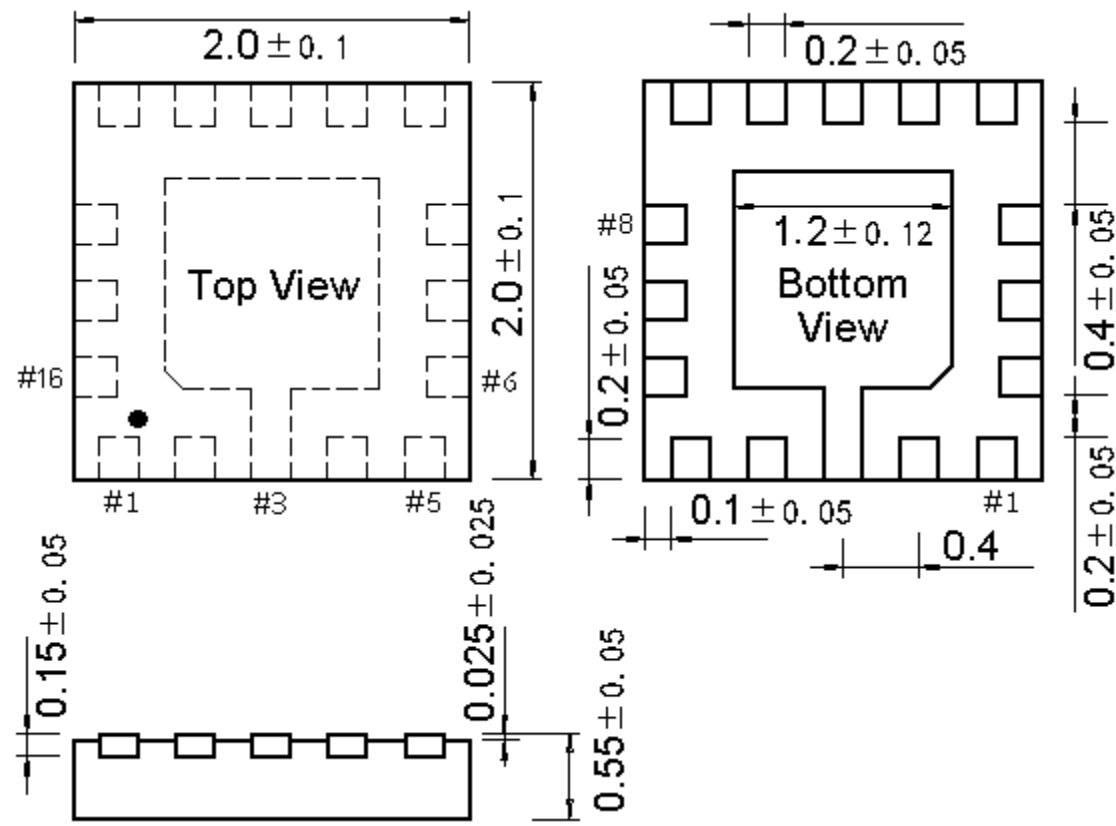


· QFN14X25 (QFN14-2.5\*2.5\*0.75-0.5)

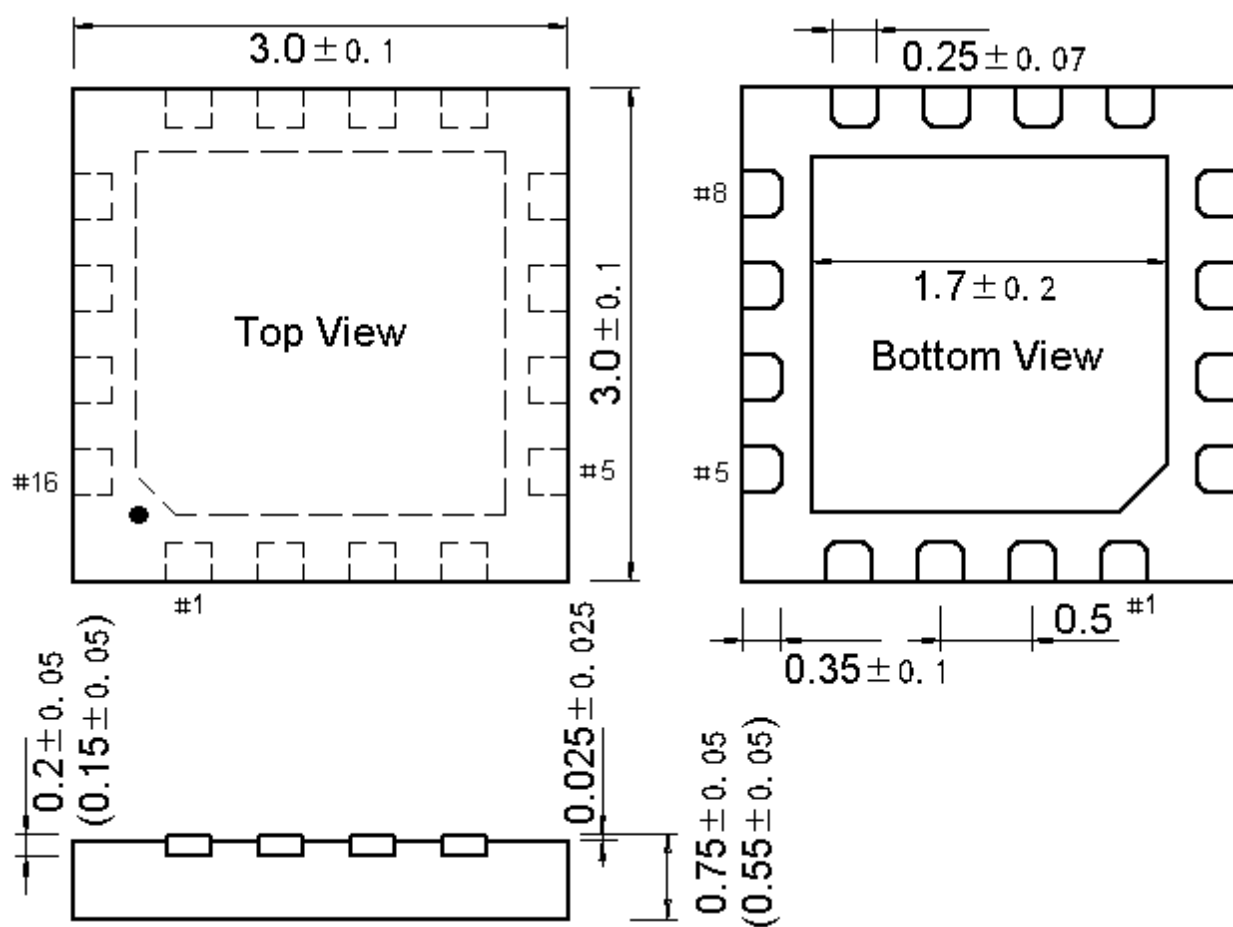




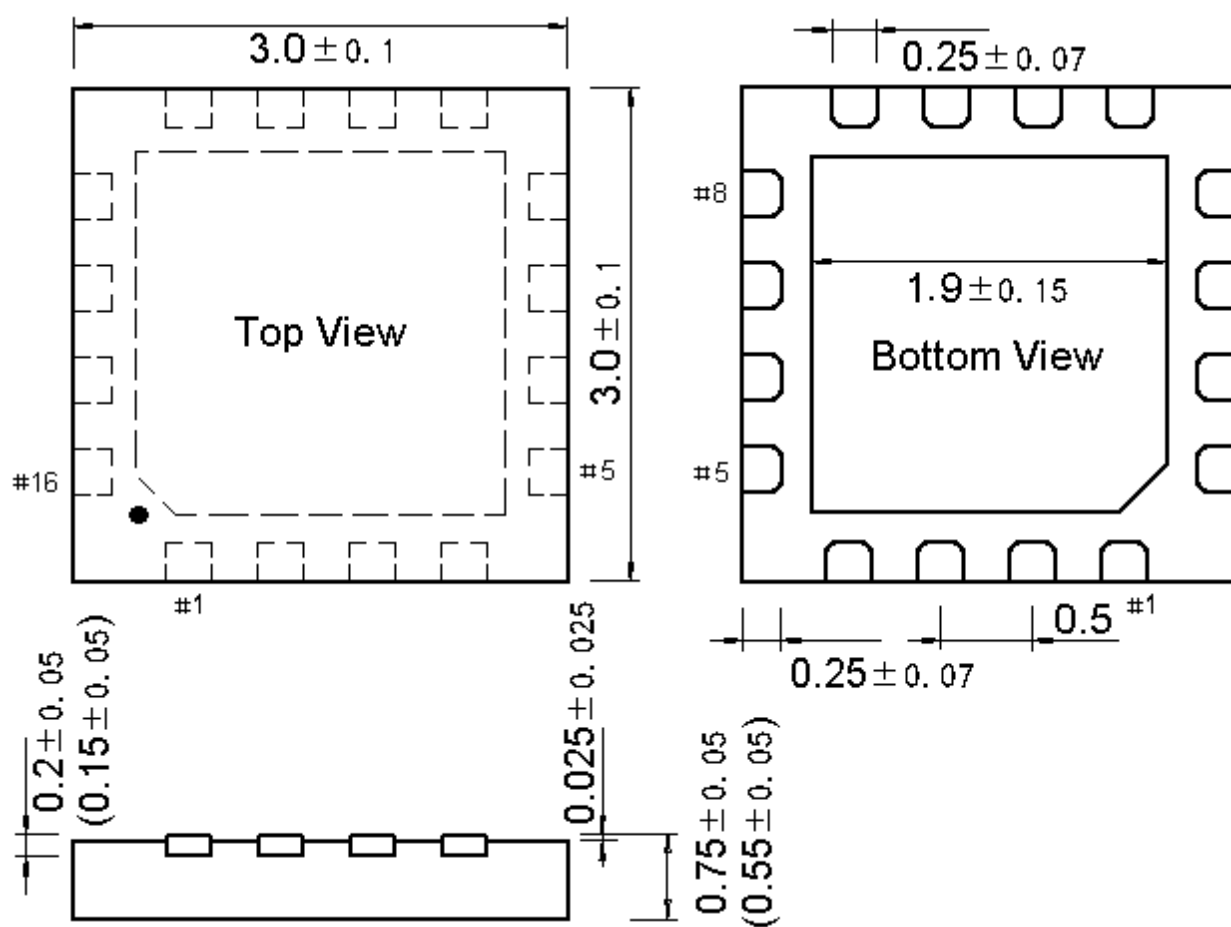
· QFN16C2 (QFN16C-2\*2\*0.55-0.4)



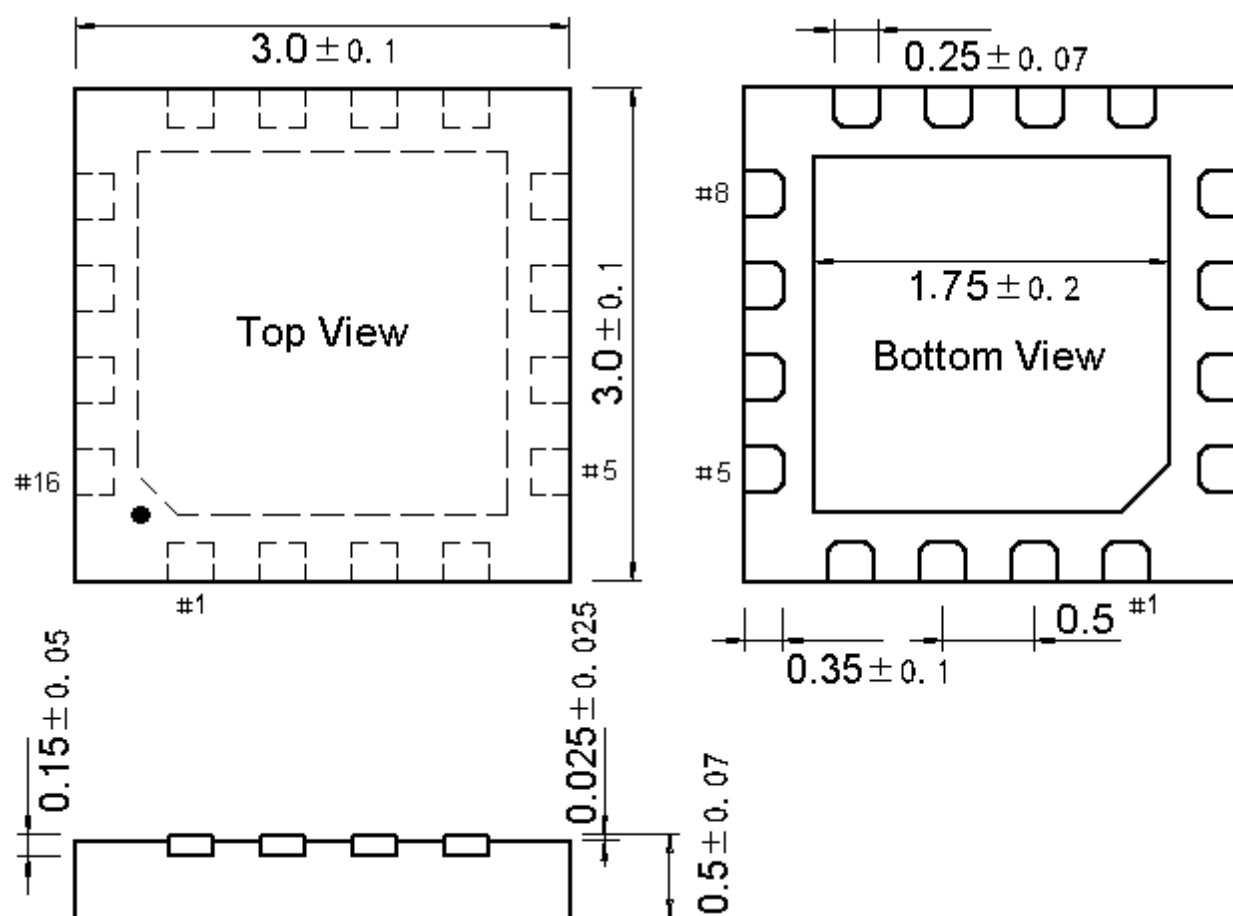
## . QFN16 (QFN16-3\*3-0.5)



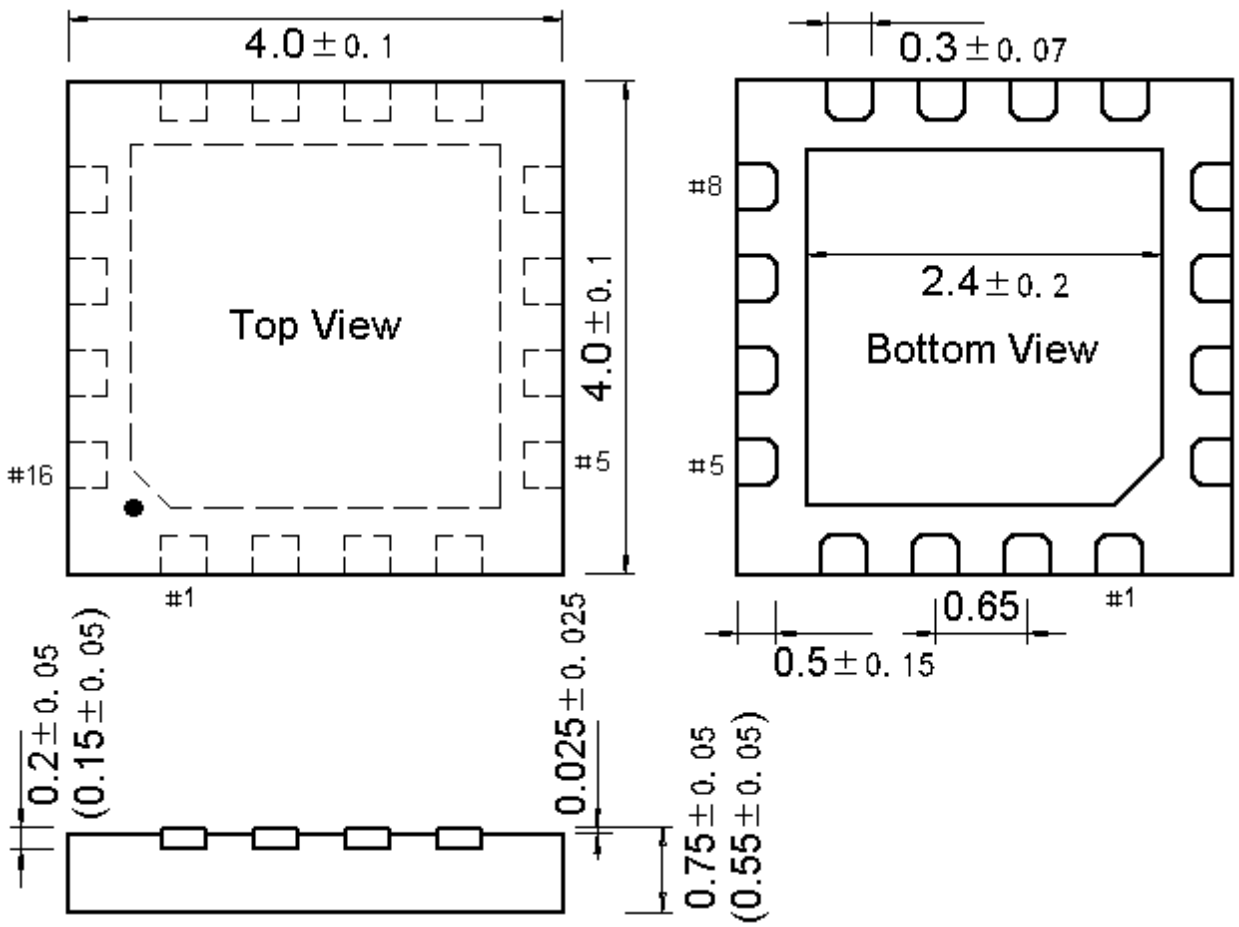
. QFN16L (QFN16L-3\*3-0.5)



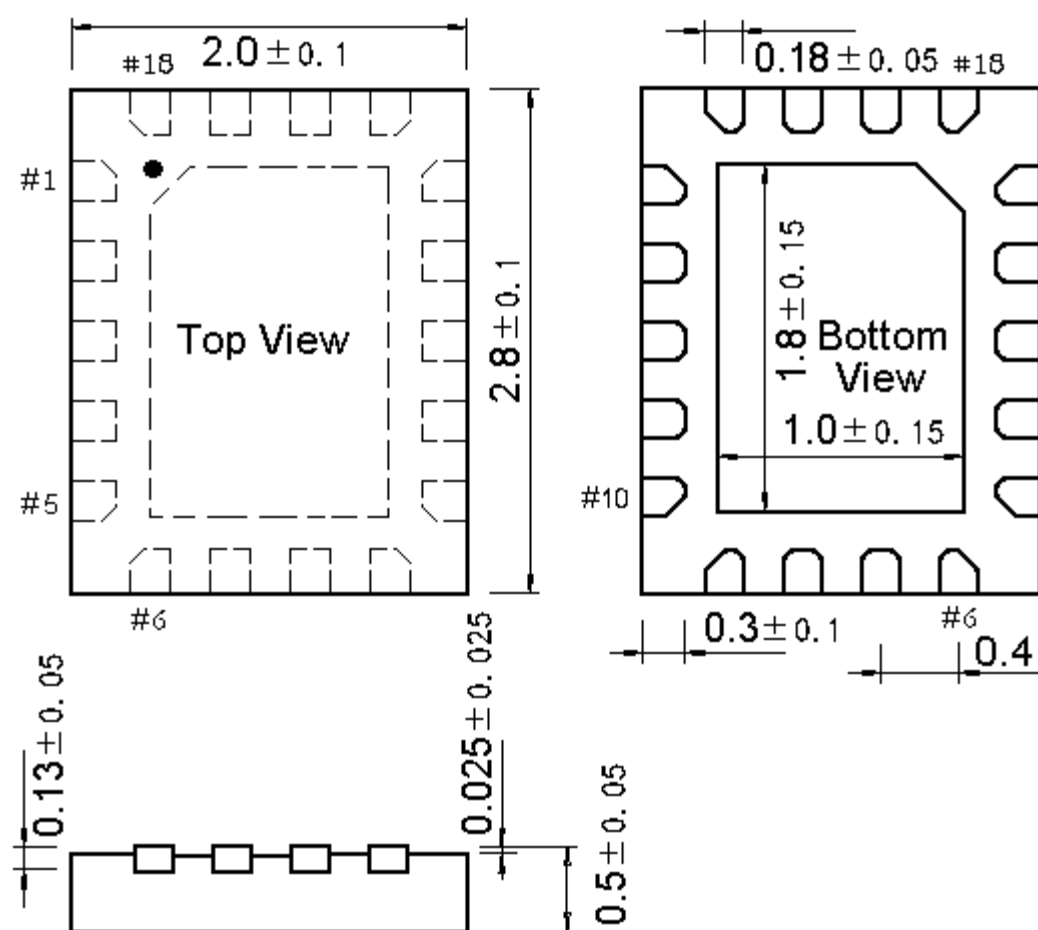
. QFN16-3\*3\*0.5-0.4



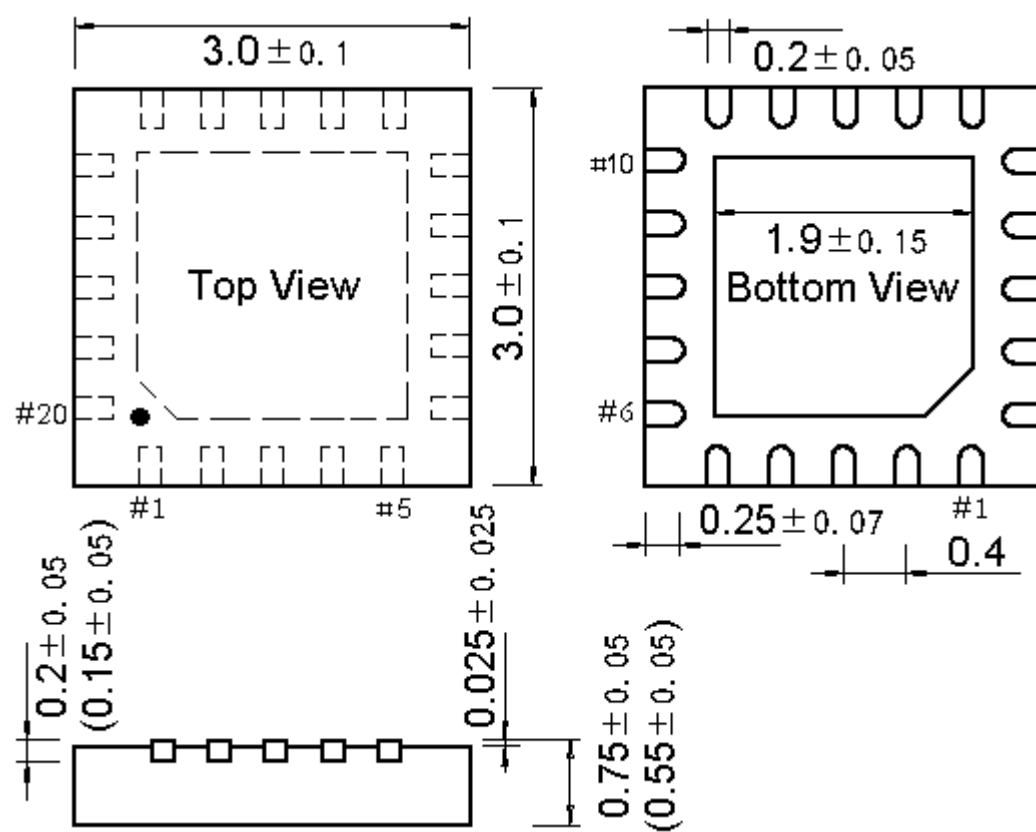
· QFN16X4 (QFN16-4\*4-0.65)



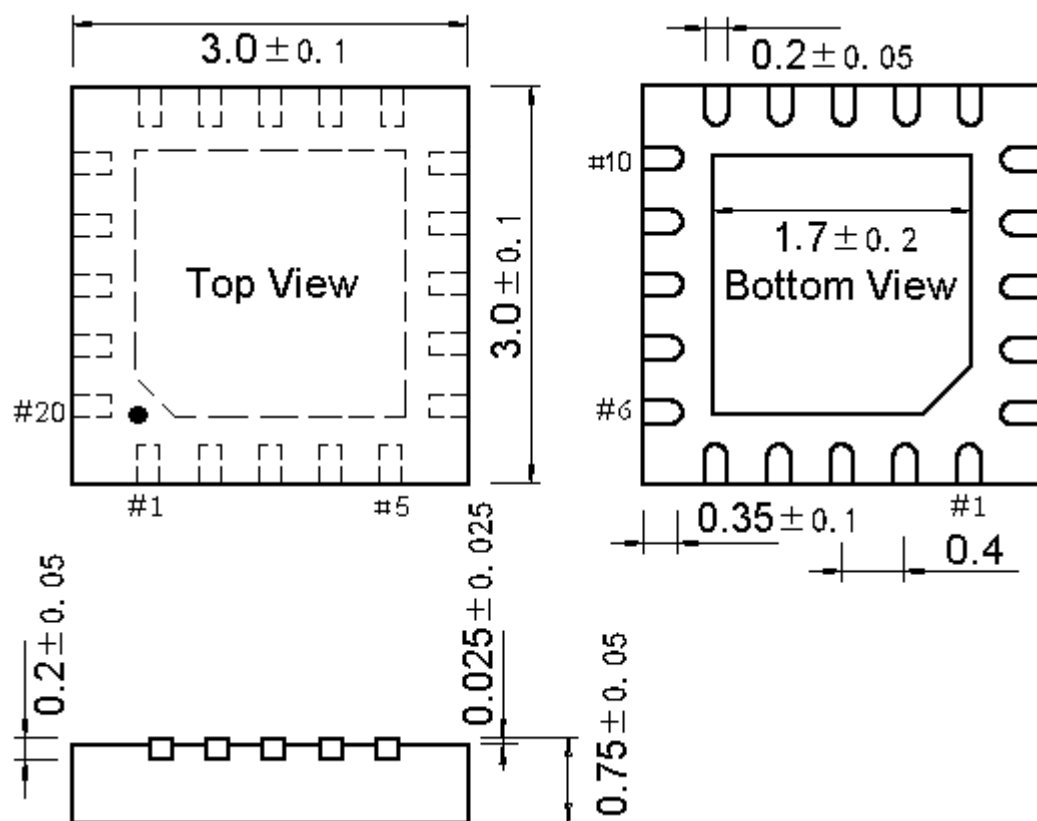
## . QFN18X2X28 (QFN18-2\*2.8\*0.5-0.4)



## . QFN20 (QFN20-3\*3-0.4)

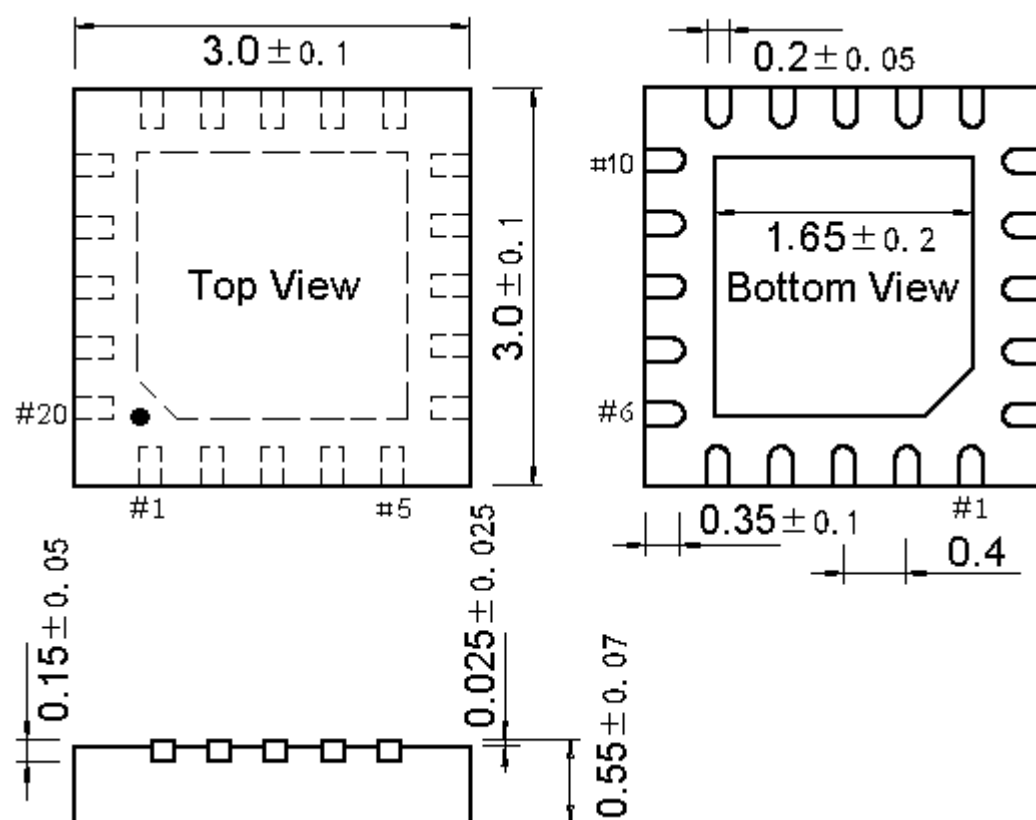


## . QFN20L (QFN20L-3\*3-0.4)

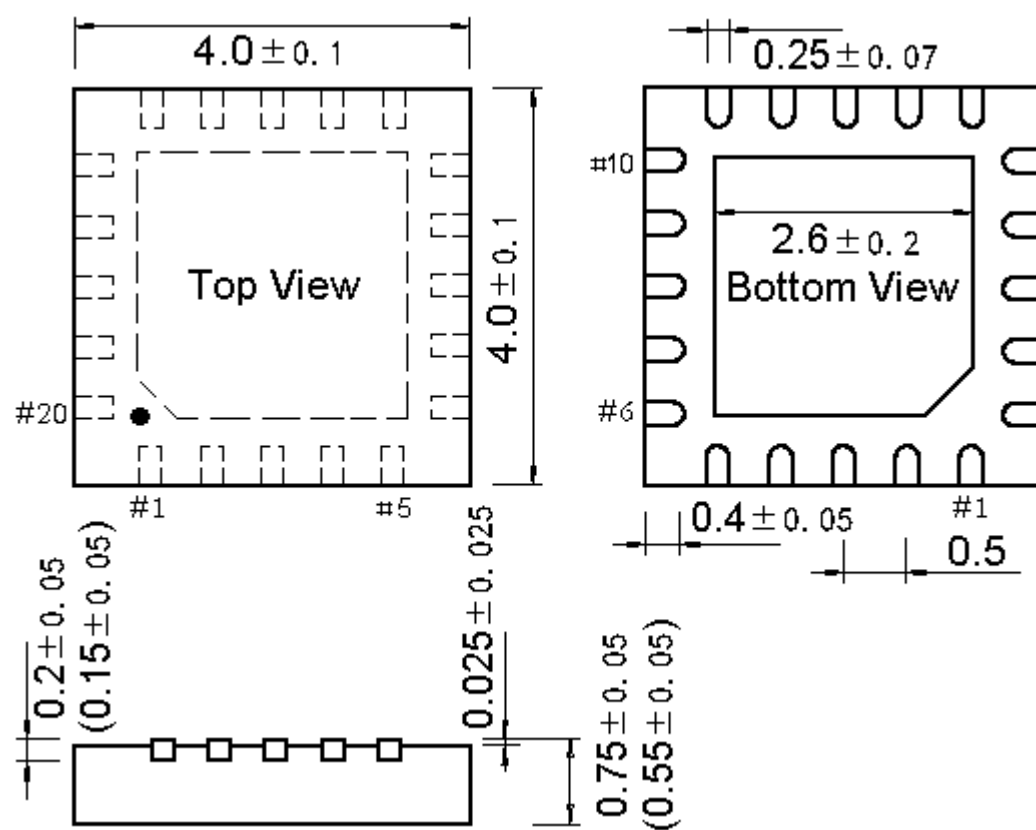




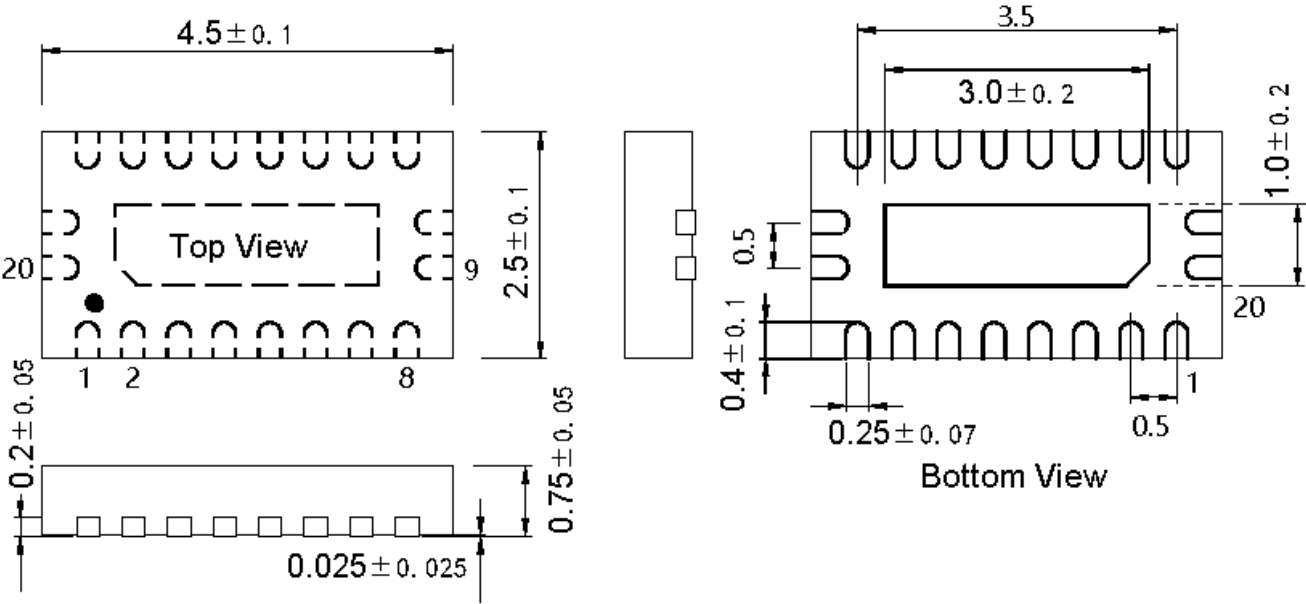
· QFN20-3\*3\*0.55-0.4



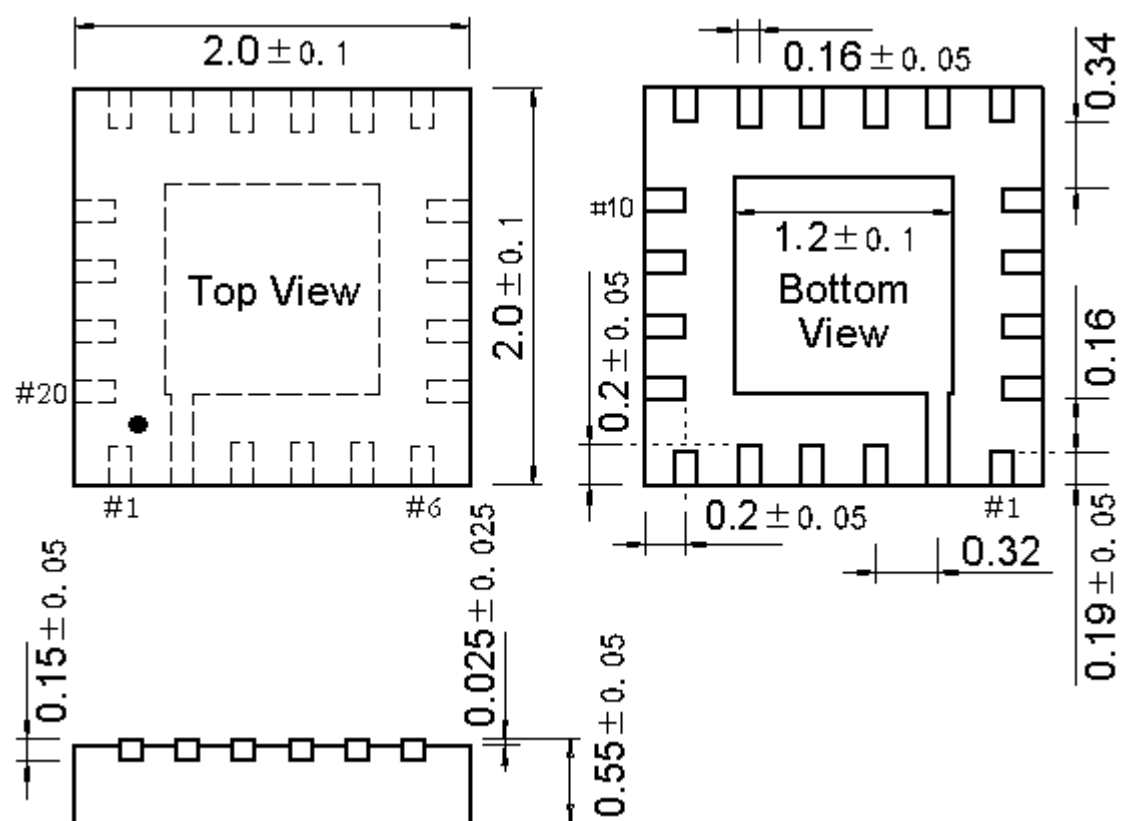
## . QFN20X4 (QFN20-4\*4-0.5)



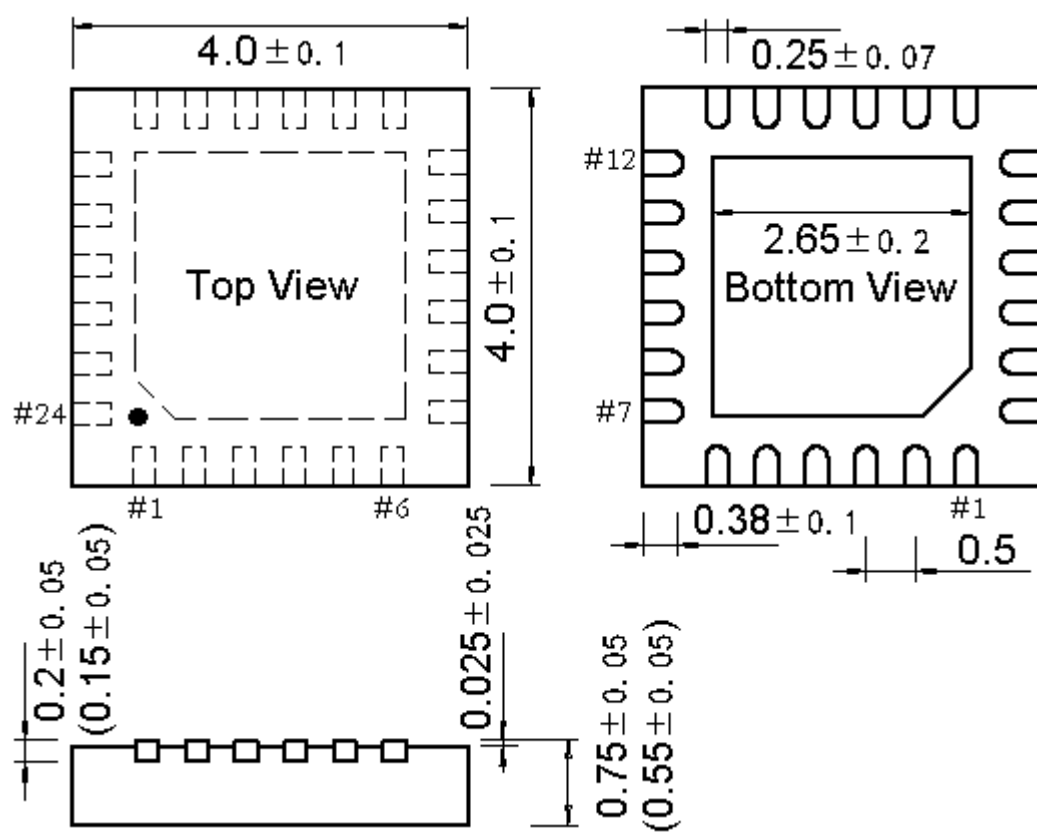
QFN20X25X45 (QFN20-2.5\*4.5\*0.75-0.5)



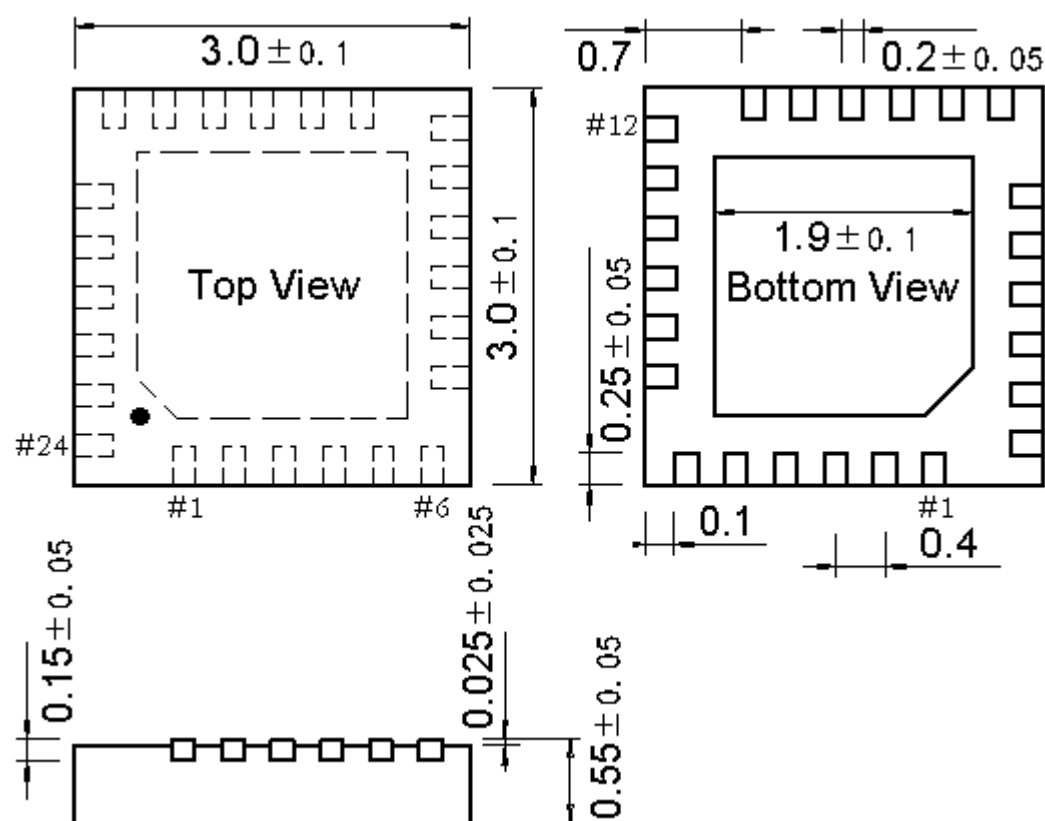
QFN20C2 (QFN20C-2\*2\*0.55-0.32)



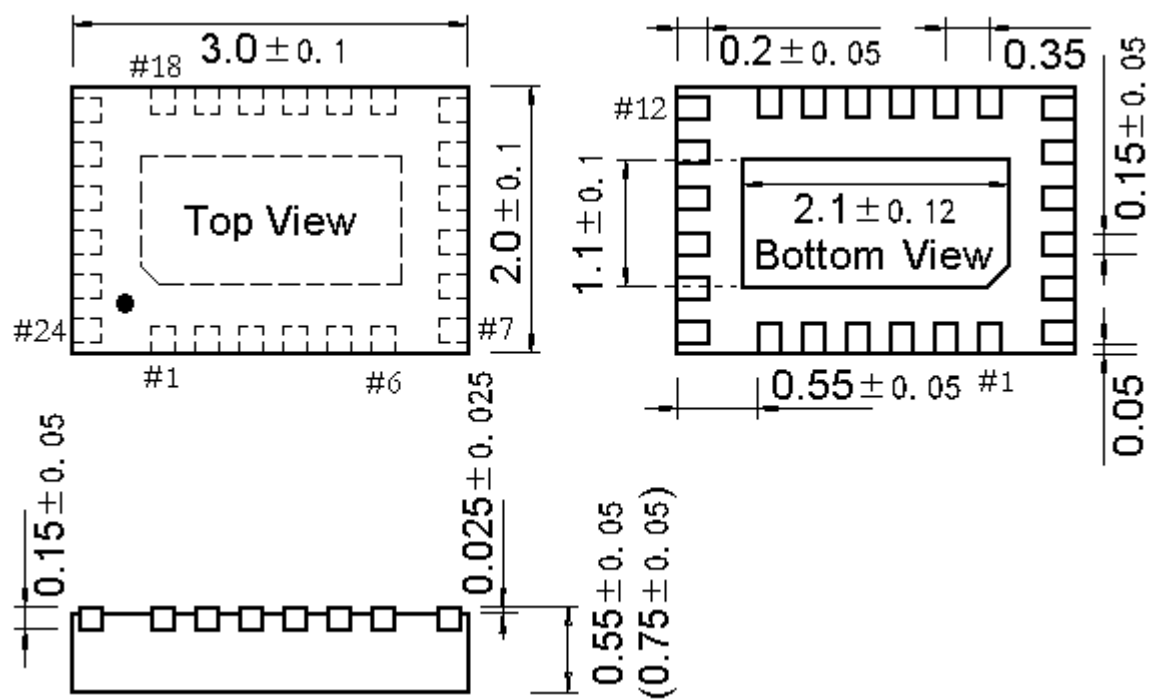
## . QFN24 (QFN24-4\*4-0.5)



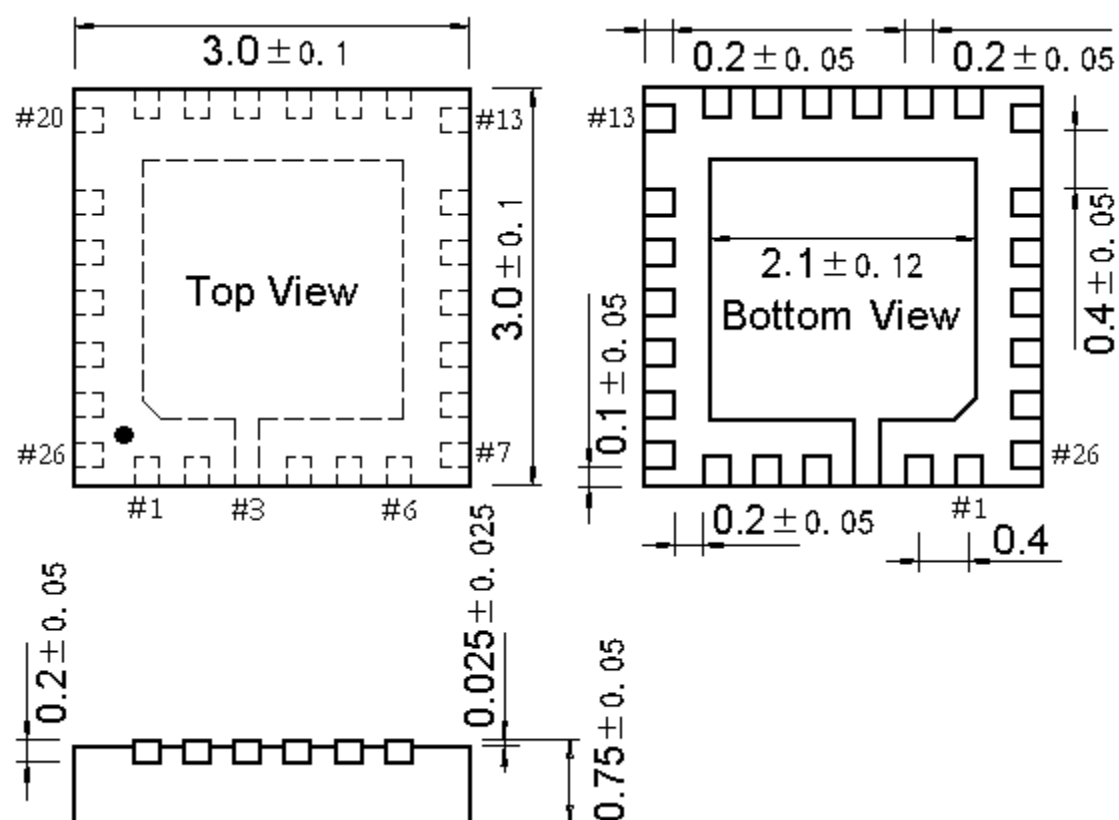
## . QFN24X3 (QFN24-3\*3\*0.55-0.4)



## . QFN24C2X3 (QFN24C-2\*3-0.35)

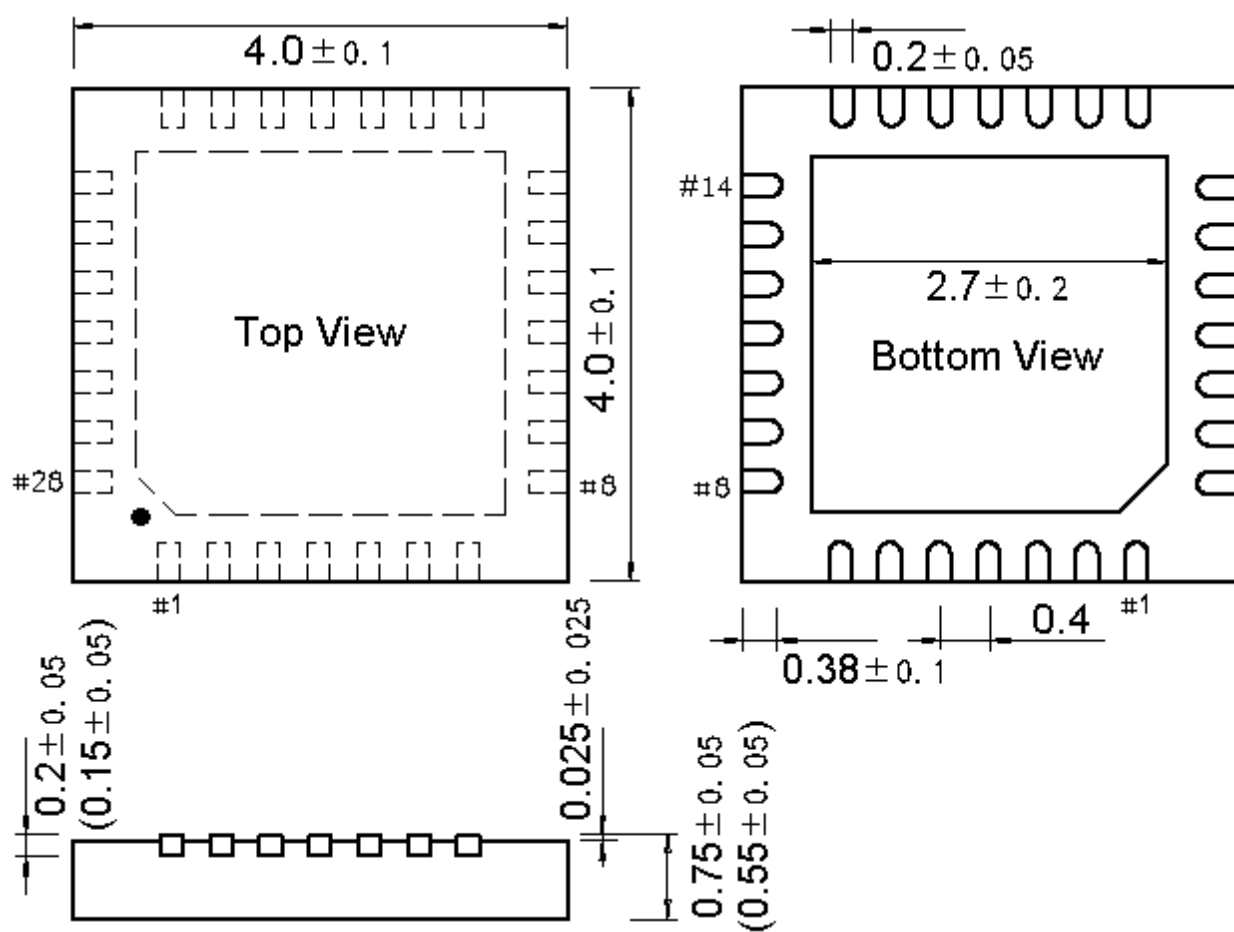


## . QFN26C3 (QFN26C-3\*3\*0.75-0.4)

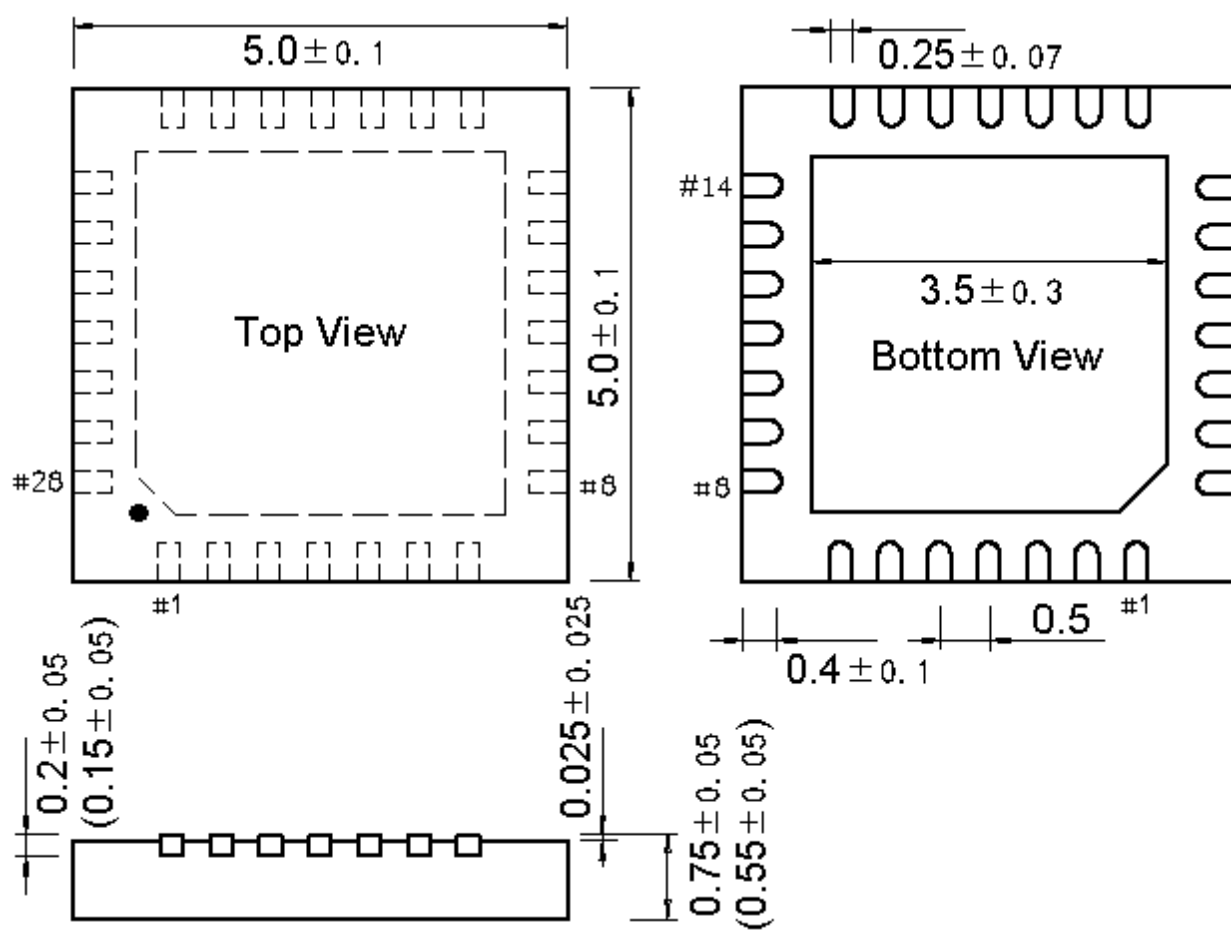




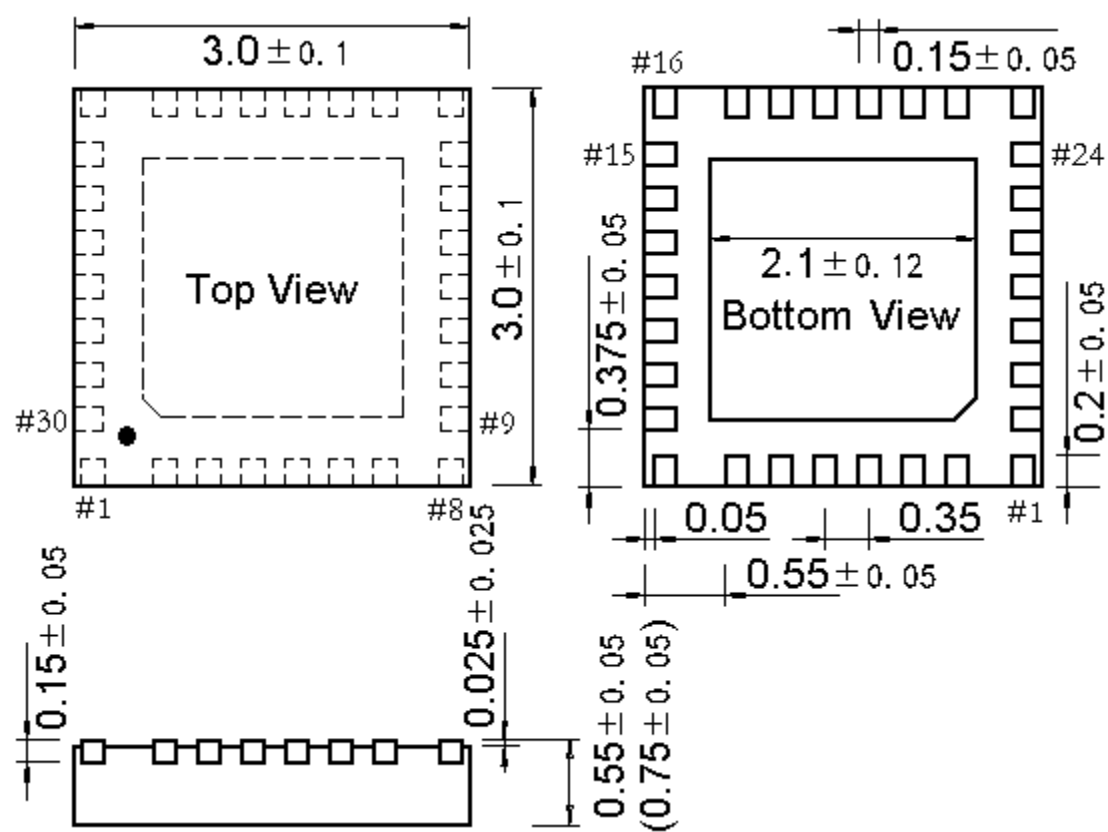
QFN28 (QFN28-4\*4-0.4)



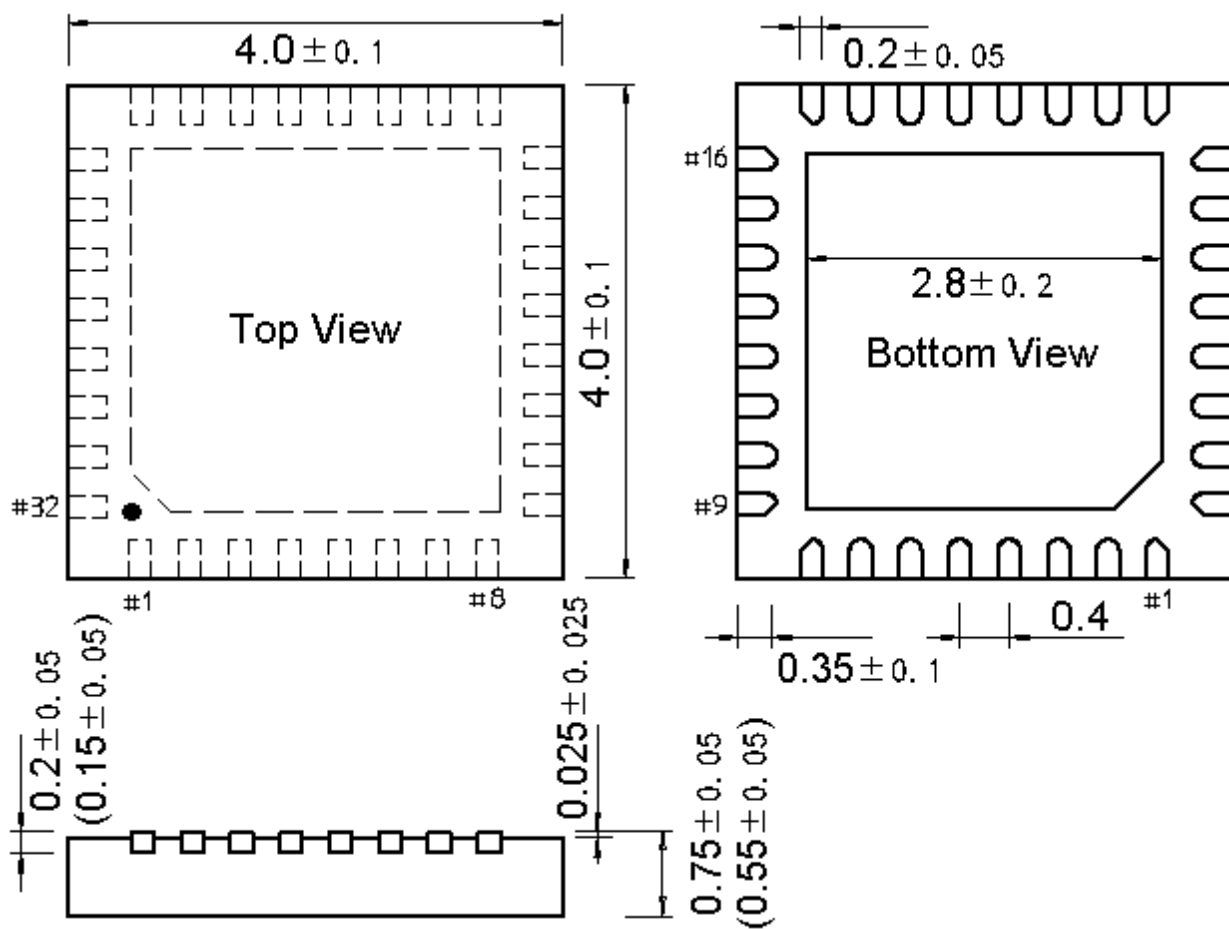
## . QFN28X5 (QFN28-5\*5-0.5)



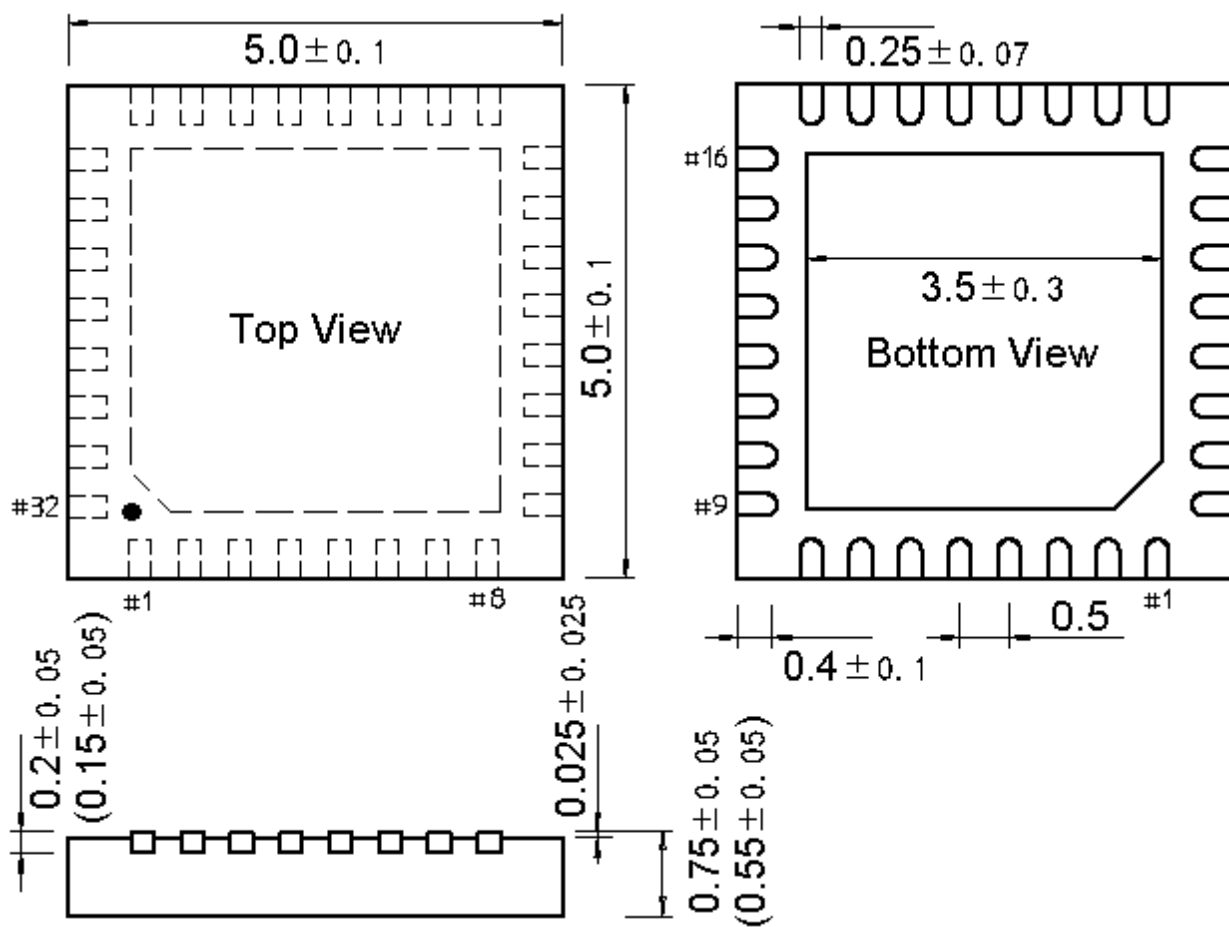
## . QFN30C3 (QFN30C-3\*3-0.35)



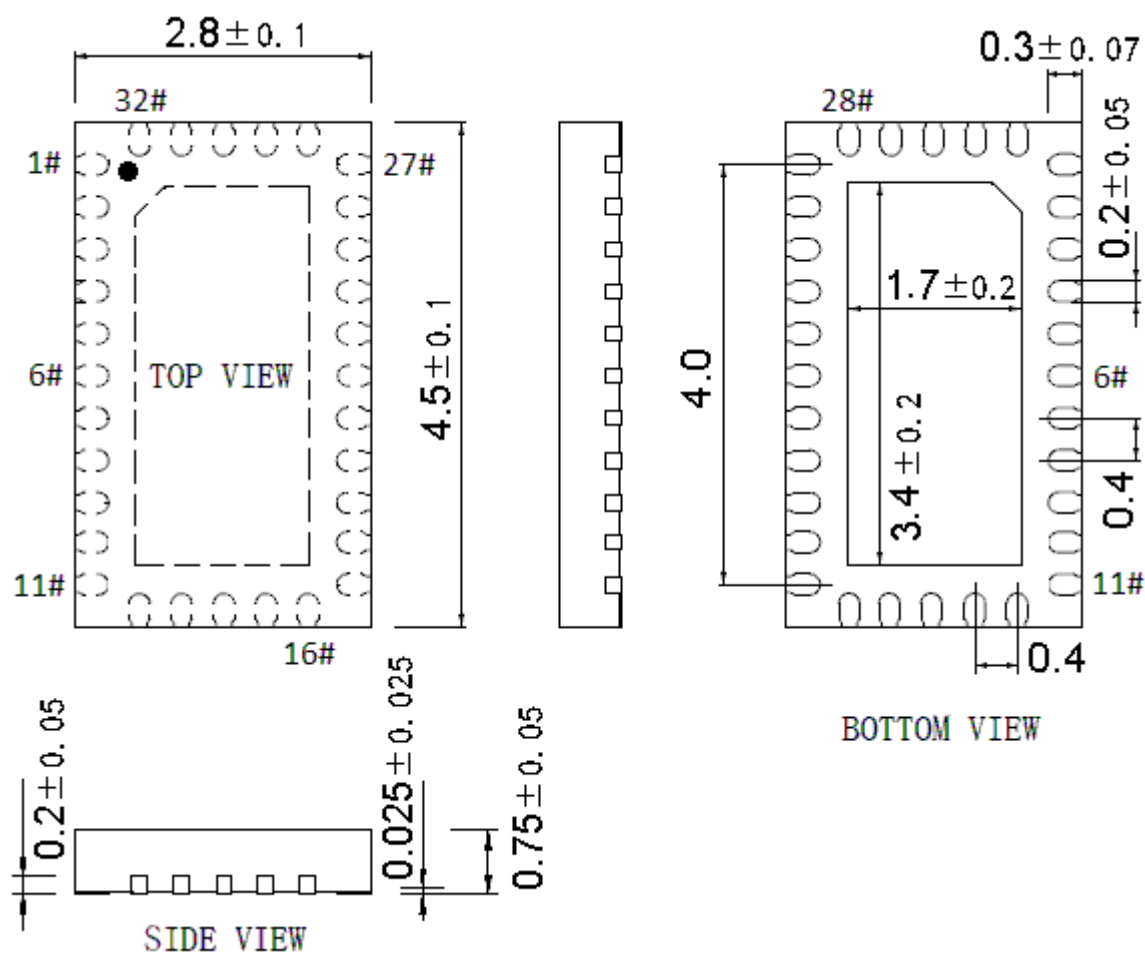
## . QFN32 (QFN32-4\*4-0.4)



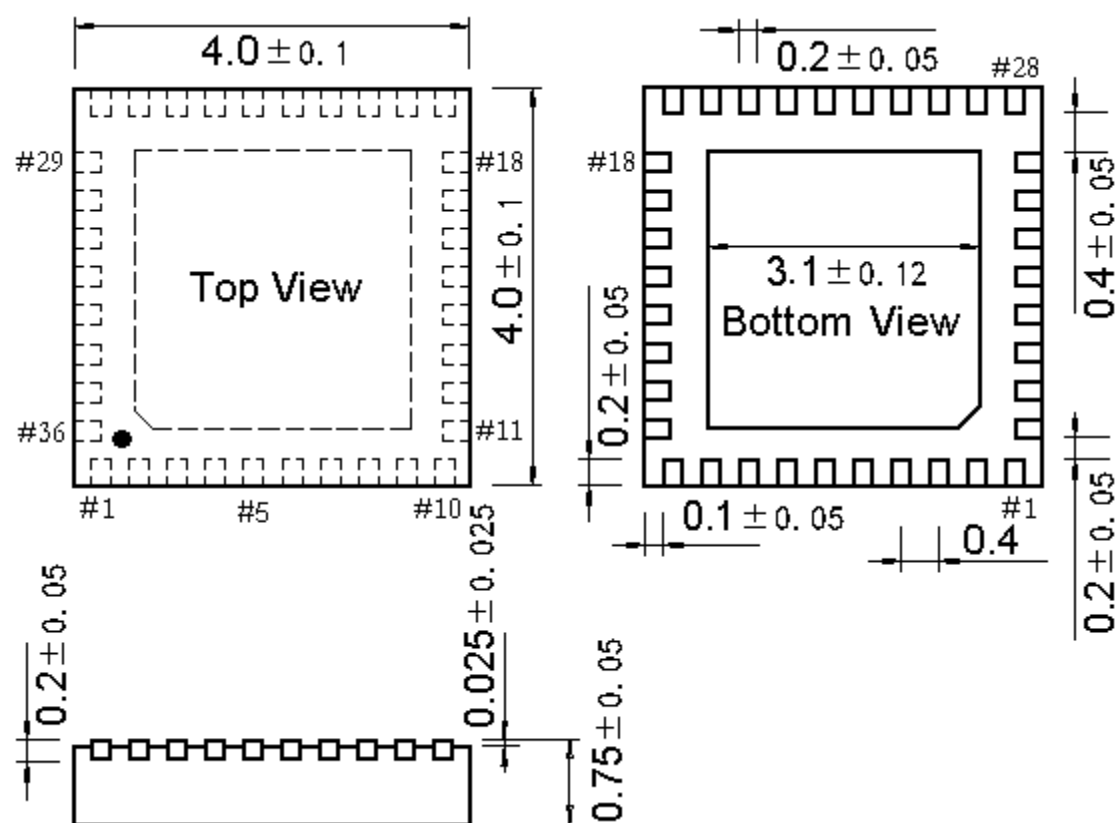
· QFN32X5 (QFN32-5\*5-0.5)



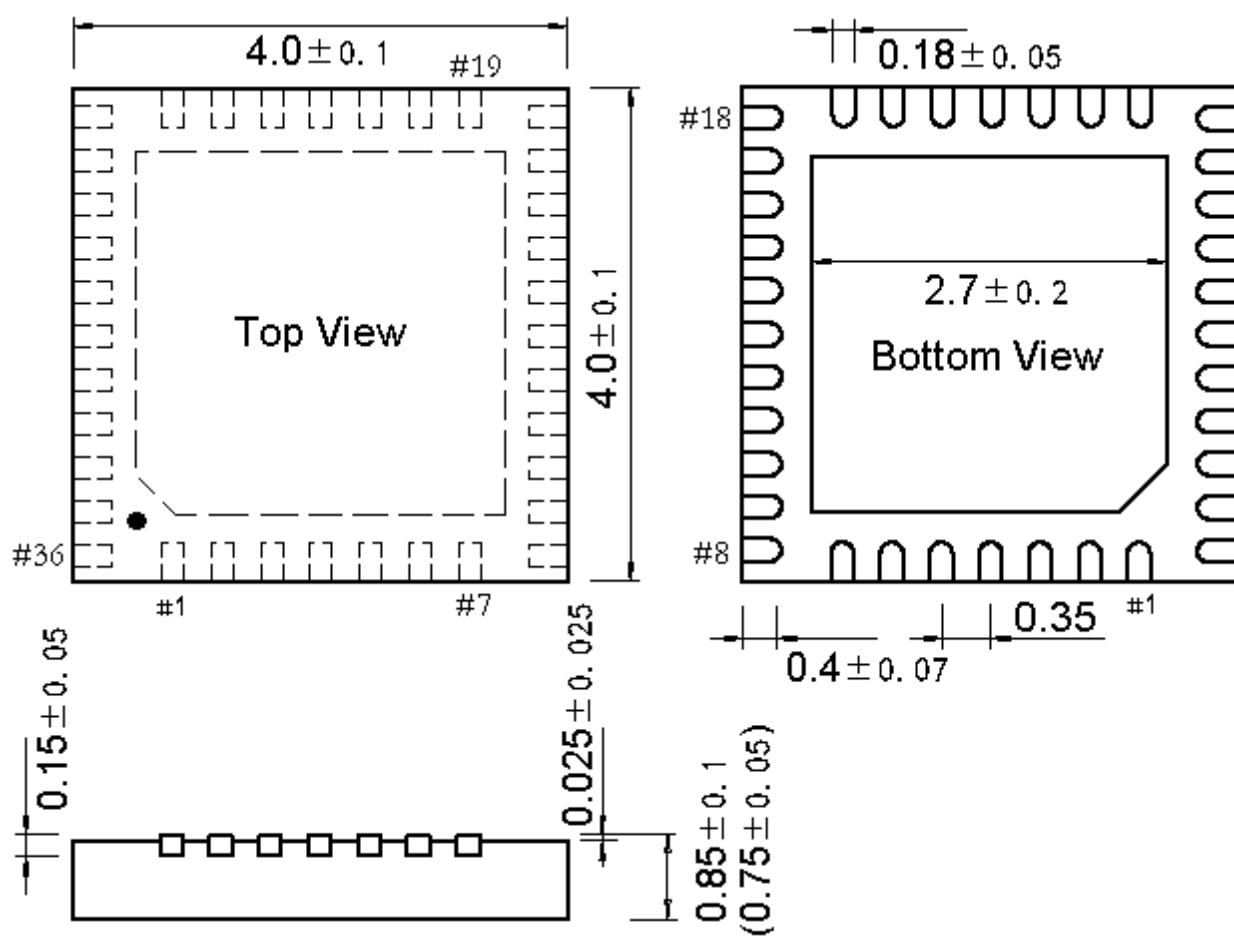
· QFN32C28X45 (QFN32C-2.8\*4.5\*0.75-0.4)



## . QFN36C4 (QFN36C-4\*4\*0.75-0.4)

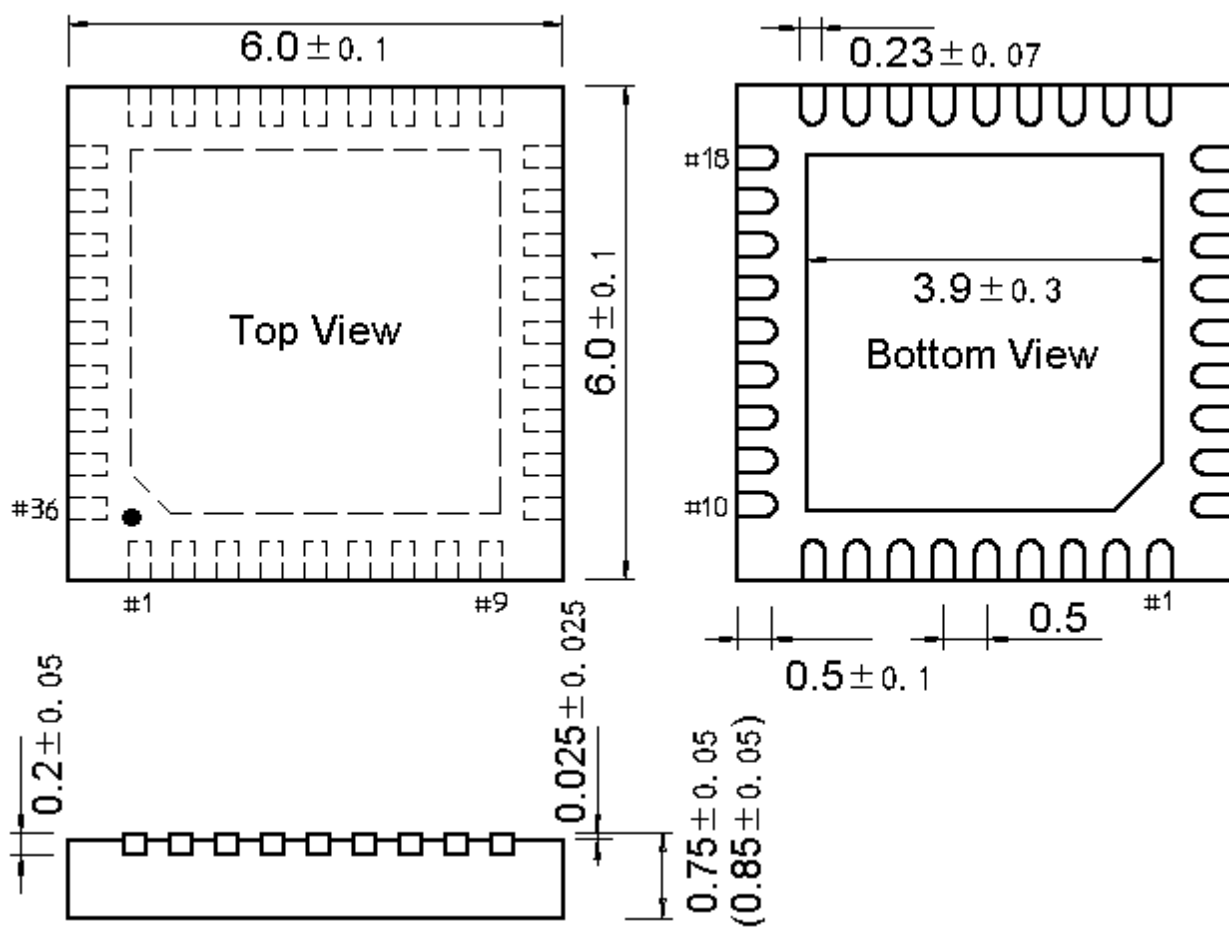


## . QFN36X4 (QFN36-4\*4-0.35)

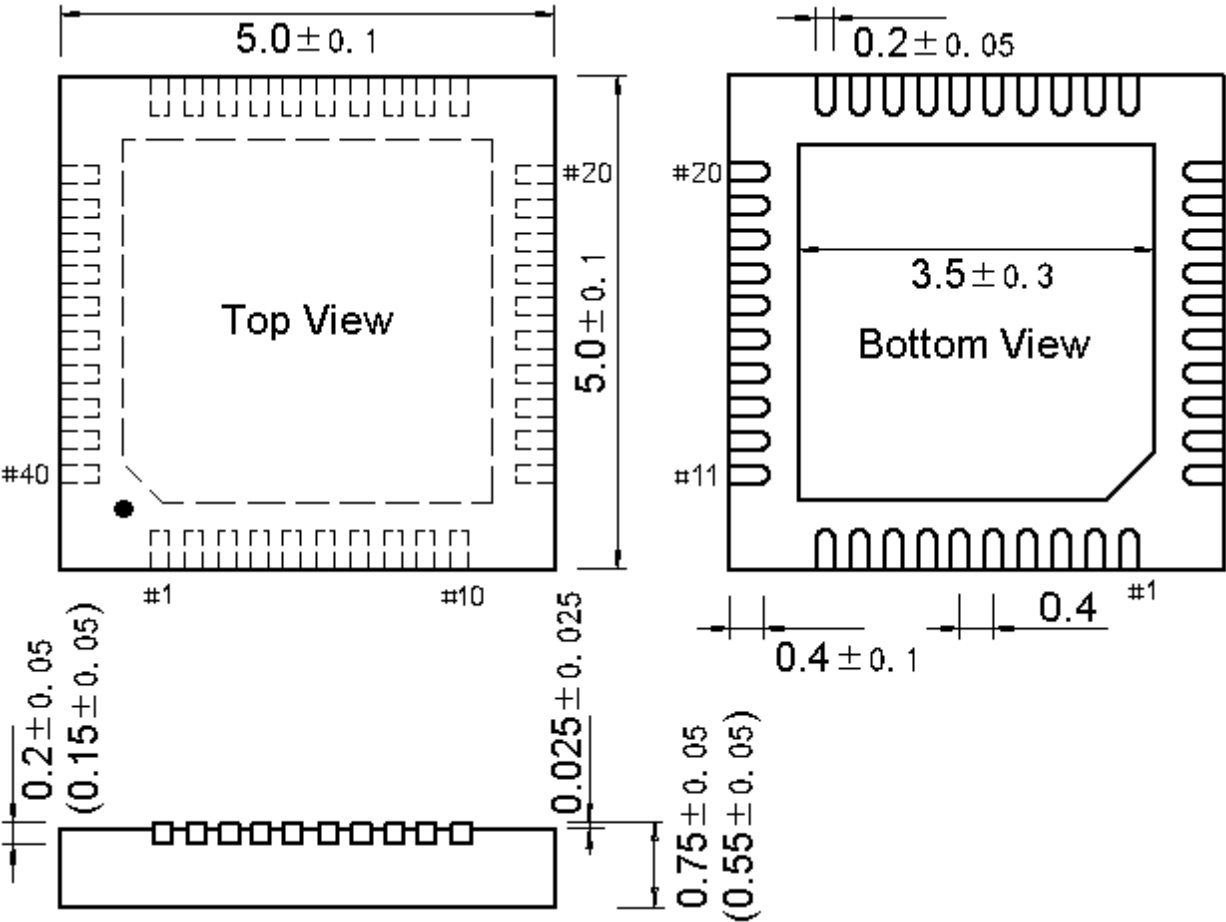




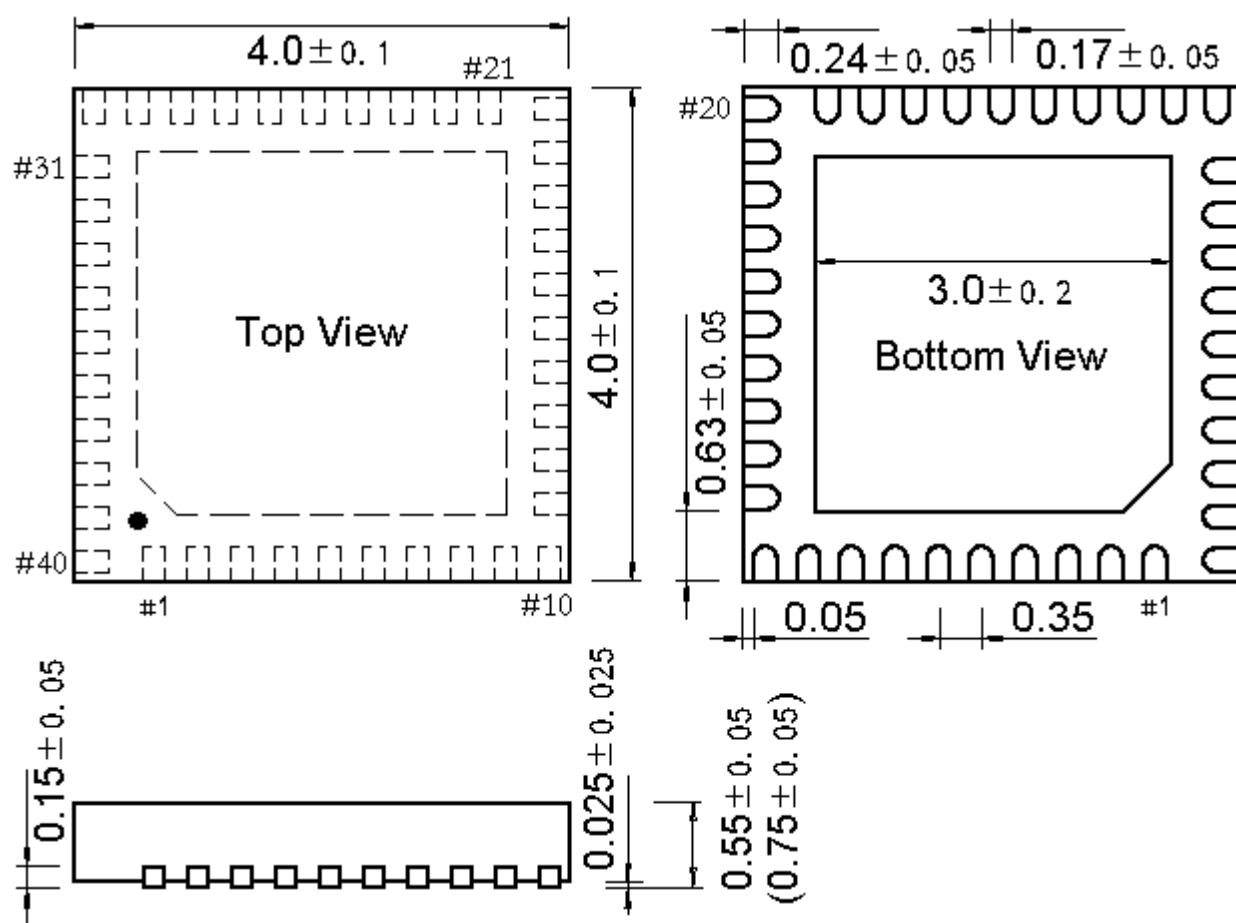
## . QFN36X6 (QFN36-6\*6-0.5)



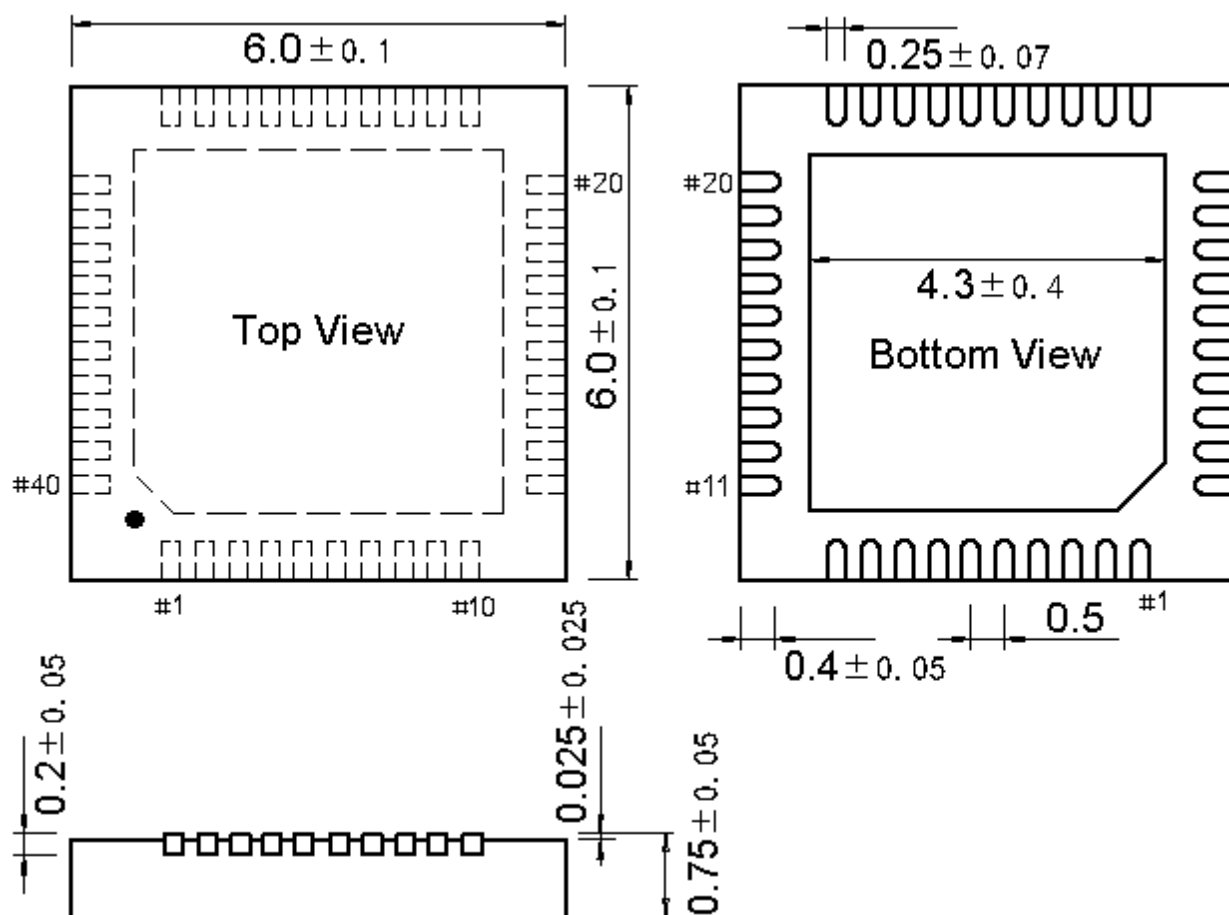
QFN40 (QFN40-5\*5-0.4)



## . QFN40C4 (QFN40C-4\*4-0.35)

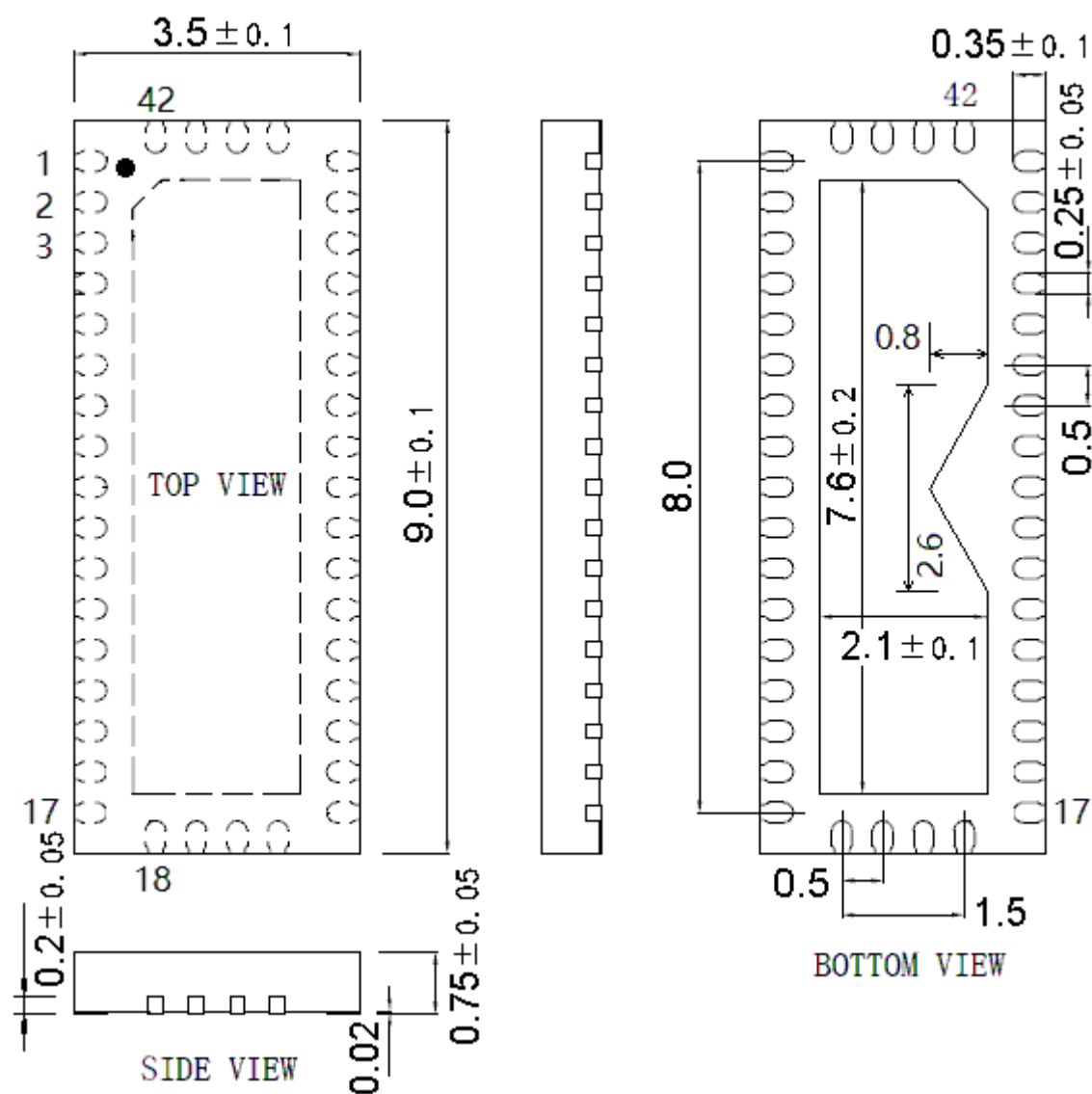


## . QFN40X6 (QFN40-6\*6-0.5)

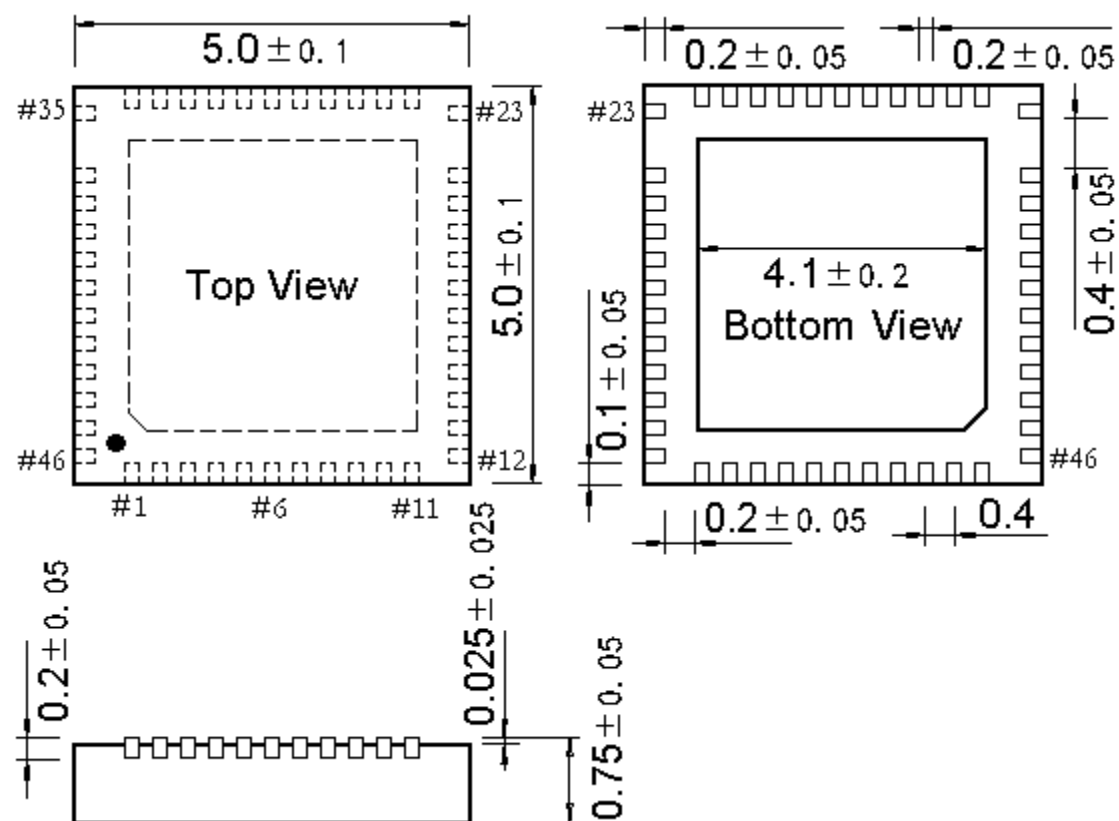




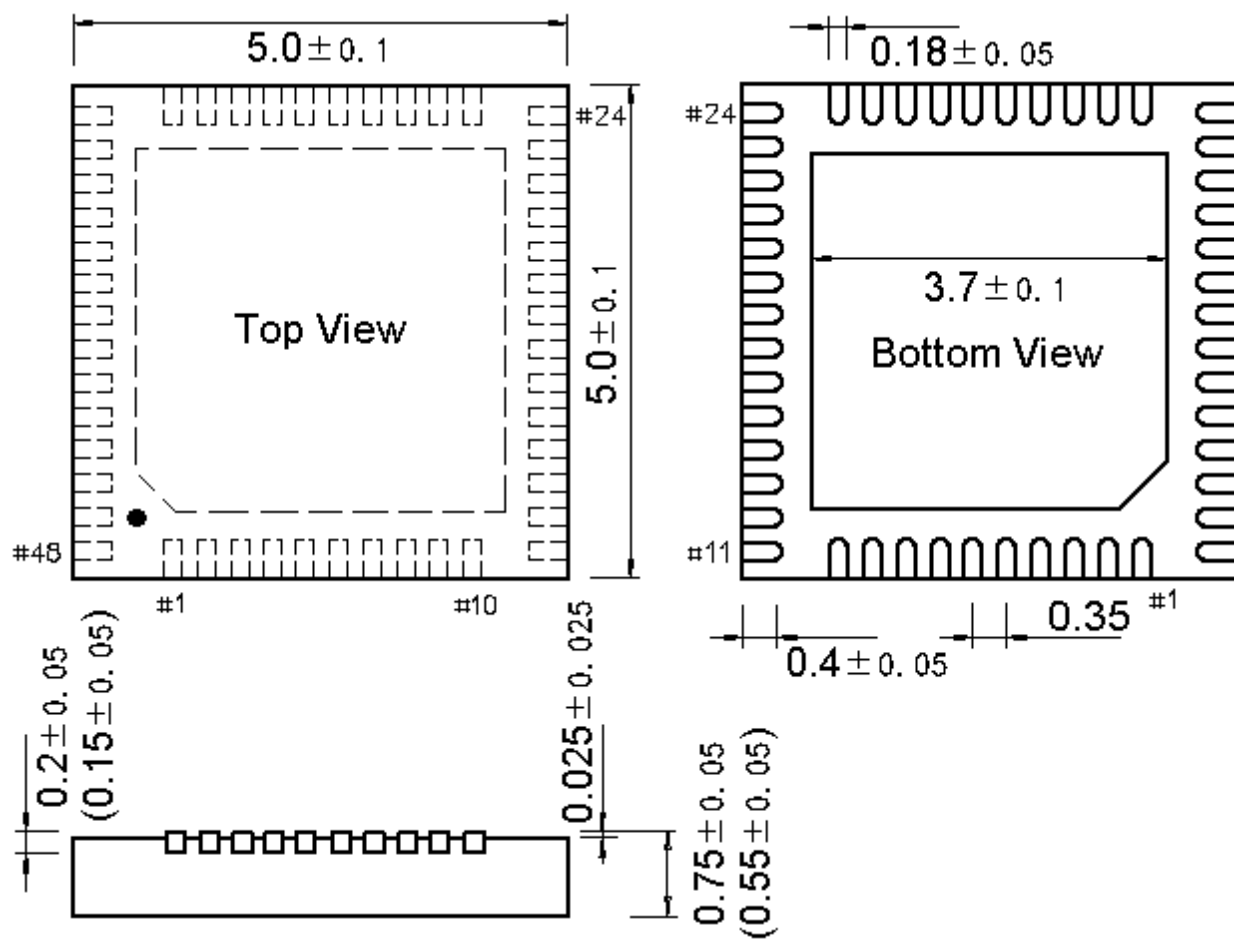
· QFN42C35X9 (QFN42C-3.5\*9\*0.75-0.5)



## . QFN46C5 (QFN46C-5\*5\*0.75-0.4)

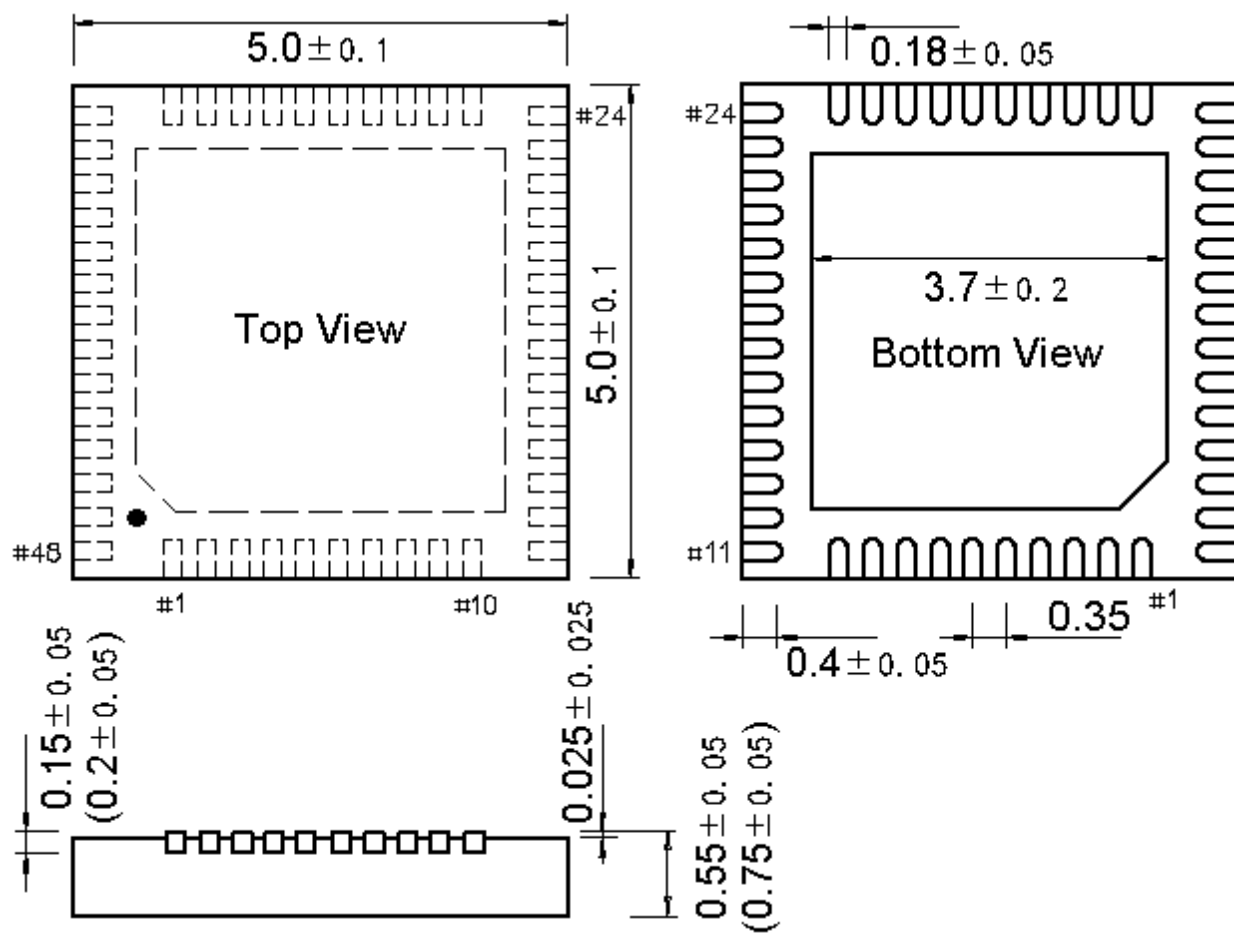


· QFN48 (QFN48-5\*5\*0.75-0.35)

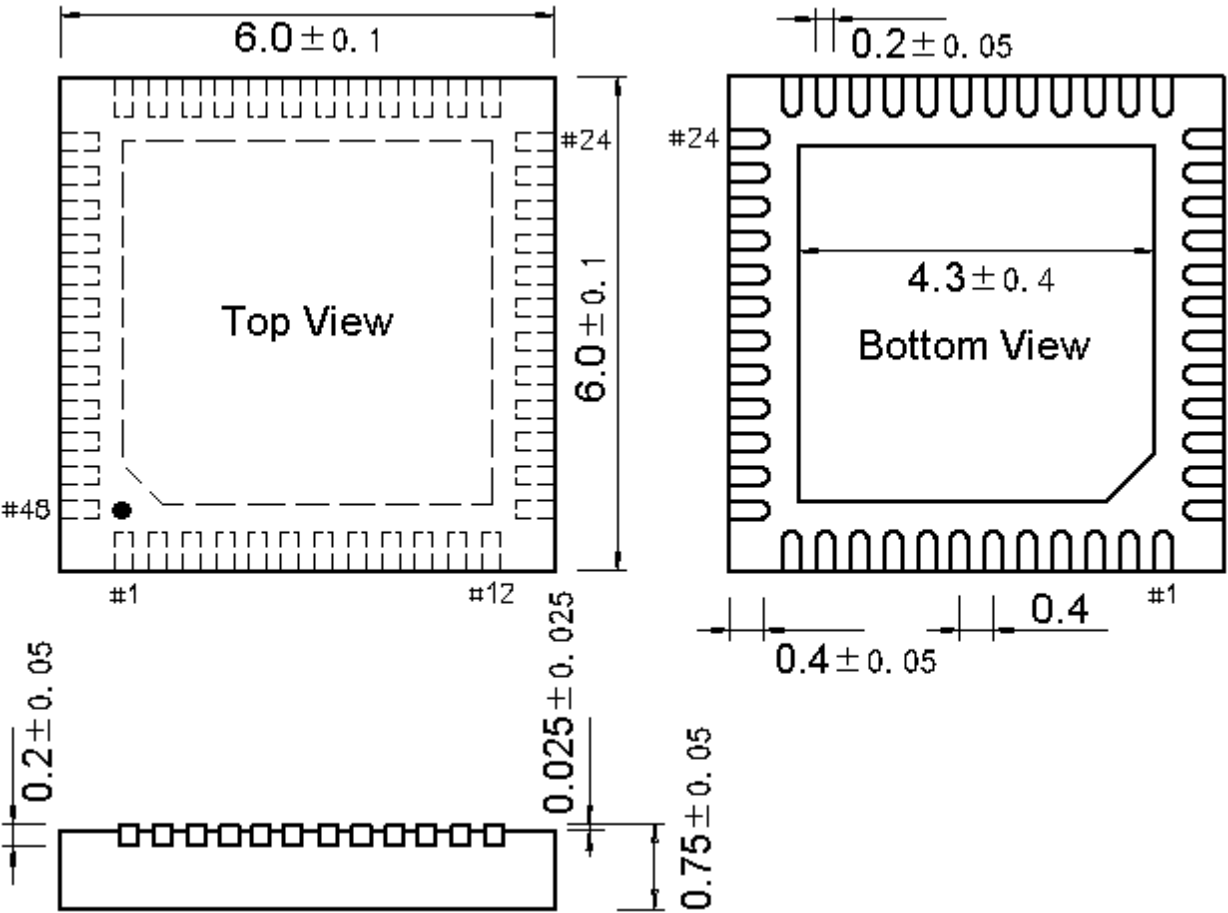




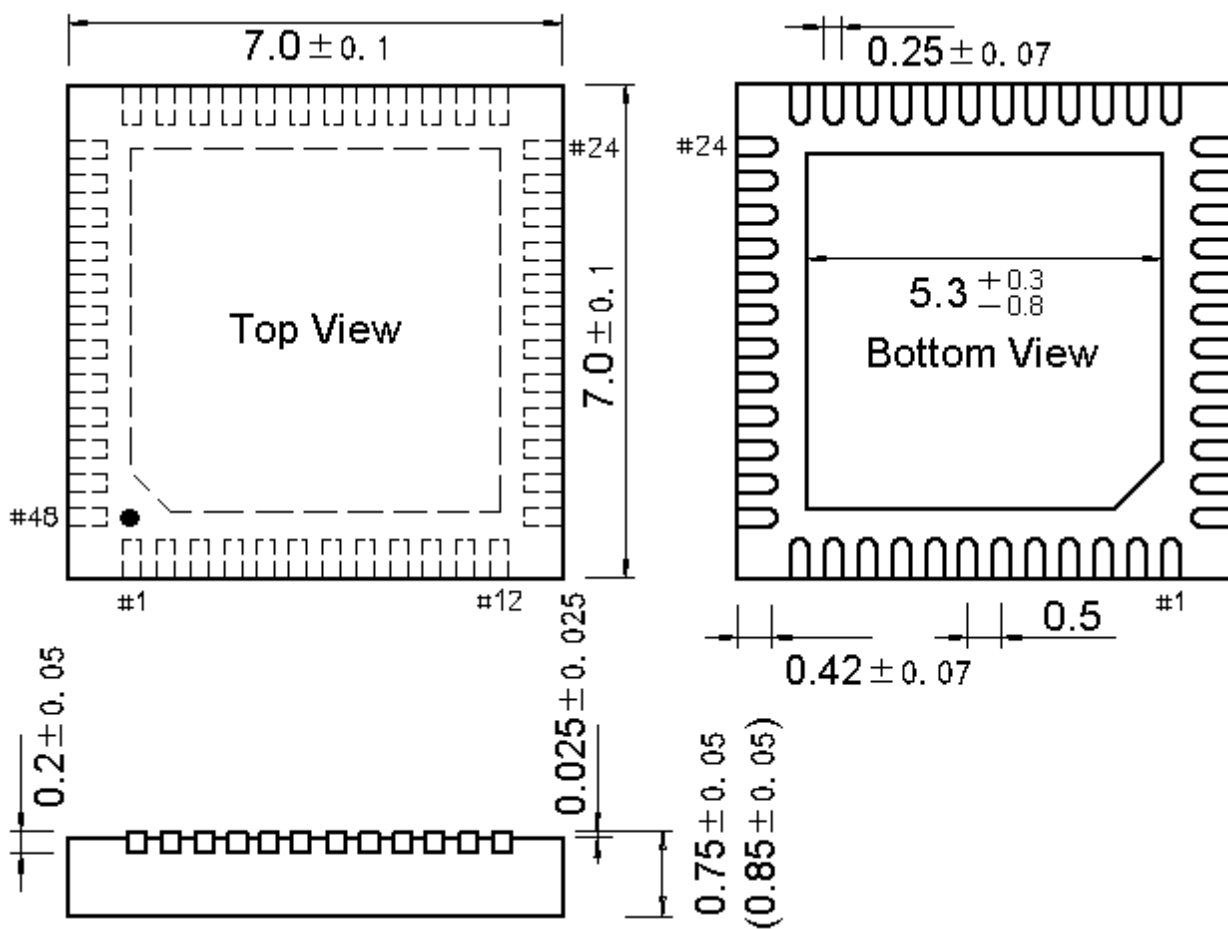
. QFN48-5\*5\*0.55-0.35



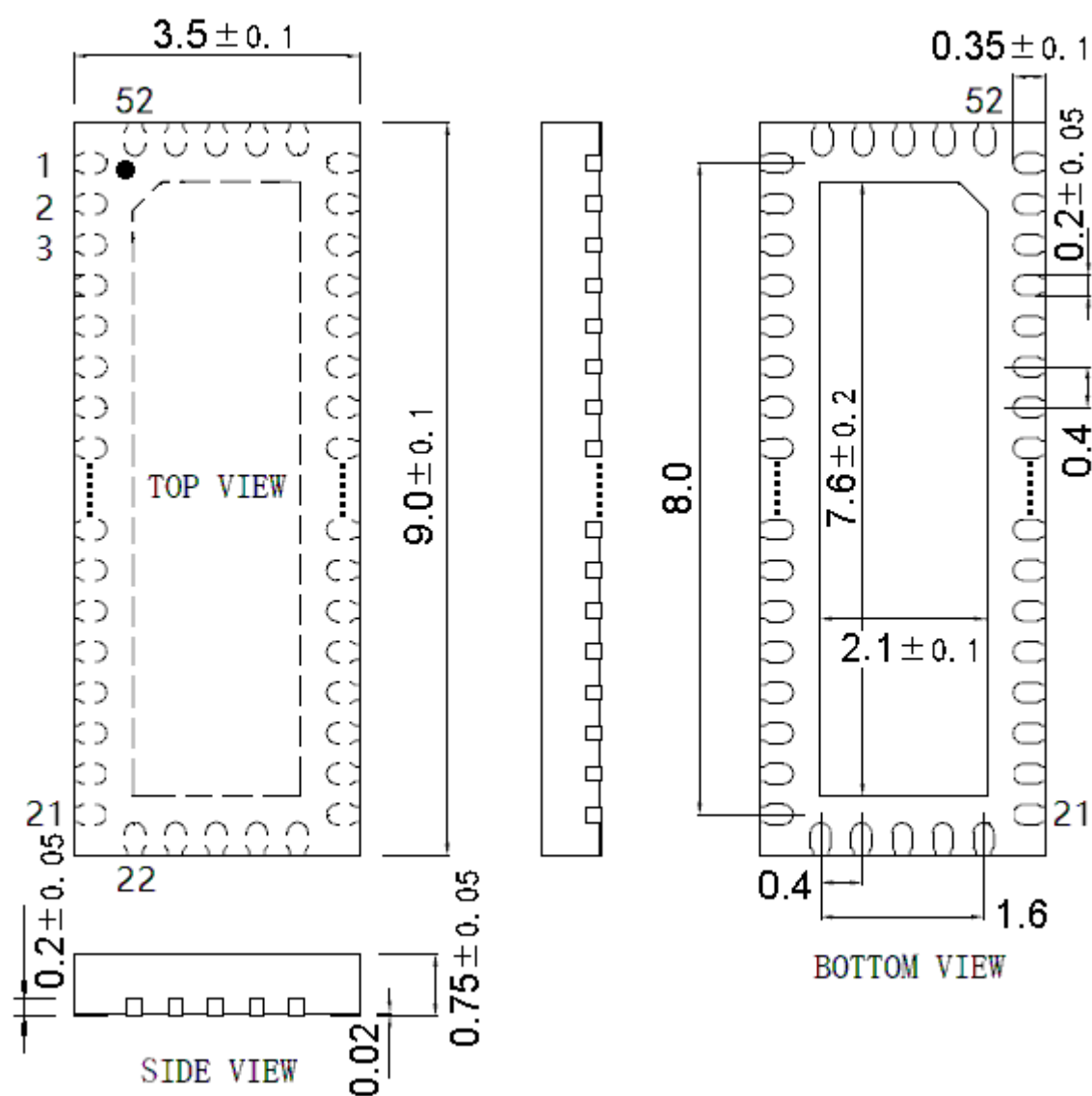
QFN48X6 (QFN48-6\*6-0.4)



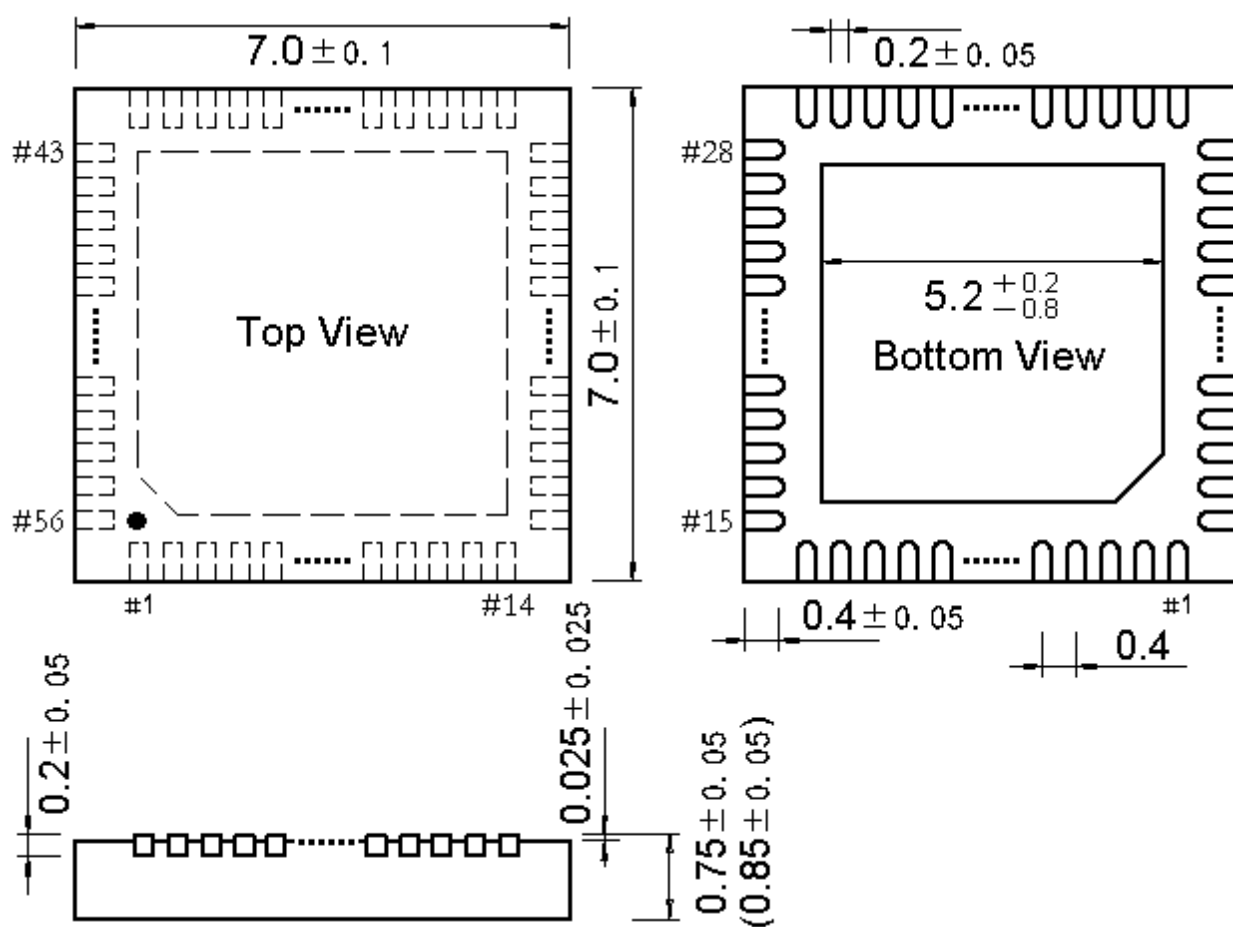
## . QFN48X7 (QFN48-7\*7-0.5)



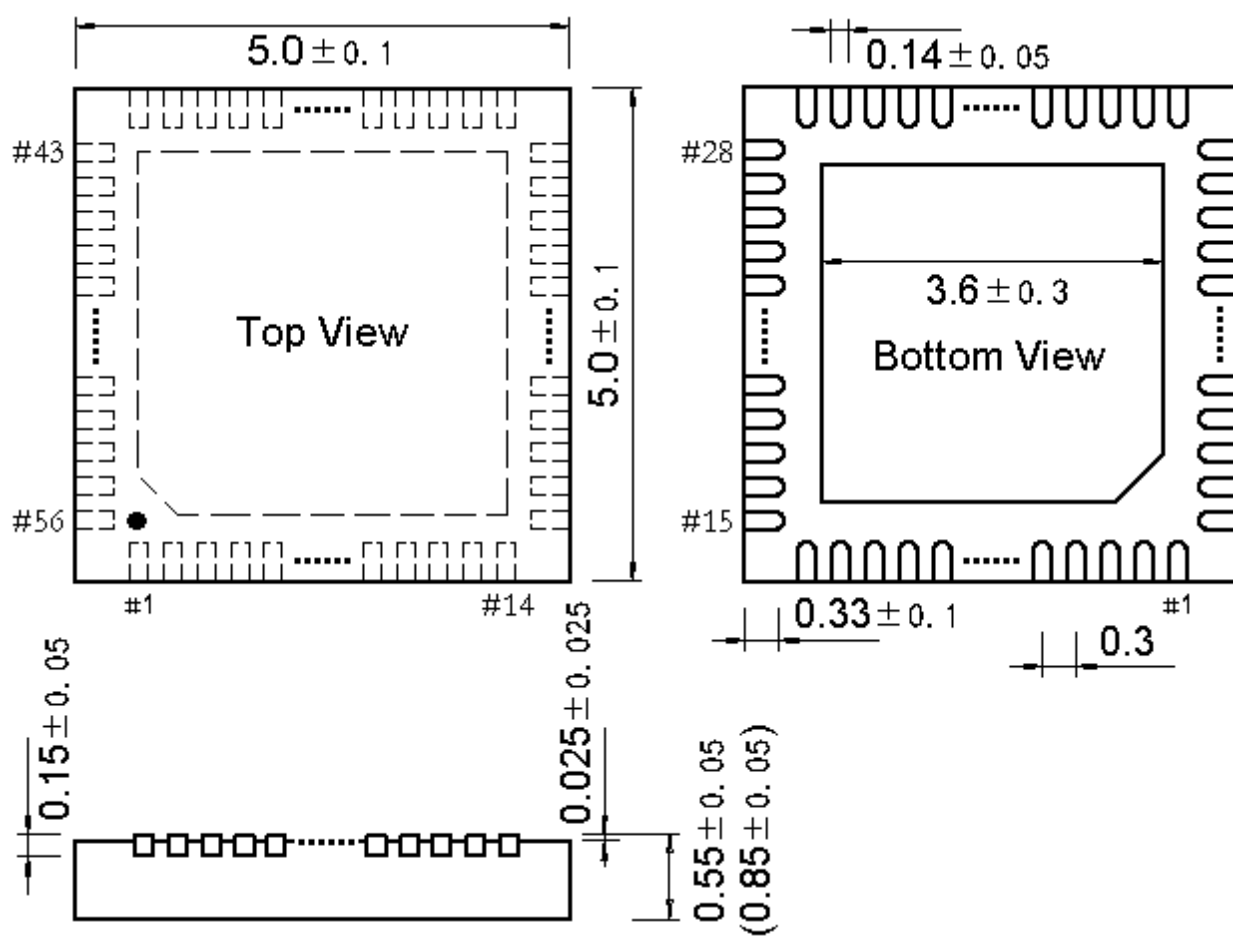
· QFN52X35X9 (QFN52-3.5\*9\*0.75-0.4)



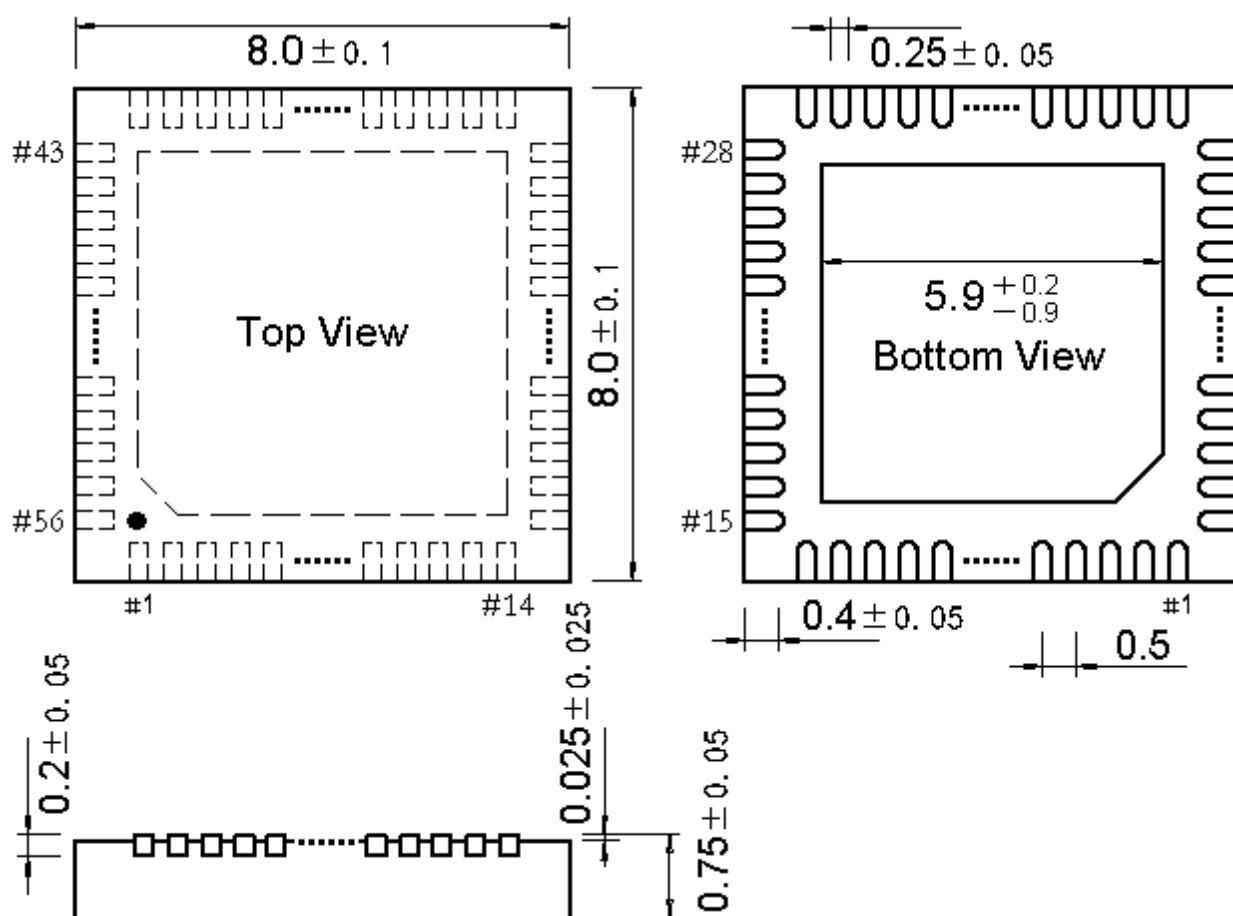
## . QFN56 (QFN56-7\*7-0.4)



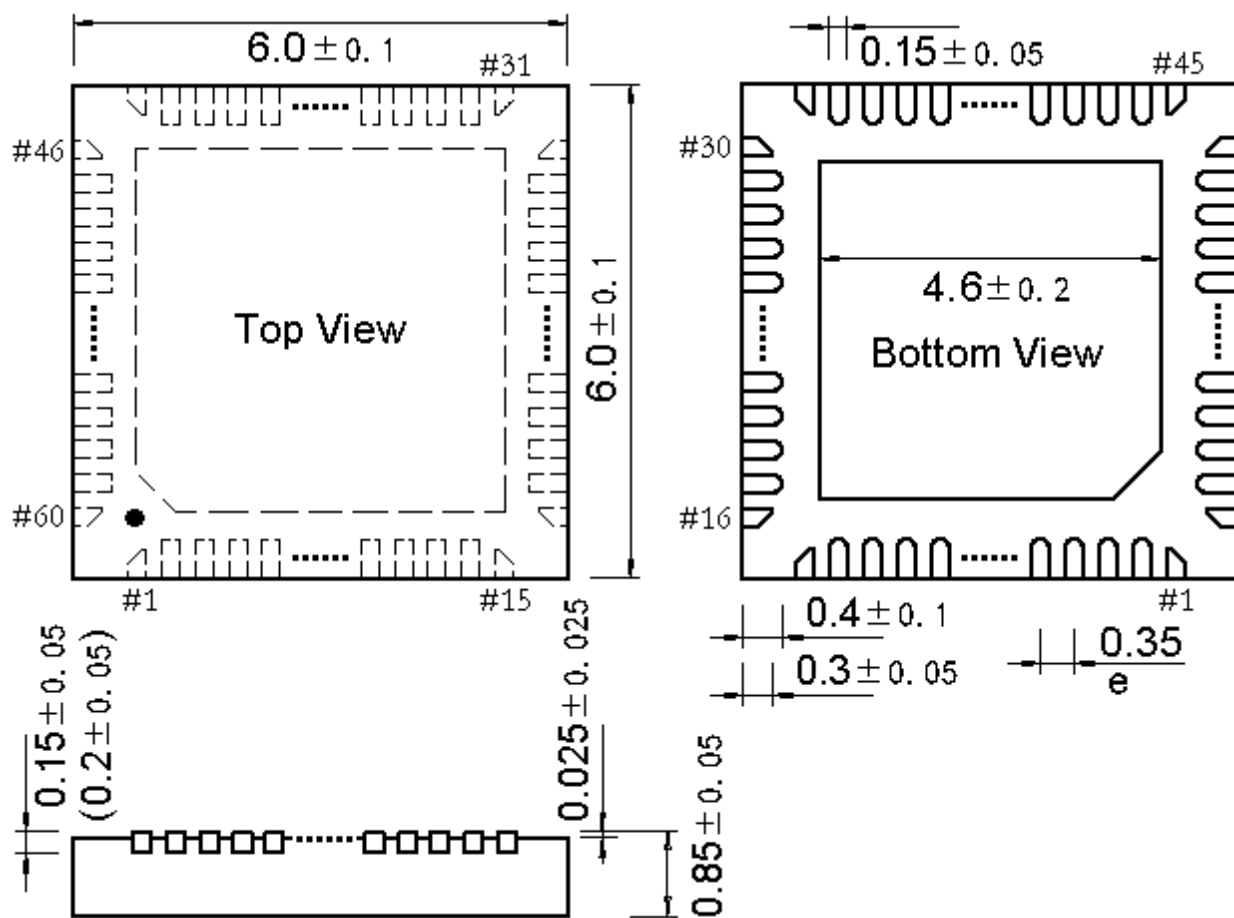
## . QFN56X5 (QFN56-5\*5-0.3)



## . QFN56X8 (QFN56-8\*8-0.5)

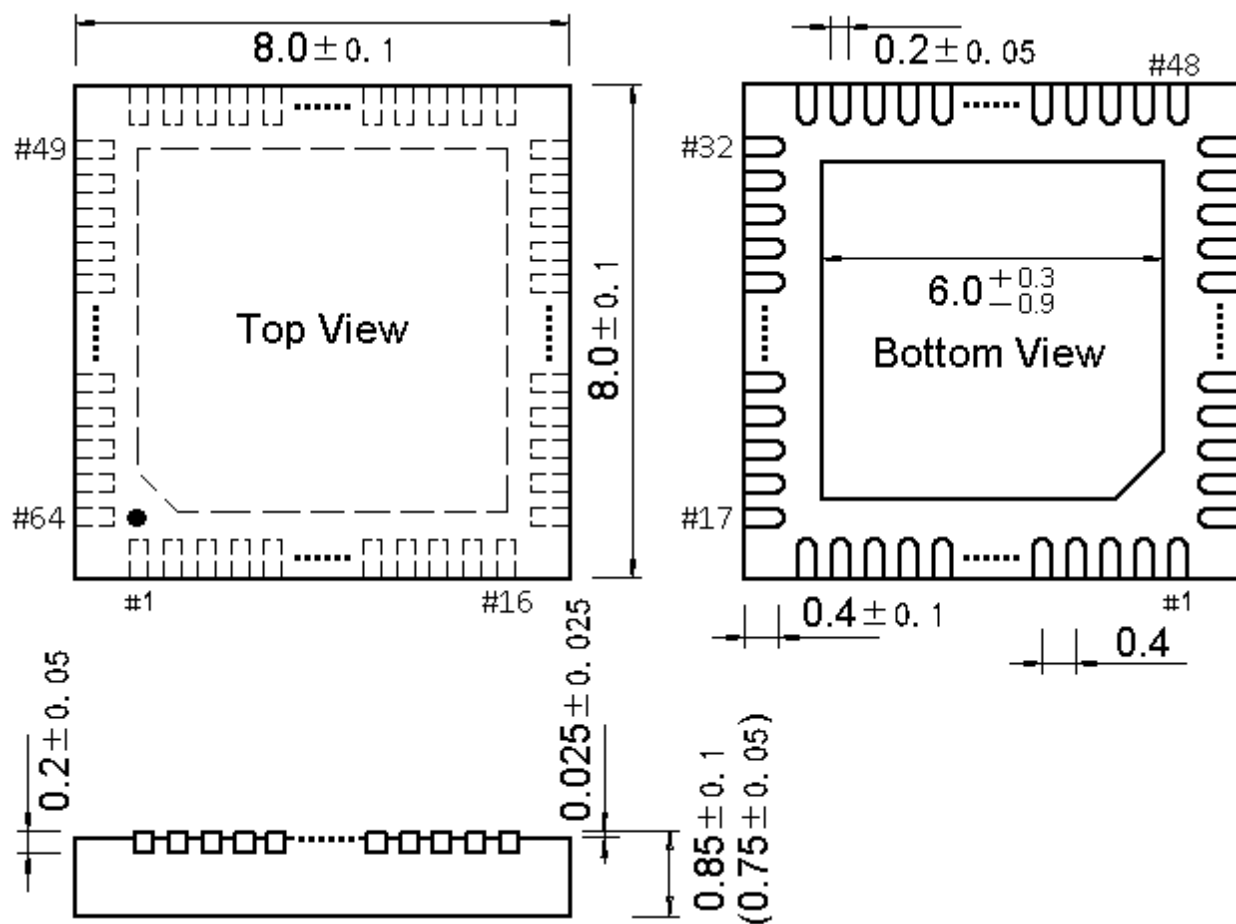


## . QFN60X6 (QFN60-6\*6-0.35)

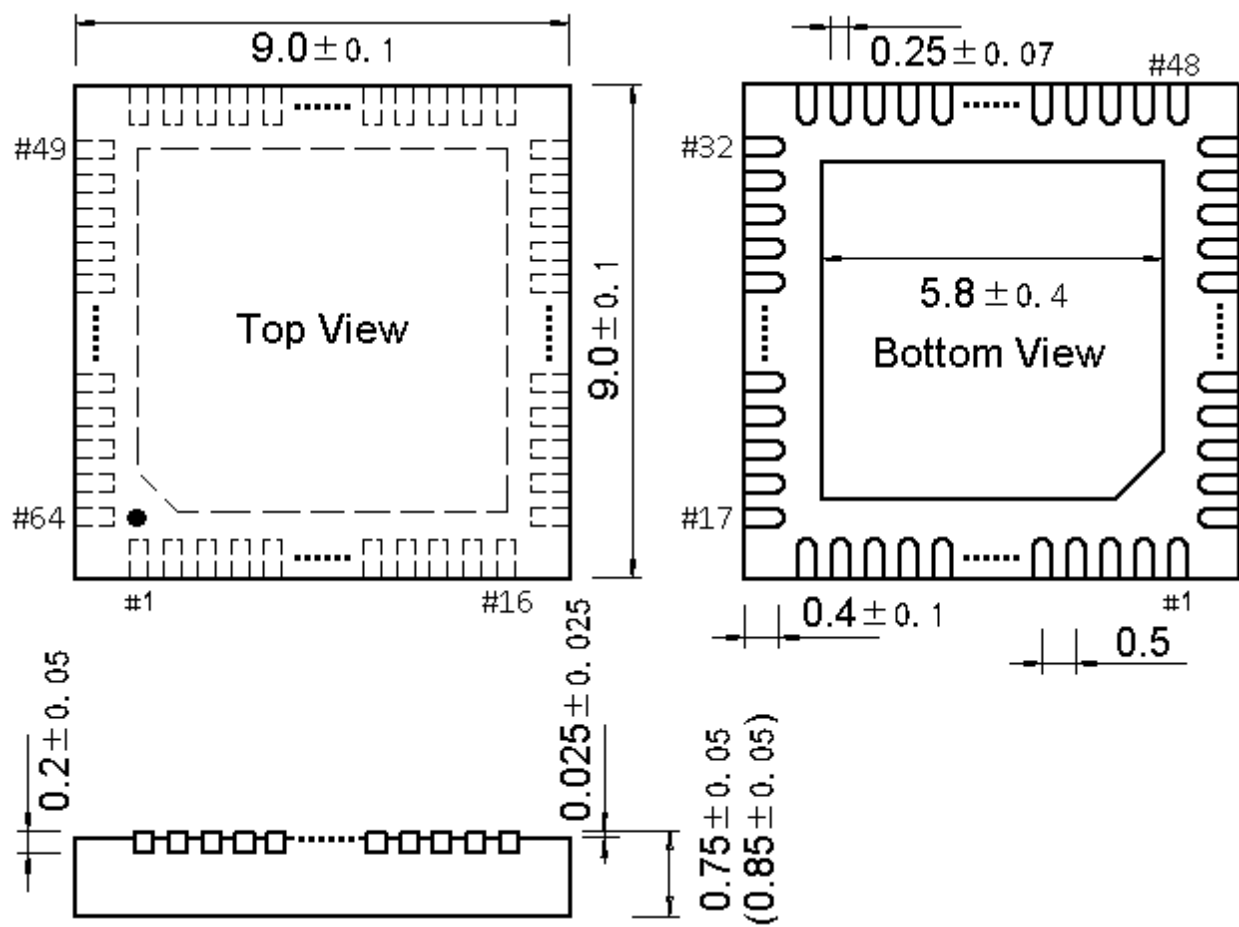




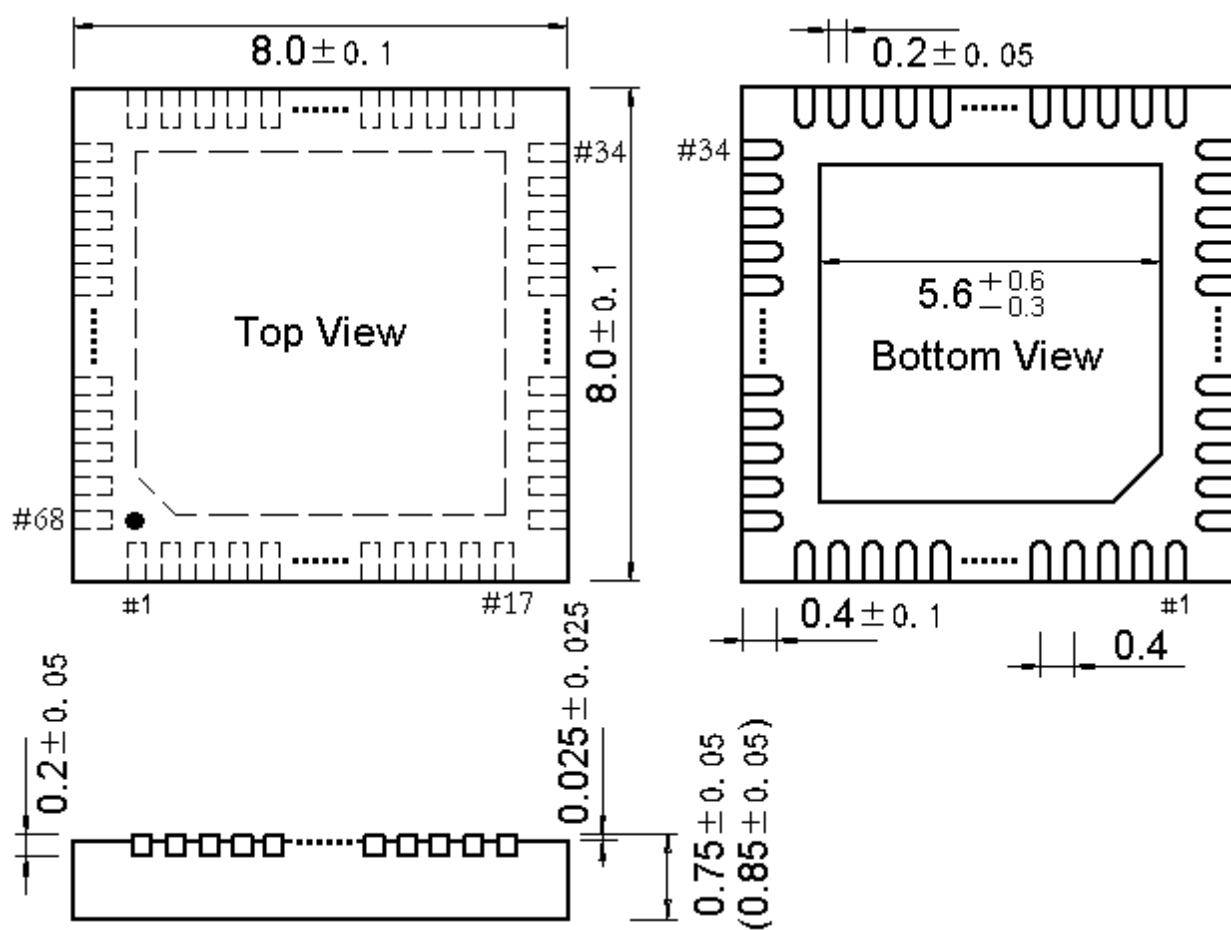
## . QFN64 (QFN64-8\*8-0.4)



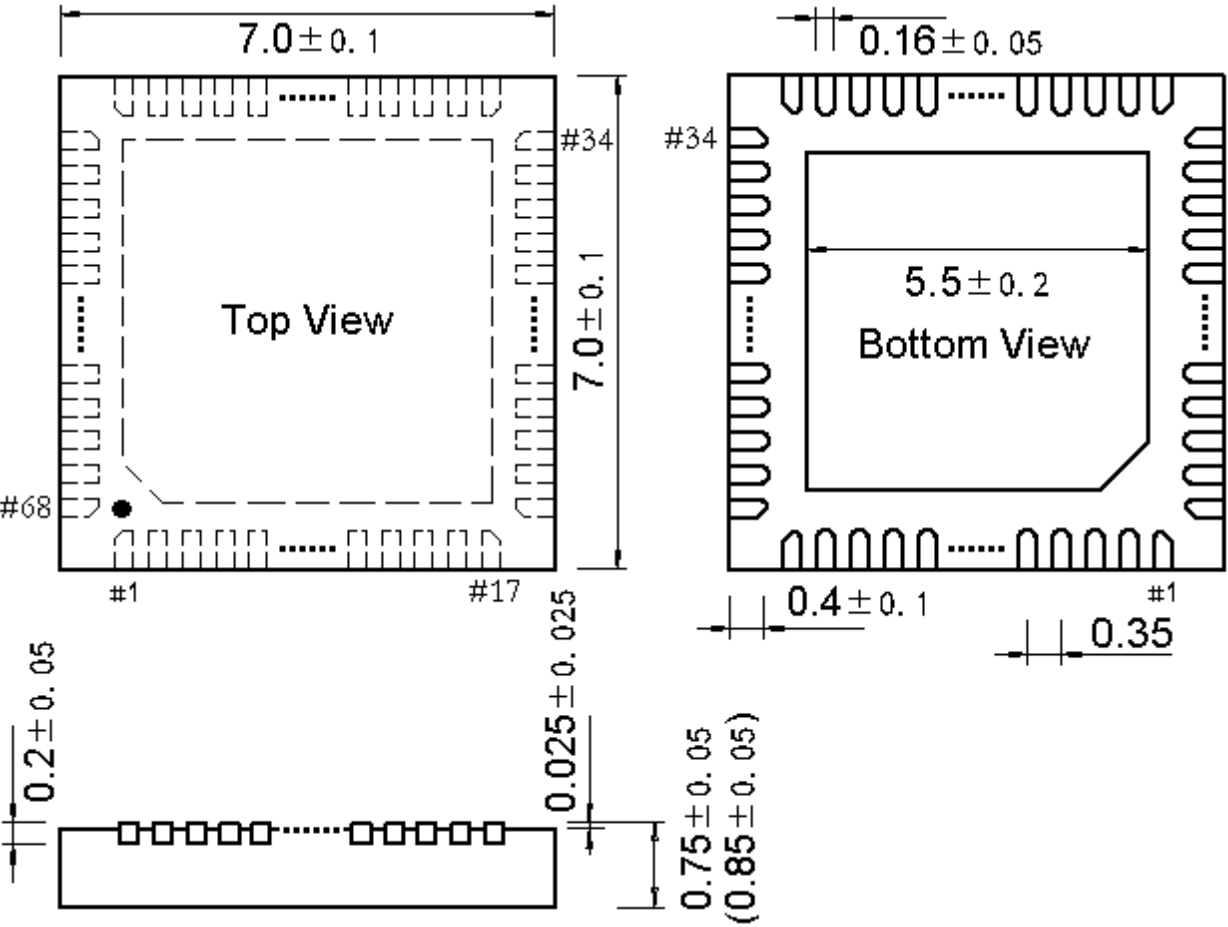
## . QFN64X9 (QFN64-9\*9-0.5)



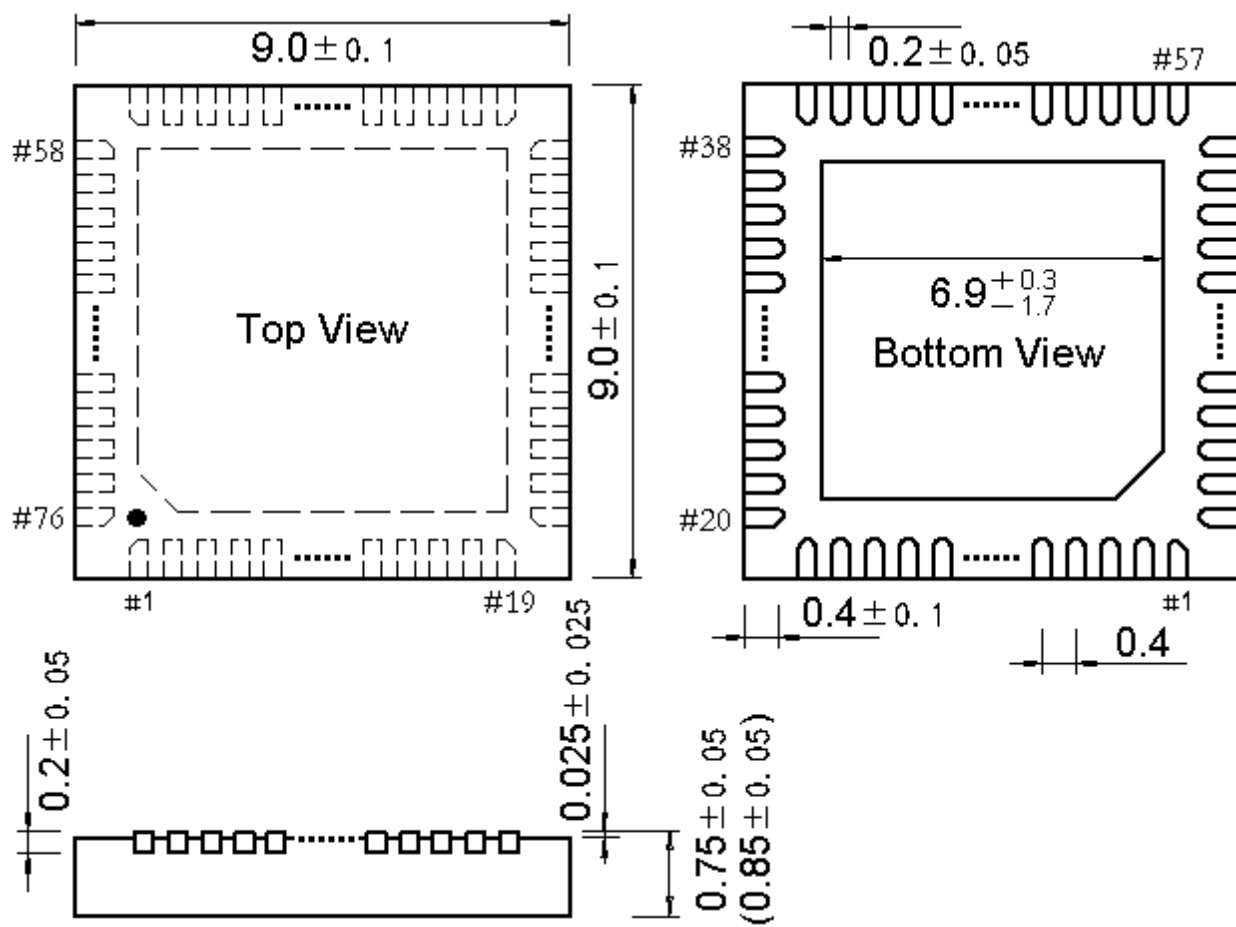
## . QFN68 (QFN68-8\*8-0.4)



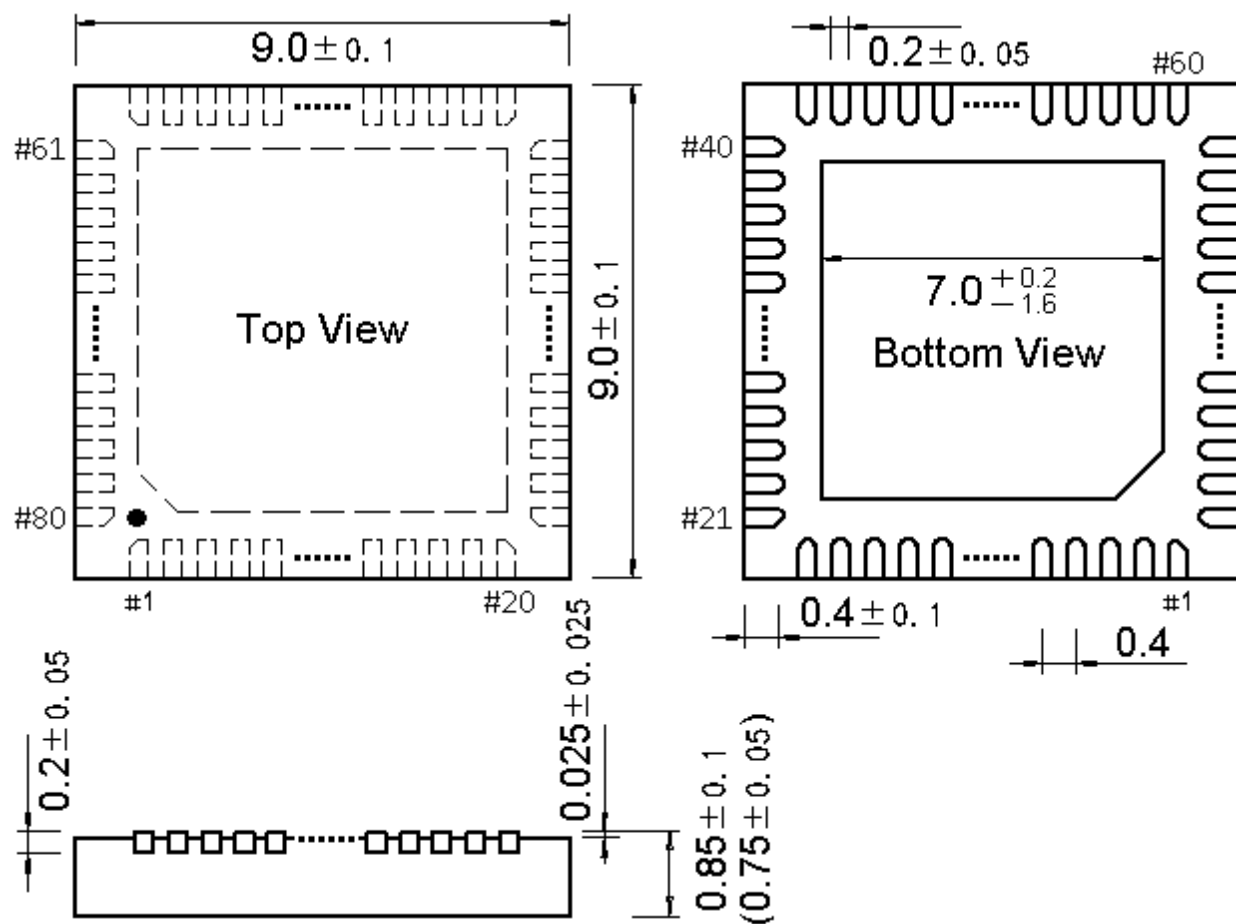
QFN68X7 (QFN68-7\*7-0.35)



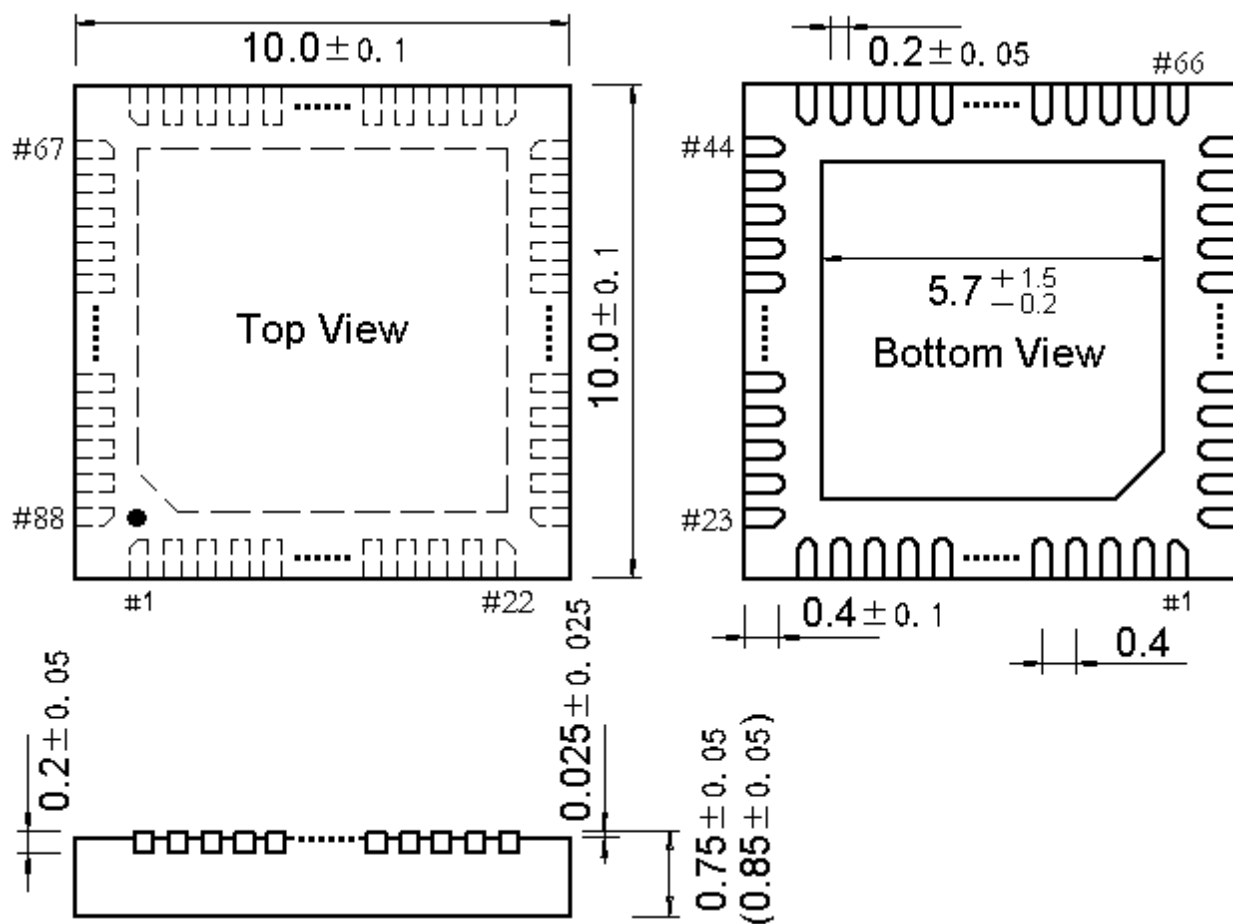
## . QFN76 (QFN76-9\*9-0.4)



## . QFN80 (QFN80-9\*9-0.4)



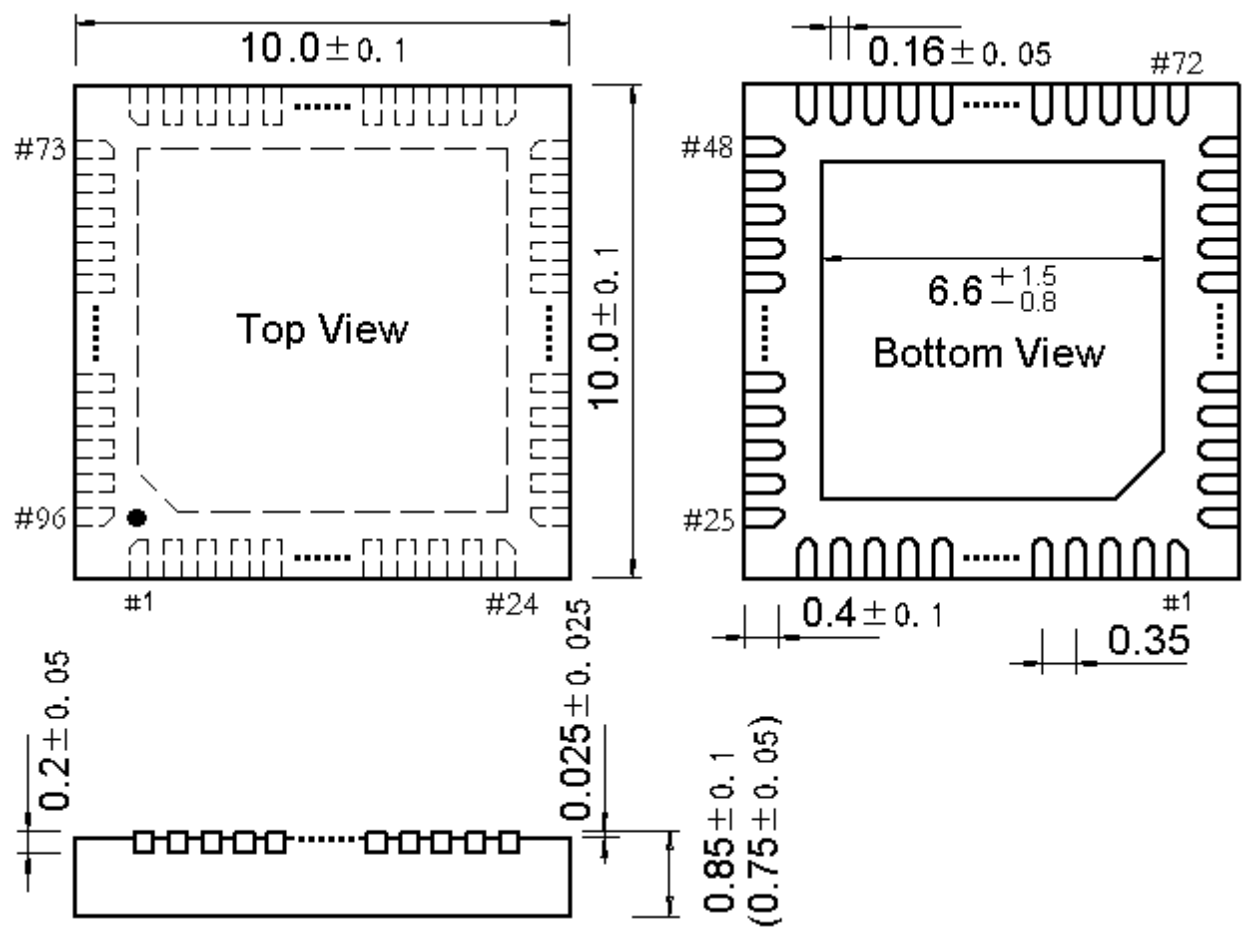
## . QFN88 (QFN88-10\*10-0.4)



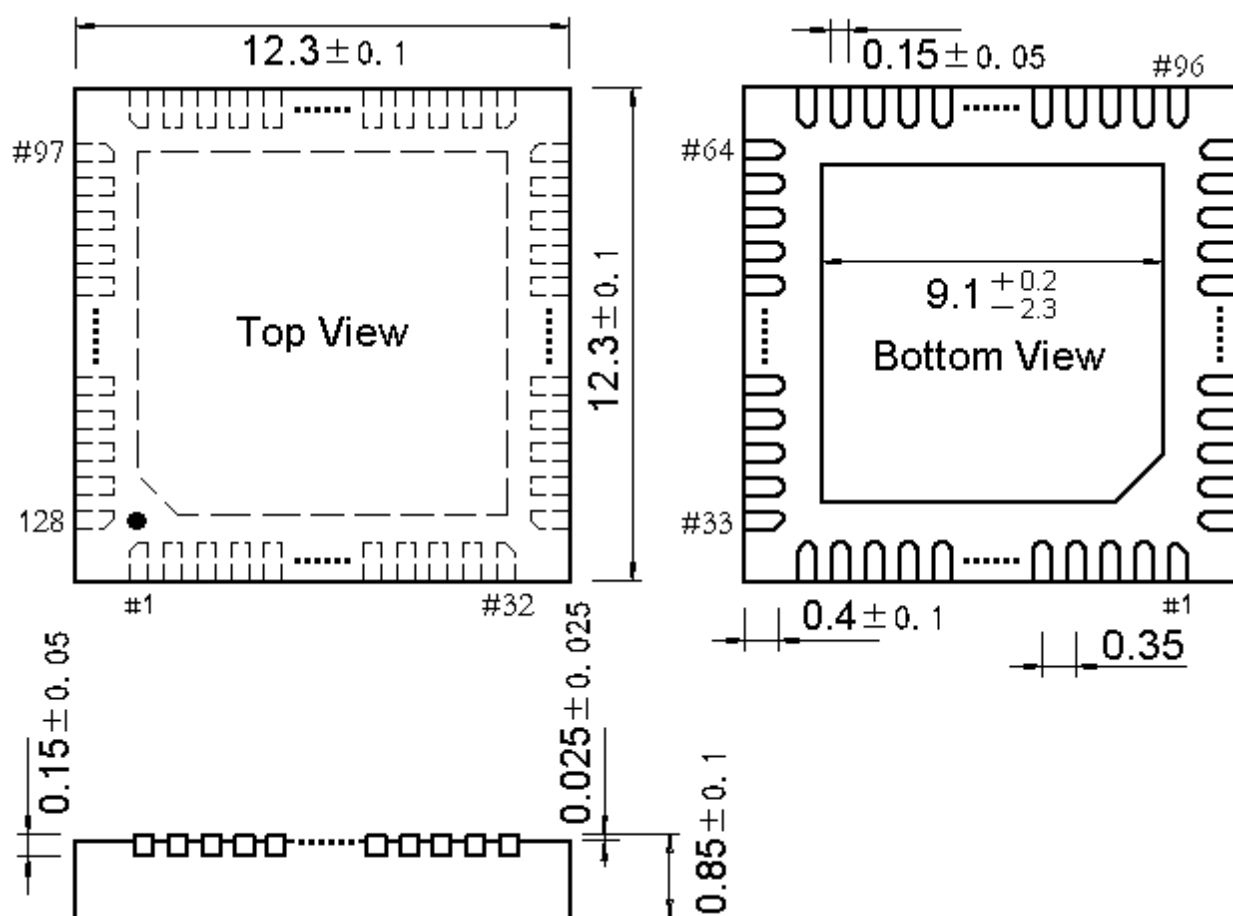




· QFN96 (QFN96-10\*10-0.35)

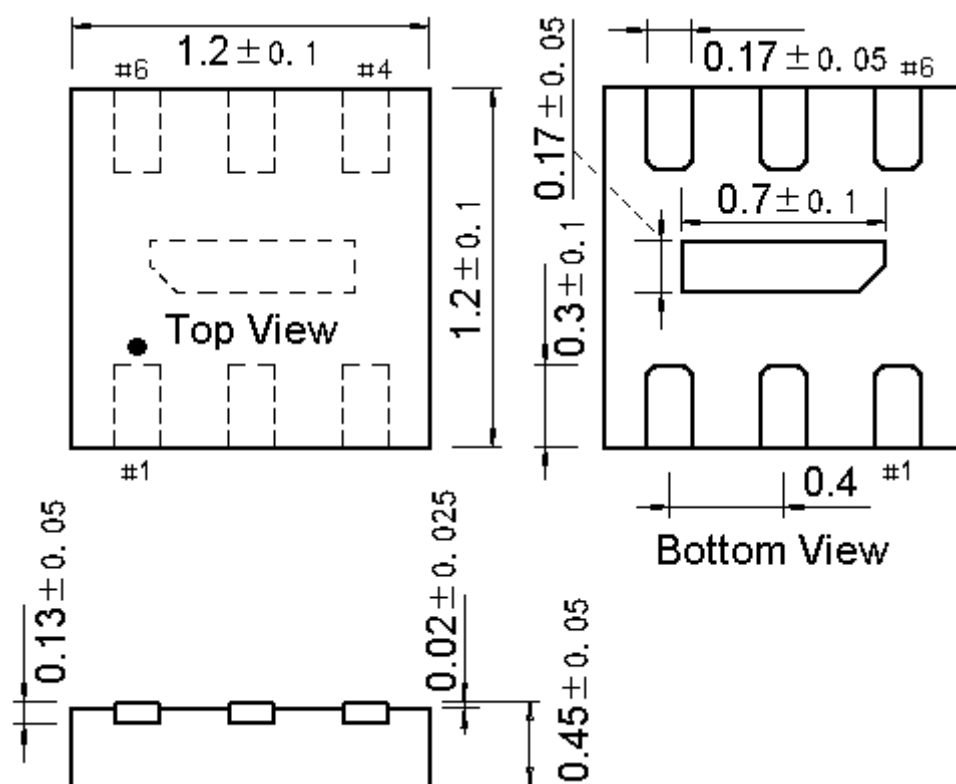


· QFN128 (QFN128-12.3\*12.3-0.35)

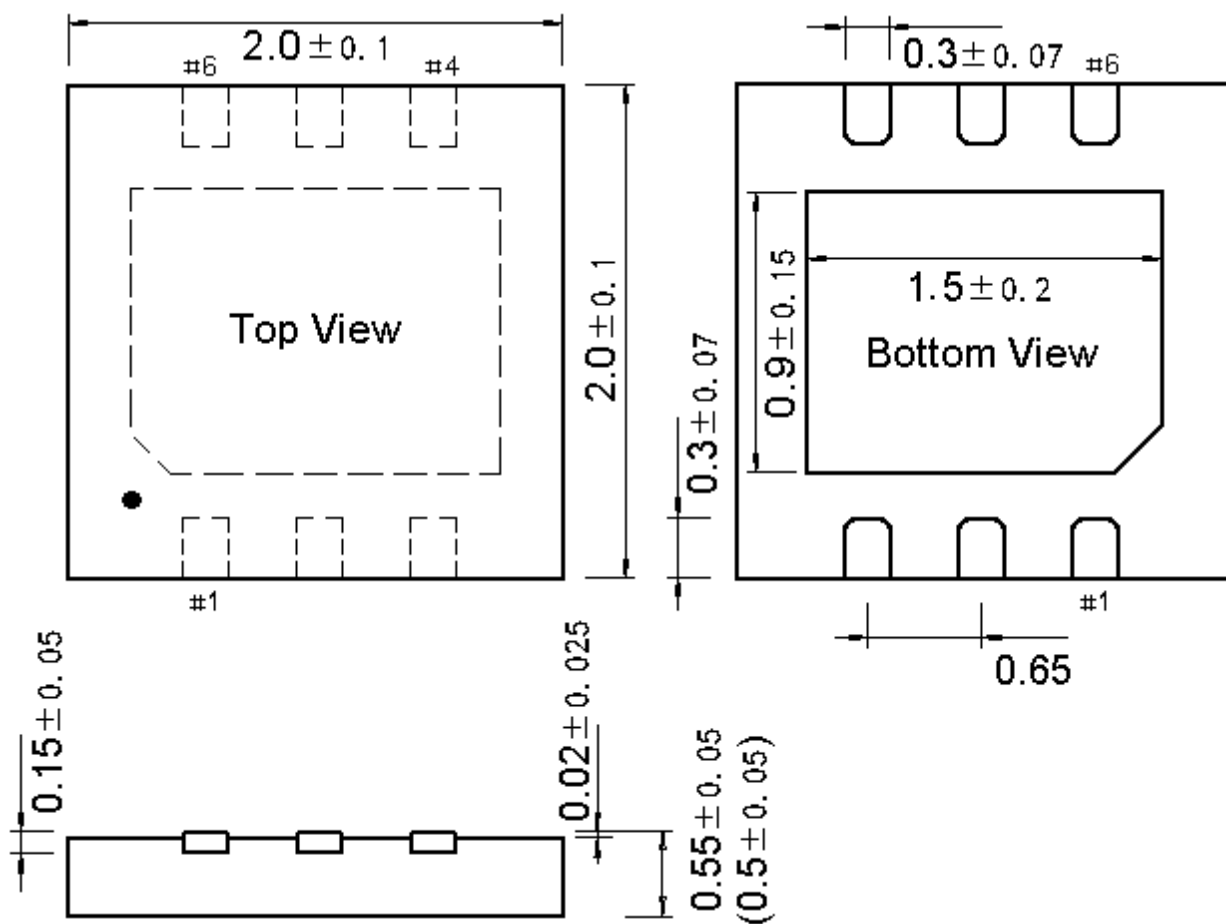


## DFN 类

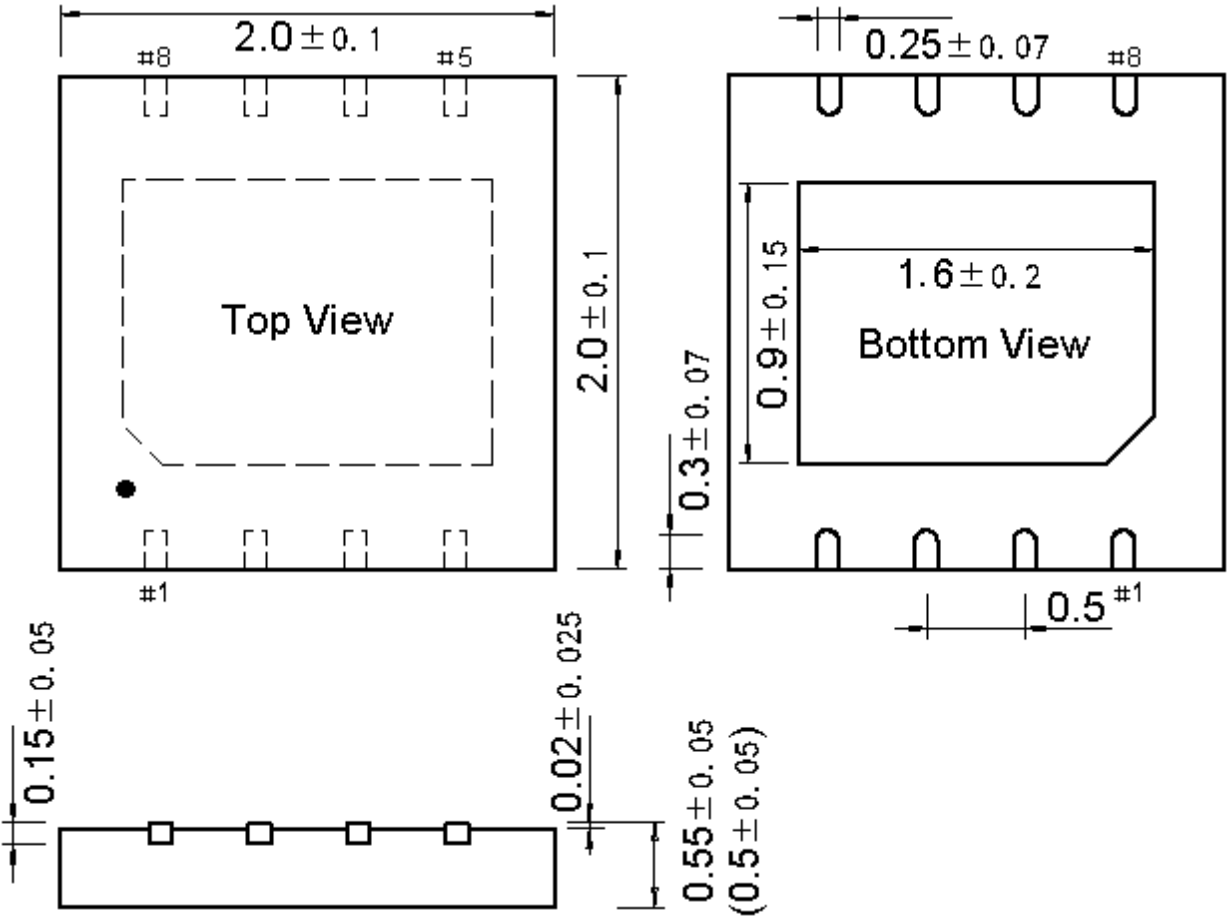
. DFN6X1.2 (DFN6-1.2\*1.2\*0.45-0.4)



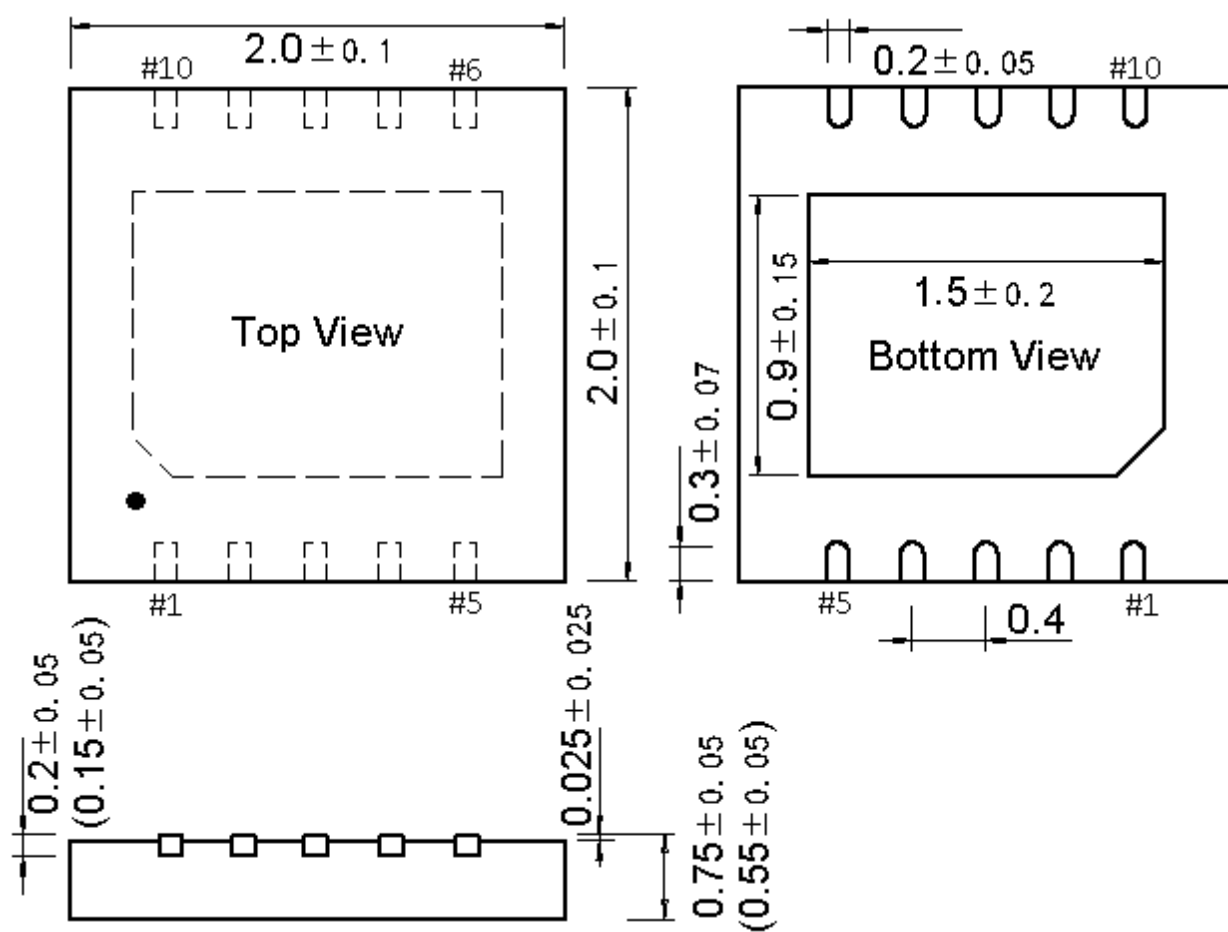
## . DFN6 (DFN6-2\*2-0.65)



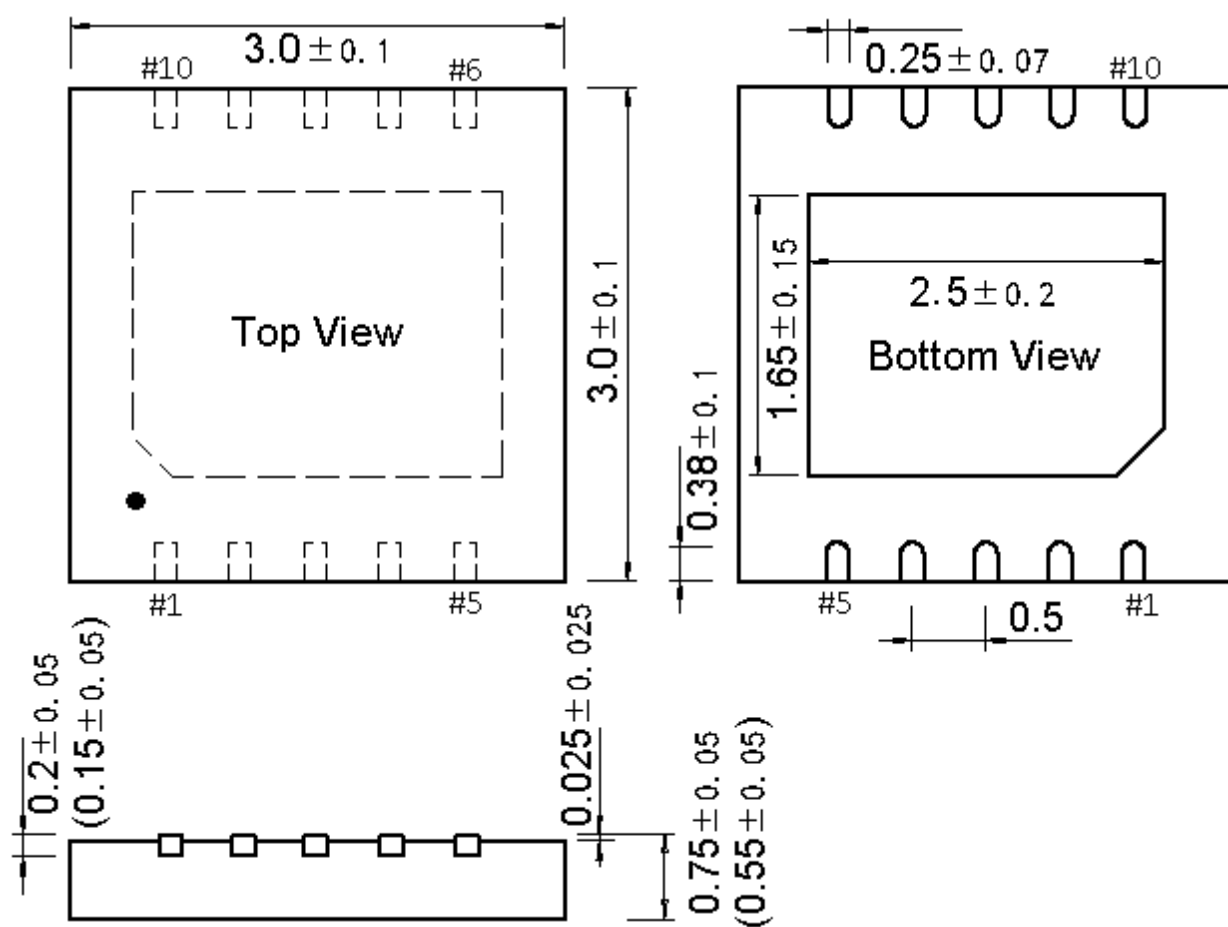
DFN8 (DFN8-2\*2-0.5)



## . DFN10 (DFN10-2\*2-0.4)

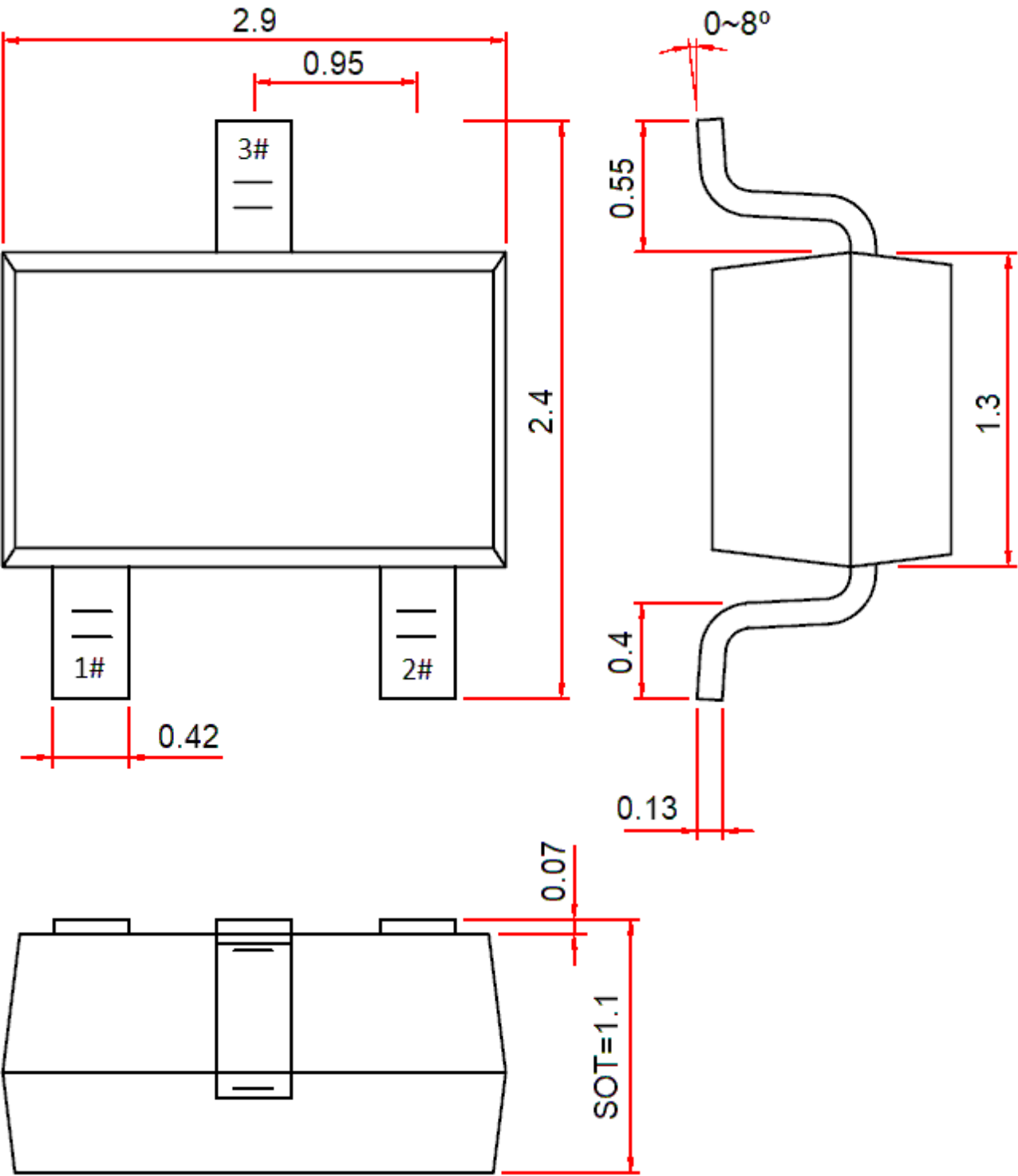


## . DFN10X3 (DFN10-3\*3-0.5)



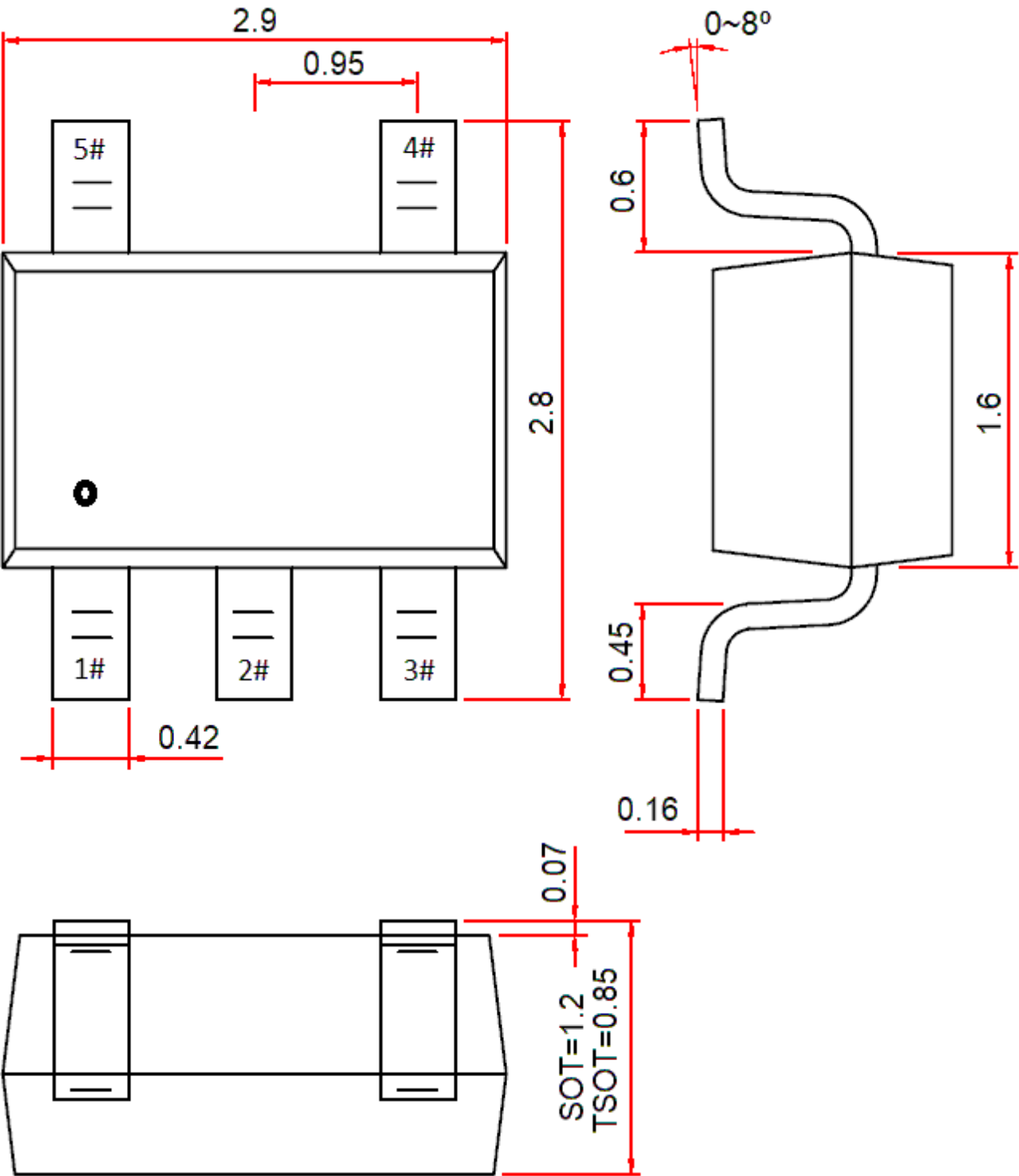
SOT 类

· SOT23-3

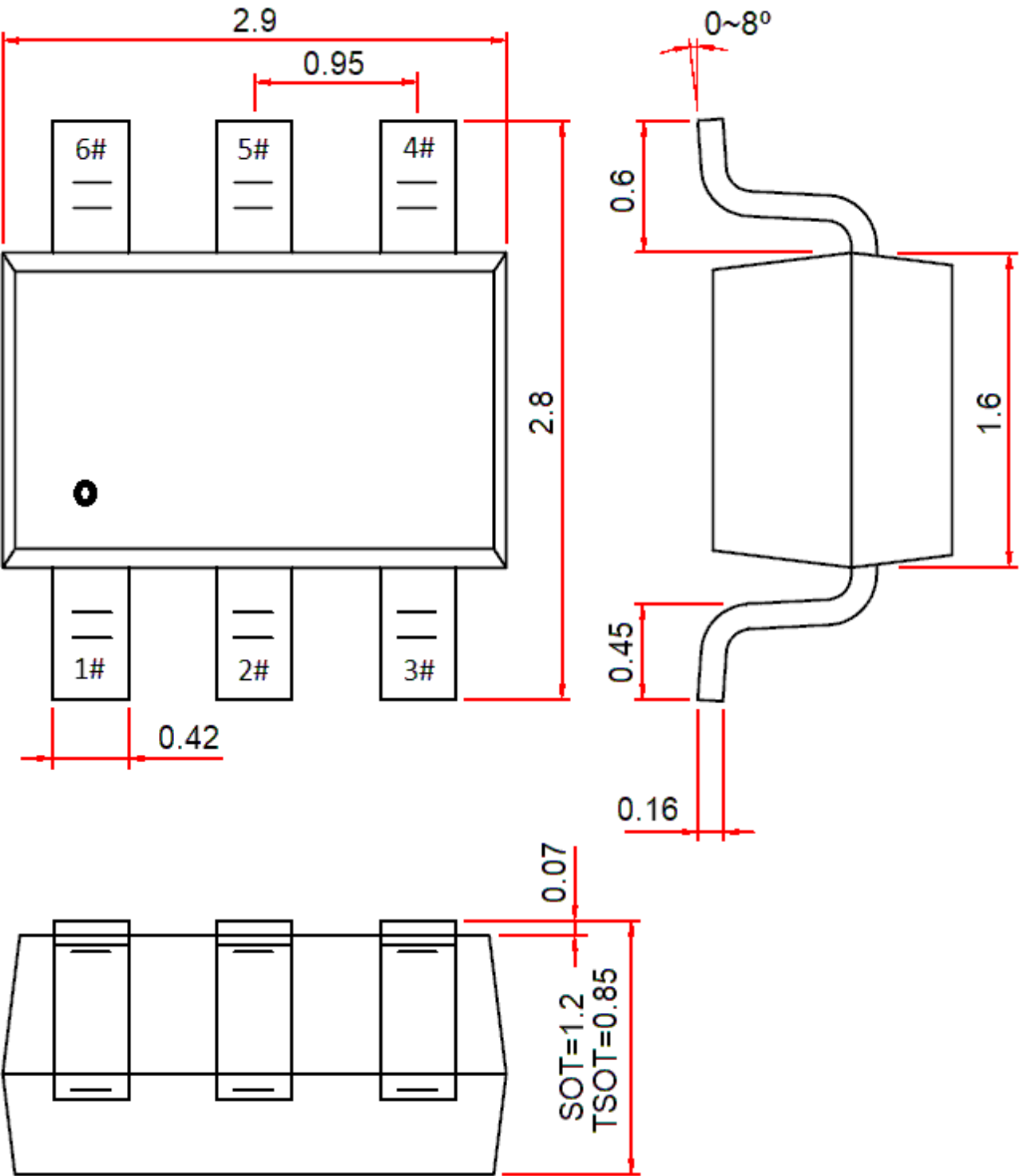




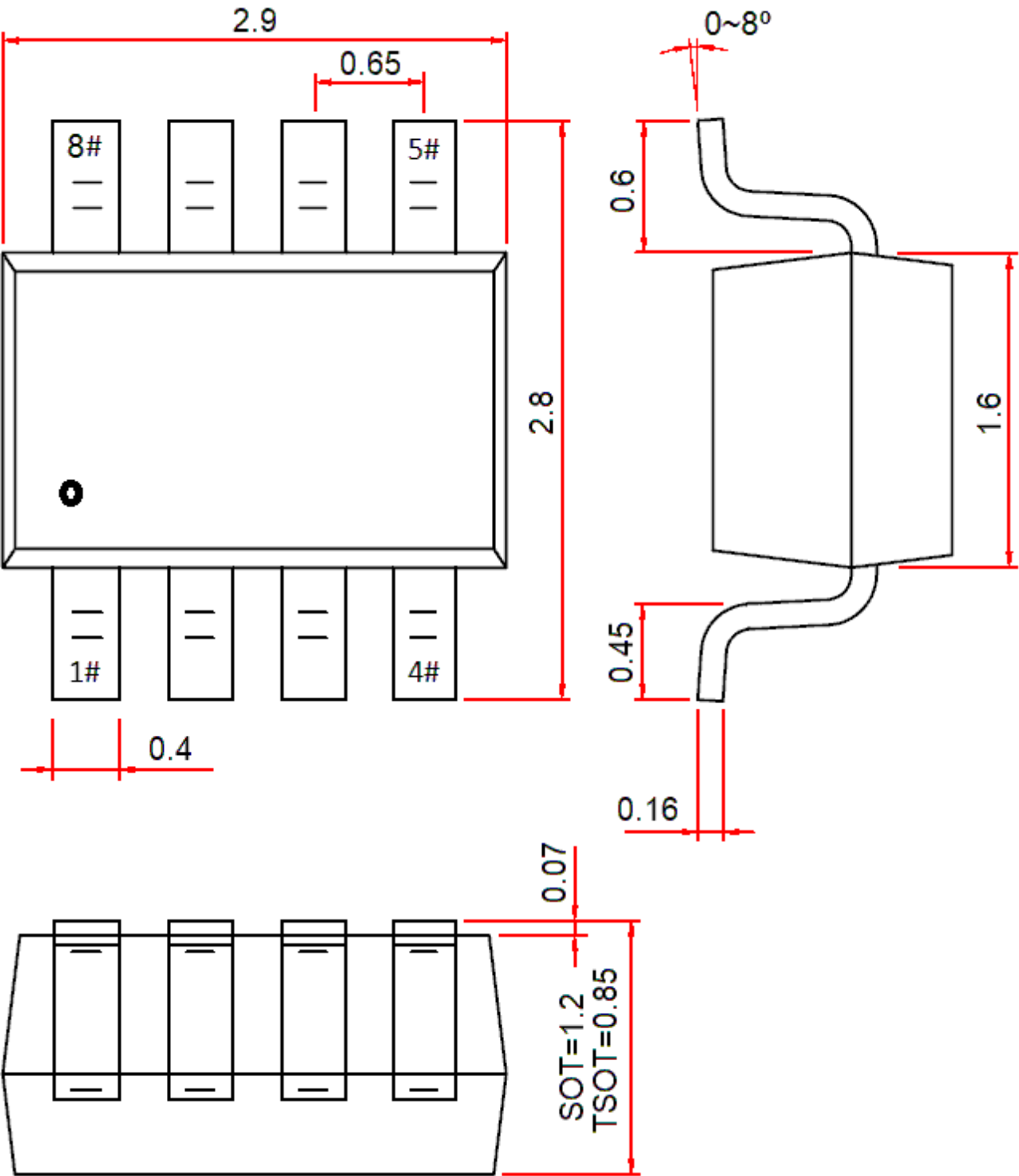
· SOT23-5、TSOT23-5L、SOT25



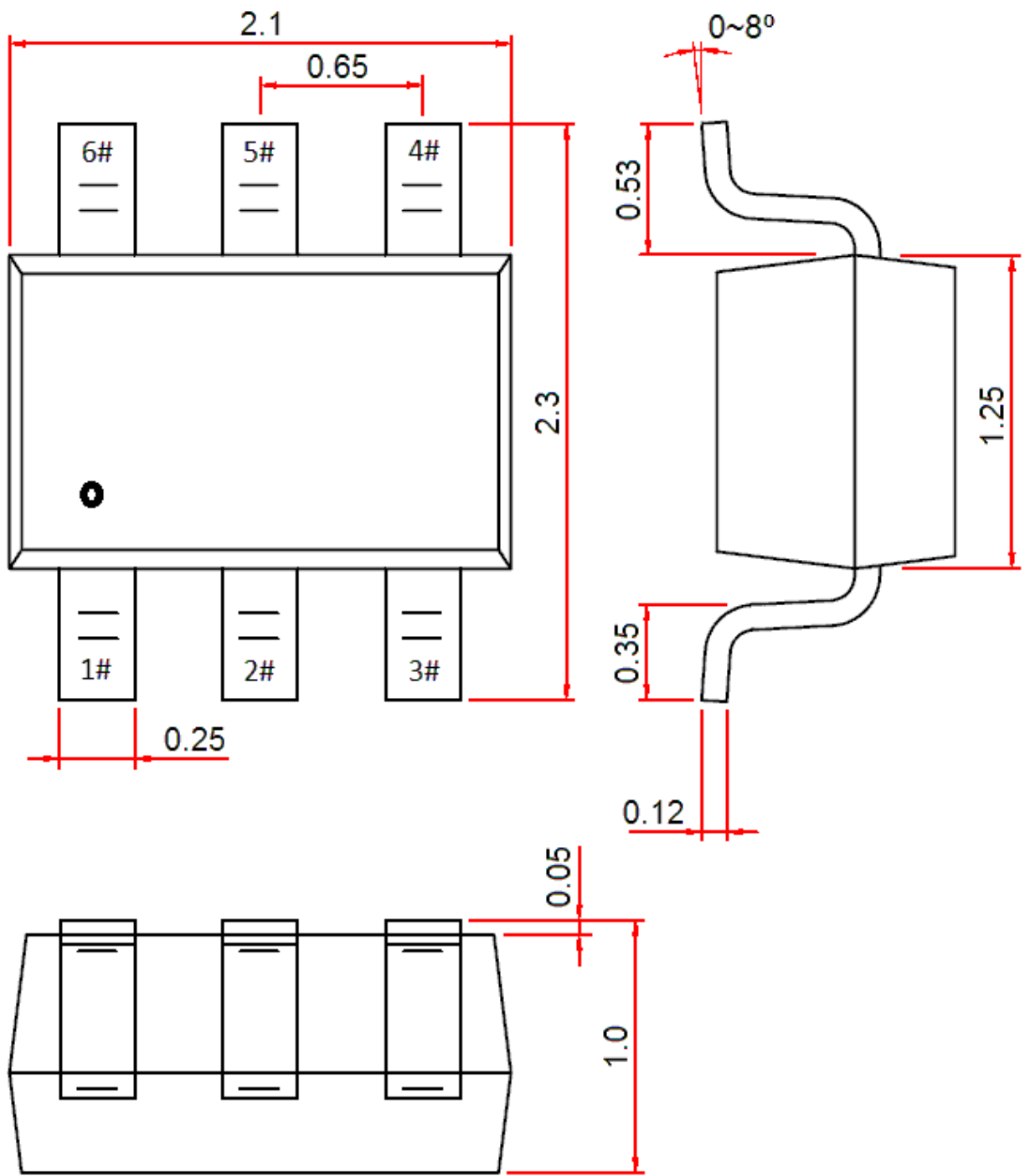
· SOT23-6、TSOT23-6L、SOT26



· SOT23-8、TSOT23-8L、SOT28



· SOT363 (SC70-6L)



SOT353 (SC70-5L)

